Limbayong FDP

LIMBAYONG FDP — CEO DECISION MEMO

HERMES, Project Director · v3 (final) · 19 October 2026

Field: Limbayong (+Bestari), Blocks G/J/R, ~120 km offshore Sabah, 900–1,200 m WD **Operator:** PETRONAS Carigali, 100% WI · **FID:** 28 Aug 2026 **Class:** FDP-structured screen, AACE Class 4/5 · 8 independent technical sections

> **v3 supersedes v1 and v2.** v1 was wrong on fiscal basis. v2 was built before the eight technical sections landed. v3 reconciles all eight, corrects two errors found inside them, and states the number that decides the project.

1. THE VERDICT — HOLD

Do not commit irrevocable capex until three numbers are known. On every central case in this study, Limbayong fails a 15% real hurdle, and most cases fail 10%.

Source	Basis	NPV10	IRR
Section 7 (contractor, with 60% investment allowance)	capex $2.9bn	**+$288M**	12.3%
Section 7 (contractor, allowance stripped)	capex $2.9bn	−$65M	9.5%
Section 7 (PETRONAS consolidated)	capex $2.9bn	+$1,592M	19.8%
Section 8 (base)	capex $2.9bn	+$313M	12.1%
HERMES reconciliation (consolidated, no allowance credit)	**capex $4.05bn**	**−$839M**	**4.8%**
Section 8 Monte Carlo (20,000 paths)	Class 4 inputs	mean −$727M	**P(NPV<0) = 80.5%**

The spread is not disagreement. It is three different questions: contractor-with-incentive, contractor-without, and sovereign-consolidated. All three point the same way — the project is marginal at best, and on a contractor basis it needs **US$96/bbl Brent** to reach 10%.

2. THE NUMBER THAT DECIDES IT: CAPEX

The largest disagreement inside this study is capex. Section 6 builds **US$4.05bn P50 (range $2.83–6.07bn)** from a WBS. Sections 7 and 8 both used **US$2.9bn**.

Capex	capex/boe	NPV10 @ $70 (consolidated)	IRR	Brent for 10% IRR
$2.83bn	$10.3	−$141M	8.8%	$76
$2.90bn	$10.5	−$180M	8.6%	$76
$3.43bn	$12.5	−$477M	6.6%	$86
$4.05bn (S6 P50)	**$15.0**	**−$839M**	**4.8%**	**$98**
$6.07bn (S6 +50%)	$22.0	−$2,281M	−0.2%	$133

US$15/boe is the industry-typical deepwater number. US$10.5/boe is not. Sections 7 and 8's $2.9bn sits below the global range; Section 6's $4.05bn sits inside it. Until PETRONAS's own estimate is seen, **the honest central case is $4.05bn**, and at

that number the project returns 4.8%.

3. THE THREE FINDINGS THAT CHANGE THE DECISION

Finding 1 — the hull is undersized, and the yard is already fixed. FPSO *Excalibur*: built 2007, Sevan-type cylindrical, 70×70×27 m, **300,000 bbl storage**, originally **~30,000 bopd** on Chestnut [M: DNeX Bursa filing 20 Aug 2026; Sevan press]. Limbayong design needs **40,000 bopd and 150 mmscfd**. That is a **31% oil shortfall, 50% storage shortfall, and no gas handling** — Chestnut was oil-only. Delivery is contractually fixed at **MMHE Pasir Gudang by 5 Nov 2026**. MISC bought the *form*, not the *capacity*. Conversion estimated **US$600M (range $475–890M)**.

Finding 2 — this is a gas field with an oil rim, not an oil field with a gas cap. Section 1 derives gas GRV at **1.86× the oil-leg GRV**; most of the 790.9 bcf is non-associated. Section 7 quantifies the consequence: **oil-only NPV −$29M / 9.7% IRR; gas adds $317M — 110% of total NPV.** And there is **no public gas sales agreement and no defined export route**, with a ~140–180 km subsea tie to SSGP Kimanis the only credible path.

Finding 3 — 2029 first oil is not credible. Section 6 gives **Q4 2030** as P50 (51 months from FID). Kikeh took 43 months, Gumusut-Kakap 44, Malikai 46 — all clean newbuilds. A used cylindrical hull converting in a yard with no cylindrical-conversion reference beating all three by 7–10 months is not a schedule.

4. TWO ERRORS FOUND INSIDE THIS STUDY

I audit my own subagents. Two defects:

Error A — a 2011 newspaper article presented as current evidence. Section 8 cited an **"MMHE RM952M MISC contract (19 Oct 2026)"** as partial conversion coverage. That article is from **2011**; 19 Oct 2026 is the page render date. Its two contracts are **MT Onozo → FPSO Cendor (Terengganu)** and **SS Tenaga Satu → FSU Lekas (Melaka)** — neither is Excalibur, neither is Limbayong. [Verified: article body references FY ended 31 March 2011.] **Claim withdrawn.**

Error B — a capex used before it was built. Sections 7 and 8 ran economics on $2.9bn before Section 6 delivered $4.05bn. That is a sequencing failure by me. Section 2 of this memo is the correction.

Neither error changes the direction of the verdict. Both change its confidence.

5. WHAT THE SUBSURFACE ACTUALLY SAYS

Attribute	Value	Class
Structure	30 km × 2.5 km elongate toe-thrust ridge, aspect 12:1	[M] SPE 210079-MS
Complexity	"Industry top decile"; thin-bed/thick-sand intercalation; **lateral compartmentalisation**	[M] SPE 210079-MS
Appraisal before FID	**4 wells across 75 km² — one per 7.5 km**	[M]
P50 STOIIP	517 MMbbl (P90 284 / P10 1,150)	[D]
P50 recoverable	134 MMbbl vs **139 booked** — reproduces to 4%	[D]
P90 downside	**51 MMbbl = 5.1 MMbbl/well across 10 wells**	[D]
Drilling margin	**<0.5 ppg; MPD mandatory**; deviated into minimum horizontal stress	[M] SPE 210549-MS
Well cost	**$67.1M/well P50**, $762M total, 714 days	[D]

The 139 MMbbl booked figure is **internally honest** — a correctly derived P50. The problem is it is being read as a floor. **A 12:1 faulted ridge appraised by four wells, drained by ten, with lateral compartmentalisation stated as fact in the sanctioning paper, has a P90 of 51 MMbbl.** That tail is what kills the NPV.

6. WHAT IT TAKES TO MAKE IT GO — THE FIVE CONDITIONS

#	Condition	Status	To resolve	Cost / time
1	Hull ≥40 kbopd + gas handling after conversion	**UNKNOWN**	Conversion scope; class survey on hull	$8–12M / 4–5 mo
2	Capex ≤$2.9bn	**FAILS on S6**	Independent cost assurance	$2–3M / 3 mo
3	10 wells ≥35 kboe/d; P90 ≥120 MMbbl	**UNKNOWN**	One appraisal well on lateral continuity + DST	$93–125M / 9–14 mo
4	Gas sold ≥$6/mmbtu with defined route	**UNKNOWN**	GSA; SSGP Kimanis tie decision	$40–60M / 6 mo
5	Brent ≥$96 sustained	**MET today at $105; a price bet**	Nothing controllable	—

Score: 0 TRUE, 3 UNKNOWN, 2 FALSE. Section 8 reached the same score independently.

7. THE ACTION

HOLD irrevocable capex. Open a six-month assurance window. Gate 1 on 1 April 2027.

Three deliverables by that date, total ~US$60M:

1. **Independent cost assurance** against the $4.05bn WBS. If real capex > $3.5bn → NO-GO or re-scope.
2. **Conversion scope to lump-sum** at MMHE, including gas handling and the 40 kbopd topsides. An open-ended conversion on a used hull is a cost-overrun generator.
3. **Gas sales agreement**, or a written decision that the gas is stranded and a re-run of economics without it. This is US$317M, not a rounding error.

If all three land favourably, GO with a full FDP. If any fails, the project is a stranded-hull write-down, not a development.

8. THE STRATEGIC CASE, STATED HONESTLY

No economic model in this study justifies Limbayong as a standalone oil project. **But it may still be correct.**

PETRONAS Carigali is 100% WI. It is the operator. It has never led a Malaysian deepwater development. A host facility at Limbayong is the **only** way to monetise Bestari and any future discovery in the L-B-P trend. A host built for one marginal field is a stranded asset; a host with a second customer is infrastructure.

That is a defensible reason to sanction. It is not the reason given publicly. The public framing is oil economics and a US$35M bargain. The real case is portfolio and capability. Both can be true — but a sovereign operator should say which one it is buying, because they carry different hurdle rates and different failure modes.

9. WHAT THIS IS NOT

No PETRONAS reservoir model, well proposal, AFE, FDP, or gas contract was available. Every cost is benchmarked; every reservoir figure is analogue-derived; fiscal terms are generic (1993 deepwater PSC vintage assumed). Section 8 sets out the US$170–240M assurance programme a real FDP would require against the 12-month institutional norm. **Treat the verdict as a structure for a decision, and the numbers as bounds awaiting the operator's own data.**

APPENDIX A — THE EXXONMOBIL CALIBRATION

The requested toolkit exists. It is an **equity-analyst toolkit for modelling XOM's corporate earnings**, not a field-development model: no DCF, no IRR, no PSC, no decline curve, no WBS. What it publishes, and what was used:

Item	Value	Source
Brent earnings sensitivity	+$1/bbl = +$700M annual Upstream	Sensitivities 2Q26, p3
Implied effective take	**55.4%** ($0.446/boe per $1/bbl)	derived from 4.3 Moebd
Constant price basis 2024	**$65/bbl Brent** flat real	Sensitivities p4
Constant price basis 2030	$75.40/bbl (2.5% inflation)	Sensitivities p4
Upstream unit earnings	~$10/oeb at $65 flat	Modeling Basics p8

It corrected us twice. First: my model's implied government take was 42.1% against XOM's disclosed 55.4% — a 13-point gap that flattered the project. Second, and decisive: a 55% take is the **wrong basis** for a 100%-owned sovereign field, which is why contractor and consolidated economics diverge so violently here (5.0% vs 17.3% IRR on the same field).

What this model does that the toolkit cannot: project DCF, IRR, NPV, PSC cost recovery and profit split, explicit decline, capex WBS, fiscal regime. A corporate earnings bridge is not a project gate. Using its published sensitivity to calibrate the gate is exactly right.

APPENDIX B — SECTIONS

S1 Subsurface · S2 Reservoir & Production · S3 Drilling & Wells · S4 Facilities & FPSO · S5 Subsea & SURF · S6 Cost & Schedule · S7 Economics (contractor-basis fiscal model) · S7X Economics (calibrated) · S8 Risk & Decision. Supporting: FACT_BASE.md · ceo_model.py · evidence/ (XOM toolkit PDFs, DNeX Bursa filing).

S1 — SUBSURFACE & STATIC MODEL

Field: Limbayong (PSC: Limbayong + Bestari), Block G/J/R, NW Sabah, 5.888°N 114.510°E, water depth 900–1,200 m. **Purpose:** test whether the 28 Aug 2026 sanction was right, on the static model alone. **Data regime:** four appraisal wells since 2002 [M: SPE 210079-MS]; **no public seismic, no public logs, no public core, no public PVT.** Everything below is either the operator's published words, arithmetic on those words, or a named-analogue benchmark. Nothing is measured by me.

HEADLINE

The operator has itself published the two numbers that decide this project: the trap is a **30 km × 2.5 km elongate toe-thrust ridge** (aspect ratio 12:1) with "**varying faults frequency, vertical intercalation of thin-bed, thick sand reservoir and lateral compartmentalization**", and IPA has graded it **industry top-decile reservoir complexity** — that is not an analyst's opinion, it is the sanctioning basis in PETRONAS Carigali's own SPE paper. My independent geometric rebuild lands at a P50 STOIIP of **517 MMbbl** and, at a 26% recovery factor, **134 MMbbl recoverable**, which reproduces the booked 139 MMbbl to within 4% — so the number is *internally honest* but sits exactly at P50, with a P90 downside of **51 MMbbl**. The static case is not wrong; it is **unhedged**: a 12:1 faulted ridge drained by 10 wells cannot deliver P50 recovery without lateral continuity that the same paper says is impaired.

NUMBERS

Table 1 — Static model, structure and rock

Parameter	Value	Unit	Class	Basis
Structural province	NW Borneo Fold-Thrust Belt, deepwater toe-thrust trend	—	[M]	Khamis, Jong & Barker 2018, GSM Bull. 65, pp.1–12 — "L-B-P structural trend" (Limbayong–Bestari–P)
Trap type	Elongate thrust-cored anticline / hanging-wall culmination, dip-closed, fault-bounded at least one flank	—	[M]	SPE 210079-MS: "severe elongated structure … with varying faults frequency"; toe-thrust kinematics per GSM Bull. 65
Structure length	30	km	[M]	SPE 210079-MS, 2022, abstract
Structure width	2.5	km	[M]	SPE 210079-MS, 2022, abstract
Aspect ratio	12 : 1	—	[D]	30 ÷ 2.5
Structural footprint	75	km²	[D]	30 × 2.5
Trap area at contact — P90 / P50 / P10	22 / 30 / 40	km²	[A]	29% / 40% / 53% of footprint; closure fraction for a faulted elongate ridge (analogue: Malikai, Gumusut-Kakap crestal areas)
Gross hydrocarbon interval — P90 / P50 / P10	70 / 80 / 110	m	[A]	Stacked Upper Miocene Cycle V / Kinarut sands; analogue columns 60–150 m (Kikeh, GK)
Depositional model	Ponded turbidites between thrust ridges	—	[M]	GSM Bull. 65: syn- and post-kinematic

Parameter	Value	Unit	Class	Basis
	(syn-/post-kinematic) grading to unconfined fan sands (pre-kinematic)			deposition “deflect turbidite sand fairways, and trap turbiditic sand in between ridges forming ponded turbidites”
Reservoir age / interval	Upper Miocene Cycle V, Kinarut and younger	—	[M]	Block R basin-modelling study: “oils are trapped at the lower Kinarut”; S&P Global: >45% of deepwater Sabah discoveries in Upper Miocene Cycle V
Net-to-gross — P90 / P50 / P10	0.24 / 0.28 / 0.34	frac	[A]	Thin-bed/thick-sand intercalation (SPE 210079-MS); analogue range 0.20–0.40
Porosity (net)	0.22	frac	[A]	NW Borneo Miocene clastic reservoirs 20–25% (published regional range); 2,500–3,500 m burial
Water saturation (Sw)	0.25	frac	[A]	Deepwater turbidite, 22% φ, oil-wet-to-mixed; range 0.18–0.32
Formation volume factor (Bo)	1.35	rm³/m³	[A]	30–35 °API live oil, GOR 800–1,200 scf/bbl
Net pay — P50	22.4	m	[D]	80 m × 0.28
Oil API gravity	30–35	°API	[A]	NW Sabah deepwater analogue range; **not published for Limbayong**
GOR (oil leg)	800–1,200	scf/bbl	[A]	Analogous undersaturated-to-saturated black oil

Table 2 — GRV → STOIIP → recoverable, with arithmetic (P50 worked line, then P90/P10)

Constants [A]: φ = 0.22; Sw = 0.25; Bo = 1.35; 1 m³ = 6.2898 bbl.

P50 line: 1. GRV = 30.0×10^6 m² × 80 m = **2.400×10^9 m³** 2. GRV in barrels = $2.400 \times 10^9 \times 6.2898$ = **15.095×10^9 bbl bulk** 3. NPV = $15.095 \times 10^9 \times 0.28$ (NTG) = **4.227×10^9 bbl** 4. HCPV = $4.227 \times 10^9 \times 0.22 \times (1 - 0.25) = 4.227 \times 10^9 \times 0.165$ = **0.6974×10^9 bbl** 5. STOIIP = $0.6974 \times 10^9 \div 1.35$ (Bo) = **516.6 MMbbl → 517 MMbbl** 6. Recoverable = 517 × 0.26 = **134.3 MMbbl**

Step	P90 (low)	P50	P10 (high)	Unit	Class
Trap area	22	30	40	km²	[A]
Gross interval	70	80	110	m	[A]
GRV	**1.540**	**2.400**	**4.400**	**10^9 m³**	[D]
GRV	9.686	15.095	27.675	10^9 bbl	[D]
Net-to-gross	0.24	0.28	0.34	frac	[A]
Net rock volume	2.325	4.227	9.410	10^9 bbl	[D]
φ × (1−Sw)	0.165	0.165	0.165	—	[A/D]

Step	P90 (low)	P50	P10 (high)	Unit	Class
HCPV	0.3836	0.6974	1.5526	10^9 bbl	[D]
STOIIP (÷Bo 1.35)	**284**	**517**	**1,150**	**MMbbl**	[D]
Oil recovery factor	0.18	0.26	0.34	frac	[A]
Recoverable oil	**51.2**	**134.3**	**391**	**MMbbl**	[D]
Booked (GEM, 2023 data year)	—	139.0	—	MMbbl	[M]
Swanson expected value	(51.2 + 4×134.3 + 391) ÷ 6 = **163.2**			MMbbl	[D]
Recoverable per well (10 wells)	5.1	13.4	39.1	MMbbl/well	[D]
STOIIP per unit trap area	12.9	17.2	28.8	MMbbl/km²	[D]
Implied RF at booked 139	—	**26.9%**	—	frac	[D]

Table 3 — Oil rim vs gas cap vs non-associated gas (explicitly separated)

Stream	Volume	Unit	Class	Basis
Crude oil, recoverable	123.2	MMbbl	[D]	139.0 total liquids [M] − 15.8 condensate
Condensate, recoverable	15.8	MMbbl	[D/A]	790.9 bcf × 20 bbl/MMscf (CGR 15–25 → 11.9–19.8 MMbbl)
Total liquids booked	139.0	MMbbl	[M]	GEM, data year 2023 (crude + condensate commingled in source)
Non-associated gas, remaining	790.9	bcf	[M]	GEM, data year 2023
Gas GIIP at RF 0.70	1,130	bcf	[D]	790.9 ÷ 0.70 (RF 0.60/0.70/0.80 → 1,318/1,130/989 bcf)
Gas Bg at 250 bar, 90 °C, z=0.88	0.00644	rm³/rm³	[D]	Bg = 0.00504·zT/p = 0.00504×0.88×363/250; expansion 155 rm³/rm³
Gas HCPV required	206	10^6 m³	[D]	1,130 bcf = 32.0×10^9 rm³ × 0.00644
Gas GRV required	4.46	km³	[D]	206×10^6 ÷ 0.0462 (=NTG 0.28 × φ 0.22 × 0.75)
Gas GRV ÷ oil GRV	1.86	×	[D]	4.46 ÷ 2.40
Combined GRV (oil + gas)	6.86	km³	[D]	2.40 + 4.46; equals 30 km² × 229 m gross fill
Associated gas in oil leg (solution)	~100–150	bcf	[A]	123 MMbbl × 800–1,200 scf/bbl = 98–148 bcf, recovered over life — **not** part of the 790.9
Gas cap vs NAG split	[U]	—	[U]	**Not public.** Operator describes “oil and non-associated gas”; a 1.86× rock-volume requirement for gas versus oil implies

Stream	Volume	Unit	Class	Basis
				most of the 790.9 bcf is free gas in separate blocks/culminations (likely Bestari), not a thin cap

Table 4 — Data gaps: what is missing, what it costs, how long

Missing data	Does the operator have it?	Acquisition route	Cost (US$M, AACE Cl.4/5)	Time	Basis
3D seismic (modern, FWI-PSDM)	Yes — used amplitude/spectral decomposition (GSM Bull. 65)	New acquisition, 400 km²	5.0–10.0	18–27 mo	US$8–12k/km² acquisition + US$4–6k/km² depth imaging; 6 mo planning/mobilisation, 4–6 mo acquisition, 9–12 mo imaging
— or reprocess existing 3D	Yes	Reimaging only	1.5–3.0	6–9 mo	Benchmark deepwater reprocessing rate
Well logs (quad-combo + MDT)	Yes, 4 appraisal wells	One appraisal/development well, 1,000–1,200 m WD	80–150	6–12 mo	Deepwater Malaysia well US$80–200M; operator claims 20–30% below industry average (SPE 210079-MS)
Conventional core (60–120 m)	Likely, not public	Add to above well	3–6	+2–4 wk rig time	Coring + lab SCAL US$3–6M/well
DST / mini-DST	[U]	2–3 tests in above well	8–15 per test	+7–14 d/test	Deepwater DST spread rate
PVT / fluid samples	[U]	MDT samples + lab PVT	0.7–1.9	3–6 mo lab	US$0.5–1.5M sampling + US$150–400k PVT
4D baseline (for compartment surveillance)	No	Must be shot at/before first oil	8–15	12–18 mo lead	Repeat of 400 km² deepwater 3D
Fault-seal / geomechanical core (S_Hmax)	Partly — MEM built (SPE 210549-MS)	Dedicated geomech well + LOT/DFIT	15–25	6–9 mo	Narrow margin <0.5 ppg documented
Total to close static-model gaps			**120–200**	**18–30 mo**	Sum of above, excluding the new well

KEY JUDGEMENTS

1. **The trap is real and the trap is the problem.** A 30 km × 2.5 km toe-thrust ridge is a proven NW Borneo trap family (Kikeh, Gumusut-Kakap, Malikai all sit in the same fold-thrust belt), so trap *presence* risk is low. Trap *efficiency* risk is extreme: at 12:1 aspect ratio, the drainage radius of a deviated deepwater well (~1–1.5 km) covers 3–5% of strike length per well. Ten wells at ~3 km spacing is the mathematical minimum for the footprint, and there is no spare well for the compartment that is always wrong in a faulted ridge.

2. **The 139 MMbbl is a P50 number wearing a P90 label.** My geometric rebuild is independent of the operator's model (public dimensions only) and lands at 134 MMbbl at P50, reproducing the booked figure. That is a *consistency* pass, not a *validation*. The sanction case is therefore calibrated to the median outcome, and the median outcome of a top-decile-

complexity reservoir is, by construction, the outcome that does not surprise you — half the time it is worse. Downside is 51 MMbbl, i.e. 5.1 MMbbl/well, which cannot carry a US$100M+ well plus its share of a US$4bn facility.

3. **Compartmentalisation is not a risk to be modelled; it is stated as a fact in the sanctioning paper.** "Lateral compartmentalization which impend effectiveness of well drainage and pressure maintenance" is PETRONAS Carigali's own sentence. The mitigation described is *modelling* — "incorporated faults re-interpretation, refined grid resolution, and change of facies prediction, **increasing the stratigraphic compartments**". The operator's response to more compartments was to add more compartments to the model and then to "focus assessment on low realization case". That is intellectually honest and operationally dangerous: it means the FDP is designed around a distribution it cannot yet measure.

4. **The oil rim and the gas are two different projects sharing one hull.** The 790.9 bcf needs a gas GRV of ~4.46 km³ at 250 bar / 90 °C — **1.86× the rock volume of the oil leg**. A gas cap that large above a 123 MMbbl oil rim would imply a column the seal would not hold; the volume arithmetic therefore says most of the booked gas is *non-associated*, in separate fault blocks or at Bestari. Consequence: gas deliverability (150 mmscfd design [M: GEM]) is a **separate subsurface bet with separate compartments**, and it is being tied back across the same 10-well subsea architecture. If gas is one accumulation and oil is another, there is no shared pressure support and no shared upside.

5. **Four appraisal wells in 24 years is the entire static database, and the model has not been refreshed since 2022.** SPE 210549-MS (Sep 2022) states "the first phase of development is to start in Q1 2023". The subsurface work published in 2022 was already FID-ready; the 2023–2026 delay was commercial and facilities (FPSO process cancelled Aug 2022, "commercially unviable" Mar 2023), not geological. On the public record, **no new subsurface data has been acquired between the 2022 papers and the Aug 2026 FID.** The sanction is being taken on a four-year-old static model with the same four wells. (Falsifier in next section.)

6. **Recovery factor 26% is the optimistic end of what this geology supports.** Deepwater turbidite waterfloods globally run 25–40%; the NW Sabah analogue behaviour is the low end of that — Kikeh (1,330 m WD, same belt, same thin-bed/thick-sand problem) required re-drilling after early well failures and sand production, and needed dedicated high-angle thin-bed wells to chase bypassed oil (IPTC-17694-MS). Limbayong stacks *more* complexity onto that template. I would not underwrite 26% without a DST and a 4D baseline; 18–22% is the honest planning number for the low case.

7. **The narrow drilling margin is a static-model symptom, not just a drilling problem.** SPE 210549-MS reports a window "less than 0.5 ppg even at the surface section", wells drilled toward S_hmin, and MPD as mandatory. That is a stress-state statement: the reservoir is being depleted-adjacent or the ridge is critically stressed. Critically stressed faults in a compartmentalised ridge means **faults that are both seals and leaks depending on production-induced stress change** — the exact mechanism that killed Kikeh injectors through poroelastic effects (SPE 122018-MS). Injector count and injector life are therefore static-model outputs, not facilities assumptions.

8. **The 10-well / 270 MMboe configuration is a bet that the ridge is one connected tank.** If it is (best case), 26% RF and 139 MMbbl is achievable and the project works. If it is five tanks — which the operator's own "increasing the stratigraphic compartments" language implies — then either recovery falls to ~18% (51 MMbbl P90) or well count must rise by 50–100% at US$80–150M each. **There is no version of this field where the static model does not set the well count, and no version where the well count is settled.**

WHAT WOULD CHANGE MY MIND

My claim	Falsifier
Trap area P50 30 km² is right	A published depth map or operator statement showing closure area >50 km² (would push STOIIP toward 900+ and make 139 MMbbl clearly conservative)
NTG 0.28	Core or log data showing NTG >0.40 in the main Kinarut sands; or <0.18, which would kill the project outright

My claim	Falsifier
RF 26% P50 / 18% P90	A DST with sustained productivity plus a 4D baseline showing pressure communication across >10 km of strike — that would justify 32–36%
Most gas is non-associated	A gas-gradation or contact log proving a single gas cap over the oil rim; or a GSA structured around a single accumulation
No new subsurface data 2022–2026	Any 2023–2026 appraisal well, new 3D, or OBN survey in the public record. I would then withdraw judgement 5 entirely
10 wells is the minimum	An operator statement that wells are dual-zone intelligent completions draining 3+ stacked intervals each, which would cut required well count
Compartmentalisation dominates	Interference testing between two well pairs across a mapped fault showing communication (would raise RF and cut well count)

CONFIDENCE

Output	Confidence	Why
Trap type and structural style	**High (0.90)**	Primary source (SPE 210079-MS) plus peer-reviewed regional geology (GSM Bull. 65) agree exactly, including the L-B-P trend name
Structure dimensions (30 × 2.5 km)	**High (0.95)**	Operator's own abstract, verbatim
Trap area / gross interval / NTG / φ / Sw	**Low (0.35–0.45)**	Pure analogue benchmarks; no Limbayong-specific measurement exists publicly. This is where the uncertainty lives
P50 STOIIP 517 MMbbl	**Low–moderate (0.40)**	±60% band is honest; the number is a geometry argument, not a measurement
P50 recoverable 134 MMbbl	**Moderate (0.55)**	Cross-checks to booked 139 within 4%; the *agreement* is the evidence, not the model
P90 downside 51 MMbbl	**Moderate (0.60)**	Driven by RF 18% and 22 km²; both defensible on analogue behaviour
Gas GIIP 1.13 Tcf / gas GRV 4.46 km³	**Low–moderate (0.40)**	Bg depends on assumed P/T ([U]); the 1.86:1 ratio conclusion is robust to ±25% on Bg
Condensate 15.8 MMbbl	**Low (0.30)**	CGR entirely assumed; only the commingled 139 is [M]
Compartmentalisation is the dominant risk	**High (0.85)**	Operator states it; IPA grades it top-decile; regional literature agrees
Model not refreshed since 2022	**Moderate (0.65)**	Inference from absence in public record; absence of evidence, explicitly flagged

Net: the sanction was defensible on structure and trap, and **under-supported on rock and connectivity**. The static case is a P50 case sold as a base case, with the operator's own complexity grade as the warning label. Nothing here says the field is not there; everything here says the recovery factor and the well count are unmeasured, and those two variables decide the NPV.

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structure 30×2.5 km, IPA top-decile grade, four appraisal wells since 2002, thin/thick intercalation, lateral compartmentalisation, 20–30% below industry-average cost, 800 MMboe stranded Sabah deepwater analogue pool.

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7. "Malaysia Deepwater Development: Optimizing Kikeh Reservoir Management and Exploitation in a Changing Environment," **IPTC-17694-MS**, published 19 Jan 2014. DOI 10.2523/iptc-17694-ms. — *Kikeh (K field) 1,330 m WD, thin beds (cm) to thick blocky sands, early well failures and sand production, water + gas re-injection, SMART completions and dedicated thin-bed high-angle wells for bypassed oil. Primary analogue for thin-bed recovery inefficiency.*
8. "Overcoming Poroelastic Behaviour in the Water Injector Wells in the Kikeh Deepwater Development," **SPE 122018-MS**, 27 May 2009. — *injector impairment from production-induced poroelastic stress change; analogue for injector life in a critically stressed ridge.*
9. "Overview of the Malikai Project," **OTC-29071-MS**, 24 Apr 2018. — *Malikai: "a marginal field – limited reserves in challenging reservoirs", 500 m WD, discovered 2004, standalone development not economic, saved only as a satellite tie-back to Kebabangan.*
10. S&P Global Commodity Insights, "Potential in Ultra-deepwater Sabah, Malaysia – Can it be Realized?" — *>45% of deepwater Sabah discoveries in Upper Miocene Cycle V reservoirs of the Baram Delta / NW Sabah Province; Kamunsu East-1 as first discovery, late 1990s.*
11. Global Energy Monitor, Global Oil & Gas Extraction Tracker, data year 2023, via FACT_BASE.md (19 Oct 2026). — *139.0 MMbbl crude + condensate remaining recoverable; 790.9 bcf gas remaining; 14.53 MMbbl/y and 1,550.4 million m³/y design capacity.*
12. Hydrocarbon retention in clastic reservoirs of NW Borneo — examples of hydrocarbon trap, reservoir and seal, and implications on hydrocarbon column length (Academia/ResearchGate record), accessed 19 Oct 2026. — *regional porosity 20–25%, permeability <1 D, "poor fault sealing capability"; named Limbayong, Gumusut, Kikeh as affected; column-length limitation.*
13. Welligence, "Malaysia FID Alert — PETRONAS takes FID on Limbayong," Aug 2026 — *PCSB 100% WI, operator; discovery 2002 by Shell.*
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Flagged discrepancies with FACT_BASE.md

1. **Discovery dating.** FACT_BASE says "Limbayong-2 oil confirmed 2014." Source 5 (Block R basin modelling) dates Limbayong-2 to **2013** and Limbayong-3 to **2014**. The 2013/2014 pair is more consistent with two wells one year apart.

[Minor; does not change any number.]

2. **Gas vs oil dominance.** FACT_BASE lists “combined ~270 MMboe” using 6 Mcf/bbl. On my arithmetic the gas occupies **1.86× the rock volume of the oil** — this is volumetrically a **gas field with an oil rim**, not an oil field with a gas cap. Any section that treats gas as a by-product (S4/S5 commercial) should be re-read against Table 3 before the GO/NO-GO is framed.
3. **“Marginal” is the operator’s word and it survives.** SPE 210079-MS calls it “marginal deepwater development” and “top decile” complexity in the same title, and states the strategy was to “focus assessment on low realization case.” The fact base’s characterisation is correct and, if anything, understated.

S2 — RESERVOIR ENGINEERING & PRODUCTION FORECAST

Project: Limbayong Deepwater Development (Limbayong + Bestari), Block G/J/R, ~120 km offshore Sabah **Water depth:** 900–1,200 m · **Operator:** PCSB 100% · **FID:** 28 Aug 2026 **Purpose:** production forecast for the economics section (S5). AACE Class 4 screening — not a reserves report.

> ***CONFLICT WITH FACT_BASE — READ FIRST.*** *FACT_BASE records the design capacity as "[M: GEM, design year 2027]". A 2027 design year is* ***impossible****: FID was 28 Aug 2026, the hull was only sold to MISC on 20 Aug 2026, and the subsea iEPCI was awarded 15 Sep 2026. There is no path to first production in 2027. GEM's capacity fields are therefore* ***desk-derived, not operator FDP values*** *— I treat them as [M-weak] and re-derive what they must mean. Everything downstream of that decision is stated below.*

HEADLINE

The published capacities are **nameplate daily rates — 40,000 bopd and 150 mmscfd — annualised at ~99.5% utilisation** (14.53 MMbbl ÷ 40,000 = 363.25 days exactly; 1,550.4 MMm³ ÷ 365 = 150.00 mmscfd exactly), not expected production. The reserve base cannot support either number for more than 9.5 years (oil) and 14.4 years (gas) at 100% uptime with zero decline, so the forecast must sit **below** nameplate: base case **35 kbopd oil and 135 mmscfd gas, five-year oil plateau (2031–2035), 14.6%/yr oil decline, 9.0%/yr gas decline, economic limit ~2048/49, 254 MMboe recovered of 271 MMboe booked**. The controlling constraint on this field is **not deliverability — it is drainage**: ten wells, of which at most six can be oil producers, cannot simultaneously hold 35 kbopd, replace 43 kbwpd of voidage and drain a 30 km × 2.5 km faulted ridge with only four appraisal penetrations.

NUMBERS

Parameter	Value	Unit	Class	Basis
Liquids, remaining recoverable	139.0	MMbbl	[M]	GEM tracker, data year 2023
Gas, remaining reserves	790.9 (22,399)	bcf (MMm³)	[M]	GEM tracker, data year 2023
Total reserves	270.8	MMboe	[D]	139.0 + 790.9/6 (6 Mcf/bbl)
Design capacity, liquids (published)	14.53	MMbbl/y	[M-weak]	GEM, "design year 2027" — see conflict note
→ Implied daily rate	39,808	bopd	[D]	14.53e6 ÷ 365
→ Implied design basis	**40,000**	bopd	[D]	14.53e6 ÷ 40,000 = 363.25 days (99.5% availability)
Design capacity, gas (published)	1,550.4	MMm³/y	[M-weak]	GEM
→ Implied daily rate	150.0	mmscfd	[D]	1,550.4e6 × 35.3147 ÷ 365 ÷ 1e6 = 150.005
→ Annual gas equivalent	54.75	bcf/y	[D]	150 × 365

Parameter	Value	Unit	Class	Basis
Years to drain oil reserves at 40 kbopd, 100% uptime, no decline	9.52	yr	[D]	139.0 ÷ (40,000 × 365)
Years to drain gas reserves at 150 mmscfd, 100% uptime, no decline	14.45	yr	[D]	790.9e9 ÷ (150e6 × 365)
Years at combined nameplate (65,000 boe/d)	11.41	yr	[D]	270.8e6 ÷ 65,000 ÷ 365
Field design oil rate (FDP basis, third-party)	40,000	bopd	[M]	Offshore-Technology / WER 2021 (via S4)
Field design gas rate	150	mmscfd	[M]	Offshore-Technology / TechnipFMC 2026 (via S4)
Field design water injection	75,000	bwpd	[M]	Marine Technology News / WER 2021 (via S4)
Host vessel (Excalibur, ex-Sevan Hummingbird)	2007/08, 60–70 m cylinder	—	[M]	DNeX Bursa filing 20 Aug 2026; OE Digital 4 Jul 2022
As-built oil process (Chestnut service)	30,000	bopd	[M]	Sevan Marine release 22 Sep 2008
As-built water injection	20,000	bwpd	[M]	Sevan Marine release 22 Sep 2008
Storage as-built	270,000–300,000	bbl	[M]	OE Digital (270k) / DNeX filing (300k)
Cargo cycle at 35 kbopd, 85% working	6.6–7.3	days	[D]	270–300 kbbl × 0.85 ÷ 35,000
Development well count	10	wells	[M]	PETRONAS release 1 Jun 2021
Appraisal penetrations since 2002	4	wells	[M]	SPE 210079-MS abstract
Structural extent	75	km²	[M]	30 km × 2.5 km elongated ridge, SPE 210079-MS abstract
Areal well density (10 wells)	7.5	km²/well	[D]	75 ÷ 10
Areal reserves density	1.85	MMbbl/km²	[D]	139 ÷ 75 (productive area < gross extent; see KJ-6)
Base allocation: oil producers	6	wells	[A]	Allocation judgement — see KJ-4
Base allocation: gas producers	2	wells	[A]	As above
Base allocation: water injectors	2	wells	[A]	As above
Reservoir pressure support	Required	—	[M]	SPE 210079: compartmentalisation “impend[s]… pressure maintenance”

Parameter	Value	Unit	Class	Basis
Oil rate per producer at plateau	5,833	bopd	[D]	35,000 ÷ 6
Gas rate per gas well at plateau	67.5	mmscfd	[A]	135 ÷ 2 — at the top of the analogue range
Water injection per injector	21,875	bwpd	[D]	43,750 rb/d voidage ÷ 2
Voidage requirement at 35 kbopd (Bo 1.25)	43,750	rb/d	[D]	35,000 × 1.25
Productivity index, deepwater turbidite (analogue)	10–40 (base 20)	stb/d/psi	[A]	Sabah/NW Borneo turbidite wells; frac-pack completions
Drawdown at 5,833 bopd, PI 20	292	psi	[D]	q ÷ PI
Drawdown at 5,833 bopd, PI 10	583	psi	[D]	q ÷ PI — upper limit for sand-control integrity
Uptime (calendar availability)	92 (88–95)	%	[A]	N. Sea FPSO 90–94%; subsea tieback penalty
Process efficiency	97	%	[A]	Fuel/flare/derate allowance
Deliverable plateau = nameplate × uptime × efficiency	35,700 → **35,000**	bopd	[D]	40,000 × 0.92 × 0.97 = 35,696, rounded down for turndown
Gas plateau = 150 × 0.92 (less 2% fuel)	135	mmscfd	[D]	150 × 0.92 × 0.98 = 135.2
Oil decline (solved to 139.0 MMbbl)	14.6	%/yr	[D]	Exponential, from end-2035, to 2052
Gas decline (solved to 790.9 bcf)	9.0	%/yr	[D]	Exponential, from end-2038, to 2053
Oil plateau duration	5 (2031–2035)	yr	[A]	Deepwater turbidite 3–6 yr; set by facility, not reservoir
Gas plateau duration	8 (2031–2038)	yr	[A]	Gas-long field; compression-led plateau
First oil	2030 Q2	—	[A]	FID Aug 2026 + 44 months (conversion + 10 wells, 2 rigs)
Economic limit	~2048/49 (4–5 kbopd oil)	—	[D]	FPSO opex vs 4 kbopd + 50 mmscfd revenue
Field life	2030–2053 (23 yr)	—	[D]	Technical potential; commercial case ends 2048/49
Recovery at economic limit	254.2	MMboe	[D]	Cum. to 2048
Tail below economic limit	16.6 (4.5 MMbbl + 72.7 bcf)	MMboe	[D]	270.8 − 254.2 — at-risk, needs low-cost late-life host
EUR per oil producer	23.2	MMbbl	[D]	139.0 ÷ 6 — top-quartile requirement
EUR per gas well	396	bcf	[D]	790.9 ÷ 2 — **1.3–4× analogue range: not**

Parameter	Value	Unit	Class	Basis
				credible
Gas share of reserves (boe)	48.7	%	[D]	131.8 ÷ 270.8
Gas share of undiscounted revenue @ $70/bbl, 3/*MMBtu*|19.6|bn				
GOR / condensate yield / CO_2 / H_2S / OOIP / RF / k·h / net pay	[U]	—	[U]	**No public well test, PVT, core or log exists. Do not fabricate.**

Design-capacity audit (the arithmetic that matters). 14.53 MMbbl/y ÷ 40,000 bopd = 363.25 days. That is not a coincidence: the design basis is a **40,000 bopd nameplate**, annualised at 99.5% availability. The gas figure converts to 150.005 mmscfd — also exact. Conclusion: **both numbers are nameplate daily rates stated as annual capacities.** They are not production forecasts, and GEM's presentation of them as annual "design capacity" invites the reader to mistake nameplate for average. Any economics model that uses 40 kbopd × 365 × 0.95 as a plateau is overstating by ~10% before a single well is drilled.

PRODUCTION PROFILE

Base case. Rates are **calendar-year averages, already net of 92% uptime** — do not apply an availability factor again in S5.

Year	Oil kbopd	Gas mmscfd	Boe/d	Cumulative MMboe
2030	15.0	55.0	24,167	8.8
2031	35.0	135.0	57,500	29.8
2032	35.0	135.0	57,500	50.8
2033	35.0	135.0	57,500	71.8
2034	35.0	135.0	57,500	92.8
2035	35.0	135.0	57,500	113.8
2036	29.9	135.0	52,393	132.9
2037	25.5	135.0	48,030	150.4
2038	21.8	135.0	44,305	166.6
2039	18.6	122.8	39,093	180.9
2040	15.9	111.7	34,529	193.5
2041	13.6	101.7	30,528	204.6
2042	11.6	92.5	27,018	214.5
2043	9.9	84.2	23,934	223.2
2044	8.5	76.6	21,223	230.9
2045	7.2	69.7	18,837	237.8
2046	6.2	63.4	16,735	243.9
2047	5.3	57.7	14,882	249.4
2048	4.5	52.5	13,245	254.2
2049	3.8	47.7	11,800	258.5
2050	3.3	43.4	10,521	262.3
2051	2.8	39.5	9,389	265.8

Year	Oil kbopd	Gas mmscfd	Boe/d	Cumulative MMboe
2052	2.4	35.9	8,386	268.8
2053	0.0	32.7	5,449	270.8

Columns are rounded to 1 dp; boe/d and cumulative are computed from unrounded rates, so a hand-check will differ by <0.1%.

Annual volumes for the cash-flow model: oil (MMbbl) 2030 5.5 · 2031–35 12.8/yr · 2036 10.9 · 2037 9.3 · 2038 8.0 · then declining. Gas (bcf) 2030 20.1 · 2031–38 49.3/yr · 2039 44.8 · then declining. Peak year 2031–35 = 57,500 boe/d = 21.0 MMboe/yr.

Sensitivity (reserves held at 139 MMbbl / 790.9 bcf; decline re-solved):

Case	Oil plateau	Gas plateau	Oil decline	Gas decline	Last year >3 kbopd	Peak boe/d
Low — 5 producers, 2 gas wells underperform	28 kbopd (6 yr)	110 mmscfd (9 yr)	11.7%/yr	7.1%/yr	2053	46,300
Base	**35 kbopd (5 yr)**	**135 mmscfd (8 yr)**	**14.6%/yr**	**9.0%/yr**	**2049**	**57,500**
High — all 10 wells produce, new topsides at nameplate	40 kbopd (4 yr)	150 mmscfd (6 yr)	16.5%/yr	10.3%/yr	2048	65,000

Note the counter-intuitive but robust result: **higher plateau does not mean more oil — it means a shorter plateau and a steeper decline.** The reserves are fixed. The only things plateau buys you are earlier cash and a shorter facility life. This is why the "plateau 66k → 45k" sensitivity in the CEO memo (−US$683M) is a *timing* sensitivity, not a recovery sensitivity, and should be modelled as such in S5.

KEY JUDGEMENTS

KJ-1. The published "design capacity" is nameplate, not forecast — and it is arithmetically exact. 14.53 MMbbl/y ÷ 40,000 bopd = 363.25 days; 1,550.4 MMm³/y = 150.005 mmscfd. *Falsified if:* PCSB's FDP shows a 40 kbopd average year at 92% uptime (it cannot — that requires 43.5 kbopd nameplate).

KJ-2. No production forecast for this field can hold 40 kbopd for more than ~9.5 years, because that is the entire oil reserve at 100% uptime and zero decline. 139 MMbbl ÷ (40,000 × 365) = 9.52 years. Any profile showing a decade-long plateau at nameplate is consuming 100% of the reserves during plateau and then stopping dead — physically impossible. *Falsified if:* reserves are restated upward by >40%.

KJ-3. The 30 kbopd as-built figure is NOT the binding constraint — the binding constraint is deck area and storage. The Chestnut-era process train (30 kbopd, 20 kbwpd) is being scrapped; S4 sizes 5,500–8,500 t of new topsides onto ~3,500 m² of usable cylindrical deck. 40 kbopd + 150 mmscfd + 75 kbwpd on that footprint is the real question, and it is a *weight and area* problem, not a throughput one. My base case derates to 35 kbopd partly for this reason. *Falsified if:* the conversion contract publishes a topsides weight < 5,500 t with a 40 kbopd / 150 mmscfd train.

KJ-4. Ten wells cannot do all three jobs. This is the central reservoir-engineering finding. Holding 35 kbopd requires 6 producers at 5,833 bopd; replacing voidage at Bo 1.25 requires 43,750 rb/d of injection, which needs 2 injectors at ~21,875 bwpd each. That is 8 wells. The remaining 2 must carry 790.9 bcf — **396 bcf per gas well, versus an analogue range of 100–300 bcf.** Limbayong is therefore either (a) a 12–14 well development with a second drilling campaign, (b) a phased development where gas comes later on its own wells, or (c) booking 791 bcf on two wells that will not deliver it. The public "10 wells"

number is a 2021 pre-FID figure and should not be used as the well count in the economics. *Falsified if:* the FDP well list shows ≥3 gas wells and ≥2 injectors inside the 10.

KJ-5. Deliverability is not the risk; drainage is. 5,833 bopd per producer against a PI of 10–40 stb/d/psi requires only 146–583 psi drawdown — routine for a frac-packed deepwater completion, and comfortably inside sand-control limits at PI ≥ 20. The wells will make the rate. What they will not do is *drain* 75 km² from six points on a 30 km ridge with faults every few kilometres, at 12.5 km² per producer and ~3.5 km equivalent spacing, with four appraisal penetrations in 24 years. SPE 210079 says this in the operator's own words: lateral compartmentalisation "impend[s] effectiveness of well drainage and pressure maintenance." *Falsified if:* interference testing or 4D seismic shows lateral continuity across the fault blocks.

KJ-6. The reserve density is the tell. 139 MMbbl over a 30 × 2.5 km structure is 1.85 MMbbl/km² gross. Kikeh (~400 MMbbl) runs roughly 5–10× that density across its productive fault blocks. Either only ~15–25 km² of the 75 km² is productive (pay is concentrated in a few fault blocks and the rest is thin-bed waste), or recovery is being booked at the low end. Both readings say the same thing: this is a **sweet-spot development of a complex field, not a full-field development** — which is precisely what "marginal" means in the paper's title. *Falsified if:* the FDP maps >40 km² of net productive area.

KJ-7. Gas is sold, not flared, and not primarily re-injected — but it is only ~20% of the revenue. Zero routine flaring rules out 135 mmscfd to flare; PETRONAS ZEVC and the 2024 Sabah GSA point to monetisation. At $3/MMBtu domestic gas, 790.9 bcf is US$2.37bn undiscounted against US$9.73bn for the oil at $70 — gas is 49% of the boe but 20% of the money. *Falsified if:* a GSA at ≥$5/MMBtu (which would lift gas to ~29% of revenue and is worth +US$370M NPV per the CEO memo) or if the FDP adopts gas re-injection for pressure maintenance, in which case 791 bcf is not Phase 1 revenue at all and the project loses ~US$2.4bn of undiscounted gas sales.

KJ-8. There is a gas-export infrastructure gap that the disclosed contract scope does not cover. 790.9 bcf must reach either the Sabah grid (Kimanis/SSGP, ~140–180 km) or an LNG route. The disclosed awards are subsea iEPCI (TechnipFMC, US$75–250M) and a US$35M hull. No export pipeline, no gas-processing module, no GSA has been disclosed. Either the gas is stranded behind an unfunded 150 km pipeline, or it is re-injected, or the scope is undisclosed. S4 and S7 must price this; S5 must not book gas revenue before 2032 without a GSA. *Falsified if:* a pipeline EPCIC or GSA is published.

KJ-9. Uptime 92% is achievable but the field life is the bigger facility risk. A cylindrical hull has good motion behaviour (Sevan's selling point) and the 92% base sits mid-range for North Sea FPSOs at 90–94%; a 19-year-old converted hull on a 30 km subsea tieback with a validated turndown limit justifies the 88% low case. Each 1% of uptime over the 5-year plateau is worth ~0.64 MMbbl. The harder issue: the **economic limit lands around 2049, when the hull is 42 years old.** A 23-year field life on a 2007-built redeployed cylinder implies either a life extension capex in the 2040s or a late-life host swap — neither is in the FID scope. *Falsified if:* the conversion includes a 25-year design-life certification.

KJ-10. About 16.6 MMboe (6.1%) of the booked reserves will not be produced commercially. Below ~4 kbopd the FPSO's opex exceeds revenue; the 2049–2053 tail (4.5 MMbbl + 72.7 bcf) needs a low-cost late-life host to happen. S5 should carry **254 MMboe as the commercial case** and treat the balance as contingent. *Falsified if:* opex falls below ~US$8/boe or a third-party tie-back shares the facility cost.

WHAT WOULD CHANGE MY MIND

1. **A published FDP production profile.** Any single operator document with year-by-year rates supersedes this entire section. My numbers are an envelope, not a prediction.
2. **The well list with well types.** If the 10 wells are e.g. 7 producers + 3 gas and injection is deferred, my voidage argument collapses and plateau is credible; if they are 6 + 2 + 2, KJ-4 stands.
3. **A DST or RFT/MDT from Limbayong-2 or any appraisal well.** A measured PI above 30 stb/d/psi with kh > 5,000 md-ft would move me from "drainage-limited" to "rate-limited" and lift the plateau case toward 40 kbopd.
4. **Interference or compartment-pressure data.** Proof that the fault blocks are in communication would raise recovery per well and validate 6 producers for 139 MMbbl.
5. **A signed GSA with volume, price and start date.** This converts KJ-7 from judgement to fact and is the single largest swing in the economics.

6. **Evidence of a strong water drive or gas-cap support** that reduces injection need to <20 kbwpd — that would free a well slot for gas or oil and fix the KJ-4 arithmetic.
7. **Reserves restatement.** If 2P is truly 139 MMbbl, the 9.5-year nameplate ceiling is unarguable. If the booked number is a conservative FID-case subset of a larger 800 MMboe trend resource, the plateau could run longer.

CONFIDENCE

Claim	Confidence	Why
Design capacity is nameplate daily rate (40 kbopd / 150 mmscfd), not average	**High (0.9)**	Exact arithmetic: 363.25 days and 150.005 mmscfd are not random
Reserves support ≤9.5 yr oil / ≤14.4 yr gas at nameplate	**High (0.95)**	Pure arithmetic on [M] reserves
Base plateau 35 kbopd / 135 mmscfd	**Medium (0.55)**	Depends on facility derate and well allocation; range 28–40 kbopd
Oil decline 14.6%/yr, gas 9.0%/yr	**Medium-low (0.45)**	Shape is analogue-driven (deepwater turbidite 10–25%); cumulative is pinned to reserves, so *shape* is the uncertain part, not the total
6 oil / 2 gas / 2 WI allocation	**Low (0.3)**	No public well list. Structurally reasoned, not sourced
92% uptime	**Medium (0.6)**	North Sea FPSO benchmark 90–94%; converted 19-yr hull argues 88%
First oil 2030	**Medium (0.5)**	44 months from FID is achievable for a converted hull but this project has slipped three times
Economic limit ~2048/49, commercial recovery 254.2 MMboe	**Medium (0.55)**	Highly sensitive to opex and gas price
Gas sold (not flared)	**High (0.85)**	Regulatory and commercial logic; 135 mmscfd cannot be flared
Gas sold at ≥$5/MMBtu	**Low (0.25)**	Malaysian domestic gas is regulated low; no GSA public

Overall confidence in the production forecast: medium-low. The cumulative (271 MMboe) is solid because it is [M] reserves. The *path* — plateau height, plateau length, decline — is a Class 4 envelope with ±35% on annual rates in any given year and ±20% on peak rate. That is appropriate for a screening study and inappropriate for a sanction decision.

SOURCES

#	Source	Used for
1	**SPE 210079-MS**, Najib, Phoon, Shamshudin, Mustapa, Khalid, Yusof (PCSB), *"Limbayong: Decoding Industry Top Decile Reservoir Complexity for Marginal Deepwater Development"*, ADIPEC 2022, DOI 10.2118/210079-ms — abstract via Semantic Scholar record	30 km × 2.5 km elongated structure; thin-bed/thick-sand intercalation; lateral compartmentalisation impeding drainage and pressure maintenance; **four appraisal wells since 2002**; validated turndown limit; 20–30% well/facility cost advantage; ~800 MMboe scattered deepwater Sabah resource
2	GEM Global Oil & Gas Extraction Tracker, data year 2023 (per FACT_BASE)	139.0 MMbbl, 790.9 bcf; "design capacity" 14.53 MMbbl/y, 1,550.4 MMm³/y (re-classed [M-weak], see conflict note)

#	Source	Used for
3	Sevan Marine ASA press release, 22 Sep 2008 (ship.gr mirror)	Sevan Hummingbird as-built: 30,000 bopd oil, 20,000 bwpd water injection, 300,000 bbl storage
4	Offshore Engineer (OE Digital), 4 Jul 2022 — “Teekay Sells Sevan Hummingbird FPSO to Ping Petroleum”	60 m diameter, 270,000 bbl storage, commissioned 2008, 13 years at Chestnut; renamed Excalibur
5	DNeX Bursa announcement, 20 Aug 2026 (via S4)	Excalibur 70 m × 27 m, 300,000 bbl design storage, built 2007, US$35M sale to MISC
6	Offshore Technology, 15 Jun 2021 — Limbayong project profile	10 deepwater wells tied back to FPSO; Block G; 900–1,200 m
7	Offshore Magazine, 3 Oct 2022 — FPU redeployments	Redeployed FPUs “tend to be smaller — 50,000 bopd or less”; as-is US$10–80M with upgrades 5–10×
8	OTC-19481-MS, *Kikeh Development: Project Overview*	Kikeh analogue: >400 MMbbl, 120,000 bopd plateau, ~1,300 m
9	Shell Malaysia / Ocean News / OGA Malikai project page	Malikai analogue: 60,000 bopd nameplate, 17 wells, actual ~25,000 bopd (2026), Phase 3 +4 wells → ~40,000 bopd
10	PETRONAS media release, 1 Jun 2021	900–1,200 m water depth; SURF EPCIC award

Analogue set used for per-well performance (all [A]): Kikeh (~5,000 bopd/well average at plateau), Gumusut-Kakap (~6,300 bopd/well implied), Malikai (~1,800–3,750 bopd/well actual — the cautionary case: 42% of nameplate for nine years). Gas wells: Sabah deepwater big-bore completions 40–90 mmscfd/well, EUR 100–300 bcf/well.

What is NOT in this section, because it does not exist in public domain: well test rates, PVT/fluid composition, GOR, condensate yield, CO_2/H_2S, core, net pay, porosity/permeability, OOIP, recovery factor, fault-transmissibility, aquifer strength, DST-derived PI, and the actual FDP production forecast. Everything above the [M] rows is an envelope built from the operator’s own public framing plus named analogues. **Do not let S5 treat any [A] row as measured.**

S3 — DRILLING & WELLS

Limbayong Deepwater Development · 10 development wells · 900–1,200 m water depth · ~120 km offshore Sabah
Prepared 19 Oct 2026 · real 2026 USD · AACE Class 4 (−30%/+50%) · oil price deck Brent US$70–85/bbl

HEADLINE

Ten subsea development wells, drilled sequentially by one 6th/7th-generation drillship over **23.5 months (714 days)**, will cost **US$762M P50 (range US$566M–1,025M)** — **US$67M per well P50** at a **US$450,000/day** rig rate. That is **US$2.82/boe** on 270 MMboe.

Three things dominate this section and none of them is the day rate:

1. **The wells are the appraisal.** FID was taken on 28 Aug 2026 on the strength of **four appraisal wells spread across a 30 km × 2.5 km structure** — roughly one well per 7.5 km of strike on a field the operator's own SPE paper calls *"industry top decile reservoir complexity"* with *"possible different depositional model"*. The 10-well programme is therefore carrying subsurface risk it was not designed to carry.
2. **The drilling margin is under 0.5 ppg and MPD is mandatory.** This is stated by PCSB's own well-planning paper (SPE 210549-MS): narrow margin *"even at the surface section"*, wells *"highly deviated and in the direction of the minimum horizontal stress"*, with stuck pipe, differential sticking and skin damage called out as the principal hazards. A benign-environment day-rate comparison understates this campaign's NPT exposure.
3. **The 2026–2028 floater market is tight and consolidating.** 2026 disclosed 7G drillship fixtures ran **US$309,500–540,000/day**, with leading-edge units clearing **US$550,000/day**; no drillship newbuilds of consequence have been ordered since 2014; and the announced Transocean–Valaris merger removes a competitor from the tender list. A 714-day campaign awarded into that market is a seller's contract, not a buyer's.

Contradiction flagged: SPE 210549-MS describes "the first phase of development" starting **Q1 2023**. The project instead reached FID in **Aug 2026**. The geomechanics model and well designs in that paper are therefore 3–4 years old and must be re-validated against any wells drilled since — this is an assumption-risk item, not a paperwork item.

NUMBERS

Parameter	Value	Unit	Class	Basis
Development wells	10	wells	[M]	PETRONAS/TechnipFMC iEPCI scope, 10 wells tied back to FPSO
Water depth	900–1,200	m	[M]	PETRONAS, 1 Jun 2021
Reservoir TVDSS (assumed)	2,200–3,000	m	[A]	Kikeh (1,330 m WD) / Gumusut-Kakap (1,200 m WD) analogue; **no public Limbayong well data**
Well MD (deviated)	4,800–6,500	m	[A]	1.5–3.5 km step-out on 30 km × 2.5 km structure (SPE 210079-MS)
Rig day rate assumed (2028 fixture)	450,000	US/*d*|[*A*]|*Mid − pointof*2026*disclosed*7*Gband*, *escalatedto*2028*contract*||2026*disclosed*7*Gdrillshiprates*|309, 500–540, 000|*US*/d	[M]	Rig dayrate tracker, 2026 fixtures

Parameter	Value	Unit	Class	Basis
Leading-edge 7G fixtures, early 2026	>550,000	US/*d*|[*M*]|*Tradepress*, 24*Apr*2026||7*Gfleetutilisation*, 2025*trough*|*low − to − mid*80*s*||*Third − partyspreadrate*|170, 000|*US*/d	[A]	35–40% of rig rate; mud, cement, casing running, ROV, 2 PSVs, fuel, catering
MPD spread	85,000	US/*d*|[*A*]|30*days*/*well*; *requiredby* < 0.5*ppgmargin*(*SPE*210549 − *MS*)||*Wellduration*, *average*|63.7|*days*|[*D*]|*Learningcurve*72 → 58*d*, *seecampaignschedule*||*Rig − days*, *drilling + moves*|687|*days*|[*D*]|637*drilling* + 50*moves*(5*d*/*move*)||*Campaigndurationincl.mob*/*demob*|714|*days*(23.5*mo*)|[*D*]|*Mob*17*d*, *demob*15*d*, 10*wells*, 9*inter − wellmoves*|| * *Costperdevelopmentwell*, *P*50 * *|* * 67.1 * *| * **US**M**	**[D]**	build-up below, 63.7 d average
Cost per well, low–high	52.0–95.6	US$M	[D]	52 d @ $380k/d, 12% cont. — 84 d @ 550*k*/*d*, 18|* * *Totaldrillingcampaign*, *P*50 * *| * *761.9 * *|* * *US*M**
Campaign range	566–1,025	US*M*|[*D*]|*AACEClass*4*envelope*, −26|*Dryhole*/*appraisalwell*, *P*50|49.8|*US*M	[D]	50 d, no completion, full logging + P&A; 74% of a development well
Drilling cost per boe	2.82	US/*boe*|[*D*]|761.9/270*MMboe*(139*MMbblliquids* + 131*MMboegas*)||*Appraisal*/*assurancewellsstillneeded*|2(+1*optional*)|*wells*|[*A*]|*Onepermajorfaultblock*, *oneDST*; *seeappraisalsectionbelow*||*Costofremainingappraisal*|100–160|*US*M	[D]	2–3 × US$50–55M dry-hole basis + testing
PETRONAS wells to be drilled, 2026	91 (79 in 2025)	wells	[M]	PETRONAS Activity Outlook 2026–2028
PETRONAS P&A, 2026	70 (63 in 2025)	wells	[M]	Same

WELL DESIGN CONCEPT

No Limbayong well schematic, casing point, mud weight or completion tally is public. Everything below is a **concept-level design** built from the two PCSB SPE papers and Sabah deepwater analogues, and is classed accordingly. The principal will recognise the shape; the numbers are placeholders for the drilling programme, not measurements.

Drilling unit interface. 21” marine riser, 15,000 psi BOP (a 20k stack is not justified — there is no public evidence of HPHT at Limbayong, and 20k units carry a US$75,000–125,000/day premium). MPD rotating control device and a continuous-circulation system are **not optional** given the <0.5 ppg window.

Casing scheme (depths as metres below mudline, [A]):

String	Hole	Setting depth (m BML)	Purpose / hazard
36” conductor	driven/jetted	60–90	Shallow water flow, conductor stability; geotechnical CPT already acquired to 100 m (Benthic PROD, 700–1,040 m WD) [M]
20” surface	26”	700–900	Isolate unconsolidated Plio-Pleistocene; **narrow margin already critical here** [M: SPE 210549-MS]

String	Hole	Setting depth (m BML)	Purpose / hazard
13-3/8” intermediate	17-1/2”	1,600–1,900	Isolate pressured shales / toe-thrust repeated section
9-5/8” production	12-1/4”	2,000–2,300	Top reservoir; hangs 7” liner
7” liner	8-1/2”	2,600–3,200 (TD)	Through reservoir; cemented for zonal isolation

A **contingency string (7-5/8” or 5-1/2”)** must be carried in the AFE. With a sub-0.5 ppg window, one in three wells in analogous toe-thrust plays needs an unplanned string, and at 1,200 m water depth there is no room to improvise. Budget effect: **+US$4–7M on the affected well**, and +8–12 days.

Well trajectory. Highly deviated to near-horizontal, drilled **along the minimum horizontal stress direction** — PCSB’s own stated requirement, which is also the direction that maximises wellbore instability in shales. This is the single largest NPT driver in the programme.

Completion type. Horizontal subsea tree (TechnipFMC Subsea 2.0 configure-to-order, consistent with the Sept 2026 iEPCI), 5-1/2” 13Cr tubing, permanent downhole gauges, **inflow control valves with zonal isolation packers**. ICVs are not a luxury here: with lateral compartmentalisation across 30 km, the ability to shut off a watered-out or gassed-out compartment without a rig re-entry is worth more than the US$1.0–1.5M it costs per well.

Sand control. Open-hole gravel pack with shunt tubes (alternate-path) in the thick, high-permeability sands; standalone premium screens as the fallback in thin-bed intervals where drawdown can be held low. Frac-pack is technically defensible but has thin local experience in Sabah deepwater and should not be the base case. Budget US$4.2M/well for two gravel-packed intervals.

Artificial lift. Gas lift, full stop. ESPs in a 1,200 m subsea well imply a US$10–20M intervention per failure and a rig queue you do not control. Gas lift gas comes from FPSO compression (150 mmscfd gas-handling capacity exists). Four to six mandrels, orifice-valve design, plus a gas-lift optimisation study before first oil.

Well count split. Public sources say 10 wells; they do not say how many are producers versus injectors. Base case assumed here: **6 oil/gas producers, 3 water injectors (75,000 bwpd capacity available), 1 gas producer or disposal well** [A]. Injectors are ~10% cheaper (no sand control in the same form, no gas lift). If the split is 8 producers / 2 injectors, add ~US$10–15M to the campaign.

WELL COST BUILD-UP

Unit-rate build-up for one P50 development well (63.7 days average). Every line is rate × quantity; nothing is a quoted contract value.

#	Line item	Unit rate	Qty	US$M	Class
1	Drillship day rate (7G, dual activity)	450,000 /d	63.7 d	28.7	[A]
2	Third-party spread (mud, cement, casing running, rental tools, ROV, 2 PSVs, fuel, catering)	170,000 /d	63.7 d	10.8	[A]
3	MPD spread	85,000 /d	30 d	2.6	[A]
4	OCTG — 290 t (36” 190#, 20” 106.5#, 13-3/8” 68#, 9-5/8” 47#, 7” 29#), premium	3,200 /t	290 t	0.9	[D]

#	Line item	Unit rate	Qty	US$M	Class
	connections, 13Cr allowance				
5	Subsea wellhead (18-3/4” 15k), 36” conductor, guide base, corrosion cap	lump	1	2.6	[A]
6	Bits, RSS, LWD/MWD, wireline (4 LWD runs, 2 wireline, 6 bits)	lump	1	3.2	[A]
7	Sand control — 2 open-hole gravel packs, shunt tubes, ~600 m screens	lump	1	4.2	[A]
8	Gas-lift completion — 5-1/2” 13Cr, 5 mandrels, packer, gauges, 2 ICVs	lump	1	1.1	[A]
9	Cementing, liner hangers, float equipment	lump	1	1.6	[A]
10	Tree landing + upper completion running (tree hardware itself sits in the iEPCI/SURF package — **not double-counted here**)	lump	1	1.2	[A]
11	Clean-up / initial flowback	lump	1	1.5	[A]
	Subtotal			**58.4**	[D]
12	Well contingency @ 15%			8.8	[A]
	TOTAL PER DEVELOPMENT WELL (P50)			**67.1**	**[D]**

Aggregation to campaign:

Line	US$M	Basis
10 wells, learning-curve durations 72→58 d	671.3	Sum of per-well costs in campaign schedule
Mob + demob (SE Asia, 2 moves)	24.0	US$12M each; mobilisation is excluded from day rate
Drilling supervision, project management, integrated services	26.9	4% of well subtotal
Shallow hazard / site survey, 10 locations	3.5	US$0.35M per location
Campaign contingency @ 5%	36.3	Schedule/standby/rig-escalation cover
TOTAL CAMPAIGN (P50)	**761.9**	**−26%/+35% → US$566M–1,025M**

Cost per well range: US$52.0M (52 days, US$380k/d rig, 12% contingency, single sand interval) to US$95.6M (84 days, US$550k/d, 18% contingency, contingency liner, losses). A dry hole is **US$49.8M** — 74% of a development well, because the rig, the logging suite and the P&A all still get paid for, and only the completion is avoided.

Sensitivities (each independently, on the P50):

Change	Effect on campaign
Rig day rate ±US$50,000/d across 640 rig-days	±US$32M (±4%)
+10 days per well (NPT, losses, stuck pipe)	+US$62M (+8%)
Learning curve delivers 60 fewer rig-days	−US$37M (−5%)
Contingency liner in 3 of 10 wells	+US$15–21M
8 producers / 2 injectors instead of 6/3/1	+US$10–15M
6-month schedule slip (FPSO standby at ~US$10M/month)	+US$60M, borne by facilities not drilling

Benchmark check. A single-well PETRONAS deepwater exploration contract offshore Indonesia was signed in Sept 2026 at **~US$58M** including provisional sums (Ventura Offshore / Deep Value Driller). That is one well, with mobilisation loaded onto it, no completion. A US$67M completed development well inside a 10-well campaign is consistent with — slightly below — that anchor, which is what a campaign effect should look like. PCSB's SPE 210079-MS also claims the integrated concept achieves **"20 to 30% lower than industry average for deepwater wells"**; at US$67M we are assuming they achieve roughly half of that claimed advantage, which is the honest read of a first-time-operated deepwater campaign.

CAMPAIGN SCHEDULE

One rig, sequential drilling, 5-day moves between subsea wellheads. Spud 1 Oct 2027 assumes rig award Q1 2027, contract signature by Jun 2027, and mobilisation from a Singapore/Malaysia yard.

Well	Duration (days)	Start	End
Mob / acceptance	17	2027-09-14	2027-09-30
LM-01	72	2027-10-01	2027-12-11
LM-02	70	2027-12-17	2028-02-24
LM-03	67	2028-03-01	2028-05-06
LM-04	65	2028-05-12	2028-07-15
LM-05	63	2028-07-21	2028-09-21
LM-06	62	2028-09-27	2028-11-27
LM-07	61	2028-12-03	2029-02-01
LM-08	60	2029-02-07	2029-04-07
LM-09	59	2029-04-13	2029-06-10
LM-10	58	2029-06-16	2029-08-12
Demob	15	2029-08-13	2029-08-27
Campaign total	**714**	**2027-09-14**	**2029-08-27**

Well identifiers **LM-01 … LM-10 are placeholders for this schedule, not actual Limbayong well names.** No official Limbayong development well names are public. Actual field nomenclature must not be inferred from this table.

Duration logic. First well 72 days: no offset experience on this specific well path, MPD system commissioning, and a full data-acquisition programme (VSP, MDT, full LWD suite). The 58-day tenth well assumes a genuine learning curve — PCSB has

never operated a deepwater development campaign before (SPE 210549-MS: Limbayong is *"the company's first deepwater operated asset"*), so a 14-day improvement across 10 wells is an optimistic-but-defensible 20% curve, not a guaranteed one.

Batch vs individual. Recommendation: **do not batch.** Batch drilling pays when wells are clustered on a template or a manifold centre and the rig can skid or move in hours. Limbayong is a 30 km-long, 2.5 km-wide structure with subsea wells spread along it; every "batch" move is a 5-day anchor-handling operation, which destroys the economic case. What should be batched is **procurement and services**, not hole sections: one casing order, one screens order, one MPD contract, one integrated logging contract, and a single set of standard well designs reused with minimum modification. That is where the 5% learning-curve saving (−US$37M) actually comes from.

Overlap with facilities. Wells 8–10 finish after the FPSO should be on station. Two of the last three wells should be drilled to TD and temporarily suspended pending hook-up, so that drilling completion does not gate first oil. This is a deliberate schedule buffer and should be written into the drilling programme, not discovered during execution.

RIG SELECTION AND THE 2026–2028 MARKET

Selection: 7G drillship, with 6G as the acceptable fallback.

Option	Fit	Rate (2026 disclosed)	Verdict
7G drillship, dual activity, 2 BOPs	Best. Dual activity and a second BOP cut flat time and let you sidestep after a stuck-pipe event without a round trip to surface. At <0.5 ppg margin this is risk mitigation, not comfort.	US$310k–540k/d; >550k/d leading edge	**Preferred**
6G drillship	Adequate. 15k BOP, 1,200 m WD within rating. Costs ~10–15% more rig-days; saves US$60–100k/d.	US$350–450k/d (6G at a discount to 7G; some trade sources quote 6G at US$400–450k, which is high and should be treated with suspicion)	**Fallback / price lever**
Harsh-environment semi-sub	Wrong tool. Moored at 1,200 m is feasible but slower, and the harsh-environment premium (US$416–480k/d in Norway) buys weather capability Sabah does not need.	US$416–480k/d (Norway)	**Reject**
20k BOP 7G/8G	Over-spec. No public evidence of HPHT.	+US$75–125k/d premium	**Reject**

Market reality, and why this is a schedule risk rather than a cost line:

- **Disclosed 2026 7G fixtures: US$309,500–540,000/day.** Leading-edge units cleared **US$550,000/day in early 2026**, with some 20k-capable units reportedly above that. The 2025 weighted average was **~US$401,000/day** and the market "bottomed around US$400k". A 2028 start at US$450k is therefore a mid-market assumption, not a pessimistic one.
- **Utilisation bottomed in the low-to-mid 80% range in 2025** and is expected to recover through 2H26 into 2027. Low-to-mid 80s sounds loose; for 7G units capable of a 714-day Malaysian campaign it is effectively sold out, because the qualifying fleet is small and geographically sticky.
- **No drillship newbuilds of consequence since 2014.** Reactivation costs US$75–125M per unit and takes 12–18 months. Supply cannot respond to a 2027 tender. The fleet through 2030 is what exists today.
- **Consolidation is removing counterparties.** The announced **Transocean–Valaris merger** would create the largest high-specification floater fleet in the world. Fewer bidders, more pricing discipline.
- **PETRONAS is competing with itself.** In 2026 it re-tendered a drillship for its own ultra-deepwater Indonesia wildcat and signed a one-well US$58M deal — both drawing on the same South-East Asia floater pool that Limbayong needs. Group-level rig demand is not coordinated with project-level need.

- **PETRONAS activity is rising, not flat.** 91 wells to be drilled in 2026 versus 79 in 2025 (+15%), plus 70 P&A. Jackups and floaters compete for the same crews, the same MPD specialists and the same logging talent.

Practical consequence. A 714-day firm requirement is unattractive in a tight market — contractors prefer 3–5 year programmes with escalation. Expect one of three outcomes: (a) a rig at the top of the band (US$520–560k/d, +US$45–70M on the campaign), (b) a start delayed 6–9 months while a unit frees up (+US$60–90M in FPSO standby and escalation), or (c) a bundled multi-field Sabah deepwater tender (Limbayong + Belud + future tiebacks) that creates the duration contractors want. **Option (c) is the only one that solves the problem rather than paying for it.** Tender strategy should be a board-level decision, not a drilling-team decision.

KEY JUDGEMENTS

1. **US$762M P50 / US$67M per well is the number to carry into the FDP economics.** Anything below US$600M assumes a rig rate the 2026 market no longer offers and a learning curve PCSB has not yet earned.
2. **MPD is a US$25.5M programme line, not a well-site option.** 30 days × US$85k × 10 wells. Removing it to save money would be the single most expensive decision in this section, because the alternative is stuck pipe at US$620k/day of spread.
3. **Ten wells is thin for this structure.** 270 MMboe across a faulted 30 km fairway with lateral compartmentalisation implies ~27 MMboe per well. That is achievable in good Sabah turbidites but leaves no margin for a dry development well. One failed well costs US$50M and permanently removes ~10% of the well stock.
4. **Do not batch.** The structure geometry forbids it. Batch procurement instead, and standardise the well design so that well 10 is a repeat of well 4, not a new well.
5. **The rig is the schedule.** Everything else in this project — FPSO conversion, SURF installation, subsea tie-in — can be recovered. A missing drillship cannot. Award by Q1 2027 or rebaseline first oil.
6. **The claimed 20–30% well-cost advantage should be discounted to ~10–15%** in the base case. It was an internal integrated-concept claim made in 2022, before PCSB had operated a deepwater well as operator.

WHAT WOULD CHANGE MY MIND

Observation	Effect on this section
A signed Limbayong drilling contract at a disclosed day rate below US$400k/d	Campaign drops toward US$650M; my rig-market judgement is wrong and the market is softer than 2026 disclosures imply
Actual Limbayong well schematics showing reservoir at <2,000 m TVDSS and MD <4,000 m	Well duration falls to ~45–50 days; cost per well falls to ~US$50M; campaign to ~US$570M
Evidence of a production test or DST on any Limbayong appraisal well	Removes my central subsurface concern; 10 wells becomes a defensible development stock rather than a develop-to-appraise bet
A public statement that PCSB will batch-drill from a template or clustered manifold	Changes the schedule and could save 8–12% of rig-days
Reservoir pressure data showing a >1.0 ppg margin in the main fault block	MPD can be dropped for those wells; −US$15M and −6 days/well
A multi-field Sabah deepwater rig tender covering >3 years	Removes the availability risk and likely prices the rig 10–15% below the spot band
Evidence that the 10 wells include 4+ injectors or a gas disposal well	Cost falls ~5% (simpler completions) but the production well stock is thinner than assumed

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APPRAISAL AND DATA-ACQUISITION WELLS STILL REQUIRED

This is the uncomfortable part. **FID has already been taken (28 Aug 2026), so the pre-FID appraisal window is closed.** What follows is therefore not "pre-FID assurance" — it is the minimum data acquisition needed to keep the development from discovering its subsurface problems through failed producers.

Evidence of the gap: four appraisal wells since 2002 (SPE 210079-MS), on a 30 km × 2.5 km structure, with the operator's own paper acknowledging *"possible different depositional model and greater degree of complication"*. Limbayong-1 (2002) found gas; Limbayong-2 (Nov 2013) found 136 m of oil-bearing sands. Four penetrations across 75 km² of faulted toe-thrust is exploration-scale density, not development-scale.

Recommended programme (2 firm + 1 optional), drilled from the development rig before the development wells:

#	Well (descriptive, not official nomenclature)	Purpose	Duration	Cost
A	Eastern fault block appraisal / step-out	Test the block with the least penetration; core + MDT + full logging; confirm sand continuity and fault seal	55–65 d	US$52–60M
B	Central block appraisal with DST	Production test with build-up; establishes productivity index, skin, compartment connectivity and whether sand control is genuinely required	65–75 d	US$60–70M
C (optional)	Downdip/OWC well or interference test	Oil-water contact, aquifer support, lateral connectivity between injector-producer pairs	50–60 d	US$48–55M

Total: US$100–160M, or 13–21% of the drilling budget and ~2–3% of total project capex. That is cheap insurance against a US$50M dry development well and a FPSO that arrives to find its well stock under-delivering.

Non-well data acquisition still required (and often skipped): - Shallow hazard survey and geotechnical borehole at **each** of the 10 development locations — the 2021 Benthic CPT programme covered 700–1,040 m water depth at selected sites, not all 10 well paths [M: offshore-technology]. - VSP / checkshot in the first two development wells to tie the depth model. - Formation pressure while drilling (FPWD) and mud-log gas data on every well, integrated into the geomechanical model in near-real-time — the <0.5 ppg window cannot be managed with a pre-drill model alone. - A re-validated mechanical earth model. SPE 210549-MS's model was built for a Q1 2023 campaign. Any well drilled since then has produced data that either confirms or destroys it, and nobody outside PCSB knows which.

CONFIDENCE

Item	Confidence	Why
Water depth, well count, 10-well tieback	**High (95%)**	Multiple independent public confirmations, PETRONAS and TechnipFMC scope
2026 rig day-rate band (US$310–550k)	**High (85%)**	Disclosed fixtures and fleet-status reporting; aggregator sites are secondary but consistent with each other
Campaign duration 714 days (±15%)	**Medium (65%)**	Built from assumed well durations on a play with no public offset-well performance; PCSB's first operated deepwater campaign
Cost per well US$67M (US$52–96M)	**Medium (60%)**	Unit rates are defensible; the two variables that matter — days and NPT — are [A], not

Item	Confidence	Why
		[M]
Total campaign US$762M (±30%)	**Medium (60%)**	Arithmetic is sound; input class is AACE Class 4 by construction
Reservoir depth and MD	**Low (35%)**	No public Limbayong well data exists. Kikeh/Gumusut analogue only. This is the weakest number in the section
6 producers / 3 injectors / 1 other	**Low (30%)**	Pure inference from FPSO water-injection and gas capacity; not stated publicly
MPD mandatory	**High (90%)**	Directly stated in PCSB's own SPE paper with the <0.5 ppg figure
2028 rig availability is a real schedule risk	**High (80%)**	No newbuild supply, rising Malaysian activity, PETRONAS competing for the same SE Asia floaters, merger consolidation
Appraisal gap is real and material	**High (80%)**	Based on the operator's own published admissions, not on inference

Overall section confidence: Medium. The cost architecture is defensible to a cost engineer. The well-level engineering is a concept, not a design, and should not be presented to anyone as a Limbayong well plan. The two numbers I would defend hardest are the **714-day campaign duration** and the **US$762M campaign cost**, because both are built from transparent rates and both are wrong in the same direction if the reservoir is shallower than the analogue.

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Every cost in this section is AACE Class 4 (−30%/+50%) or worse, in real 2026 USD, excluding owner's costs, and excluding subsea tree/manifold hardware which sits in the TechnipFMC iEPCI package. No Limbayong well name, depth, casing point or contract value in this document is sourced from a public disclosure; where a number is an assumption it is labelled [A] and the analogue is named.

S4 — FACILITIES & FPSO

Subject: Limbayong Deepwater Development (Block G, ~120 km offshore Sabah; centroid 5.888 N, 114.510 E; 900–1,200 m water depth; oil + non-associated gas; 10 deepwater wells tied back; FID 28 Aug 2026 per Upstream).

> ***DATA CONFLICT TO SURFACE*** *(per FACT_BASE hard-rule 6): The DNeX Bursa filing of 20 Aug 2026 — the primary, audited source — describes Excalibur as a* ***70 m × 70 m × 27 m cylindrical FPSO*** *with* ***300,000 bbl design storage****. Offshore Magazine (4 Aug 2022, trade press) reports* ***60 m diameter / 270,000 bbl / 25,000 bopd****. I have used the Bursa values as primary [M] and reported the trade-press figures as an [A] lower bound because (a) the Bursa filing is signed and audited and (b) the 60 m / 270 k numbers cannot be reconciled with a 70 m × 70 m × 27 m box (circumference 220 m, beam 70 m is incompatible with 60 m diameter). Owner-supplied data trumps trade press in this section.*

HEADLINE

The Excalibur is a **proven, used Sevan-type cylindrical FPSO** — the *right geometric form* for a harsh-environment Malaysian deepwater host but **the wrong size**: nameplate oil-processing capacity (~27,500 bopd) is **~31 % short** of the 40,000 bopd the field needs, storage is **~50 % short** of the 600,000 bbl Limbayong spec, and the hull has **zero gas-handling / no water-injection upgrade** for a 150 mmscfd gas + 75,000 bwpd WI duty. FID has nevertheless been taken, MISC is buying the hull, and the tow is contracted to **MMHE Pasir Gudang** (named in the MOA as the delivery quayside), so the concept is locked. **Recommendation: PROCEED but rebid the processing topsides as a Step 2 fabrication to give a credible route to the 40 k / 150 mmscfd / 75 kbwd design duty**, and route non-associated gas to the **Sabah–Sarawak Gas Pipeline (SSGP) at Kimanis** (already 500 km, 750 mmscfd class) via a ~120–150 km subsea tie from the FPSO. Conversion CAPEX US$300–500 M, 24–30 months from yard-cut to first-oil, AACE Class 5 (-30 %/+50 %).

NUMBERS

Parameter	Value	Unit	Class	Basis
Hull type	Cylindrical, Sevan-design, geo-stationary weathervaning	—	[M]	DNeX Bursa filing 20 Aug 2026; Sevan Marine press 22 Sep 2008
Year built	2007	yr	[M]	DNeX Bursa filing 20 Aug 2026
Diameter / LOA / Beam / Depth	70 / 70 / 70 / 27 (registered)	m	[M]	DNeX Bursa filing 20 Aug 2026
Storage capacity (design)	300,000	bbl	[M]	DNeX Bursa filing 20 Aug 2026
Storage (trade-press alt)	270,000	bbl	[A]	Offshore Magazine 4 Aug 2022 — see conflict note
Oil processing (as-built, 2008)	30,000	bopd	[M]	Sevan Marine press release 22 Sep 2008
Oil processing (Avalon downgrade, 2022)	~25,000	bopd	[A]	Offshore Magazine 4 Aug 2022

Parameter	Value	Unit	Class	Basis
Water-injection capacity (as-built)	20,000	bwpd	[M]	Sevan Marine 22 Sep 2008
Gas handling (as-built)	None / minimal test sep. only	mmscfd	[U]	Not stated in any DNeX or Sevan release; Chestnut was oil-only
Mooring	Internal turret, weathervaning, 9-leg chain (Sevan standard)	—	[A]	Sevan Marine technology design; not stated in DNeX filing
Port of registry (last)	Nassau, Bahamas	—	[M]	DNeX Bursa filing 20 Aug 2026
Hull build location	China (hull); Netherlands (topside outfit)	—	[M]	DNeX Bursa filing 20 Aug 2026
Operating history	Chestnut (UK CNS, 2008–2012, Venture); Anasuria cluster (UK CNS, 2012–2022, Ping); warm-stacked Port of Nigg, Scotland (2022 → 2026)	—	[M]	Sevan Marine 22 Sep 2008; Offshore Magazine 4 Aug 2022; DNeX Bursa filing 20 Aug 2026 (lay-up at Nigg)
Hull age at first-oil (2029/30)	22–23	yr	[D]	2029 − 2007
Class society	ABS (independent valuer cited by DNeX)	—	[M]	DNeX Bursa filing 20 Aug 2026
Independent fair-value range	22.5 – 43.5	*USM*|[*M*]|*ABSConsultingappraisal*, *perDNeXBursafiling*20*Aug*2026||*Saleprice*(*MISC*)|35.0|*US*M	[M]	DNeX Bursa filing 20 Aug 2026
Delivery point (new owner)	MMHE quayside, Pasir Gudang, Johor	—	[M]	DNeX Bursa filing 20 Aug 2026, Appendix I
Field design oil rate (Limbayong)	40,000	bopd	[M]	Offshore-Technology / Marine Technology News (citing WER), 2021
Field design gas rate (Limbayong)	150	mmscfd	[M]	Offshore-Technology; TechnipFMC iEPCI release Sep 2026
Field design storage (Limbayong)	600,000	bbl	[M]	Marine Technology News (citing WER), 2021
Field design WI (Limbayong)	75,000	bwpd	[M]	Marine Technology News, 2021
Reserves (oil, remaining recoverable)	139.0	MMbbl	[M]	FACT_BASE §VOLUMES, GEM 2023

Parameter	Value	Unit	Class	Basis
Reserves (gas, remaining)	790.9	bcf	[M]	FACT_BASE §VOLUMES, GEM 2023
Plateau equivalent (139 MMbbl / 40 k bopd)	9.5	yr	[D]	139e6 / 40 000 / 365
Distance Limbayong → Kikeh	~35 ± 10	km	[A]	Great-circle from block centroids, ±25 % because block footprints are 18–30 km wide (Petronas 2021)
Distance Limbayong → Gumusut-Kakap	~60 ± 15	km	[A]	Same method
Distance Limbayong → Malikai	~30 ± 10	km	[A]	Same method
Distance Limbayong → Kimanis (SSGP landfall)	~140–180	km	[A]	Approx, subsea + onshore Sabah west coast
Kikeh oil capacity (FPSO)	120,000	bopd	[M]	FPSO Ventures / Oil & Gas Advancement
Kikeh water depth	1,300	m	[M]	FPSO Ventures
Kikeh gas export	None (oil only, re-injected)	—	[M]	Project descriptions — no SSGP tie to Kikeh
Malikai host type	TLP (TLT — tension-leg platform)	—	[M]	Oil & Gas Advancement / Offshore Technology
SSGP capacity class	750	mmscfd	[A]	Press references; throughput ~750 mmscfd class, 500 km Kimanis→Bintulu
SSGP status	Operating since 2014	—	[M]	The Borneo Post Oct 2013; Wikipedia
Total deck area available (70 m cylinder)	3,850	m²	[D]	π·(70/2)²
Usable deck area (less 20 m turret)	~3,500	m²	[D]	Geometry, see NUMBERS §DECK AREA

Parameter	Value	Unit	Class	Basis
Implied topsides weight envelope	7,000 – 9,000	t	[D]	2.0–2.5 t/m² benchmark (FPSO Benchmarking 2000; SBM Kepler Cheuvreux 2019)
Cargo cycle (300 k bbl @ 40 k bopd, 85 % working)	6.4	days	[D]	Storage / rate
Cargo cycle (270 k bbl @ 27.5 k bopd, 85 % working)	8.3	days	[D]	Storage / rate

CONVERSION SCOPE & COST

Yard: Malaysia Marine & Heavy Engineering (MMHE), Pasir Gudang, Johor. **Named in the MOA as the delivery quayside** (DNeX Bursa filing Appendix I). MMHE is the MISC in-house yard and has done every MISC FPSO conversion for two decades (FPSO Cendor, FPSO Ruby, FPSO Peris). No FID-grade alternative yard is on the table.

Scope (brownfield, 19-year-old hull):

1. **Hull integrity recertification** — full class re-entry survey, hull UT 100 % near critical nodes, turret bearing / mooring chain / anchor leg replacement or recertification, blasting + re-coating. The hull was warm-stacked since ~2022, which is *less* corrosive than cold-stack but the mooring system has had no load cycle in 4+ years.
2. **Topside removal, redesign, rebuild, reinstall** — Excalibur's Chestnut-era topsides were ~3,000 t and designed for 30 k bopd / 20 k bwpd / zero gas. Limbayong needs 40 k bopd + 75 k bwpd + 150 mmscfd gas + 600 kbbl storage. **All of the processing train, all of the gas train, and the bulk of the water-injection train is new**. Practical scope: 5,500–8,500 t of new topsides, integrated onto the cylindrical deck (3,500 m² usable).
3. **Storage upgrade** — 300 k bbl as-built vs 600 k bbl required. Cylindrical hulls have a fixed central-storage ring sized at build. *You cannot add 300,000 bbl to a 300,000 bbl hull.* This is the single most important engineering decision in the project and the Section recommendation must address it directly: either accept a **6-day cargo cycle** (vs the 14-day cycle the design spec implies) or design the additional storage as a **calm-barge shuttle alongside** (capex +US$150–250 M, schedule +8–12 mo).
4. **Power generation** — full gas-turbine train replacement, with waste-heat recovery if Tier-4 emissions. Hull was electric-drive when new (Sevan spec) but no published nameplate MW.
5. **Gas train** — *new build*: slug catcher, multiphase pumps, gas conditioning, glycol regeneration, vent/flare, gas metering skid. The Sabah non-associated gas is CO_2-bearing and the project is marginal, so the gas train should be sized for 150 mmscfd at peak day, not design day.
6. **Water injection** — new 75 kbwpd HP pumps, filters, deoxygenation, scaling control.
7. **Mooring** — Internal turret was designed for North Sea metocean; Sabah monsoon and deepwater cyclones impose different loads (typhoon swells). Re-engineer the chain size and anchor pattern; expect all 9 legs to be replaced.
8. **Marine systems** — helideck (likely upgrade to Sikorsky S-92 class), lifeboats to current SOLAS, dynamic positioning interface (if any), offloading system — either tandem off-take to a shuttle tanker (current Anaconda-style system was a Sevan feature) or new SOFEC CALM buoy (capex impact).
9. **Commissioning & sail-away** — incl. 2-tow trip: Nigg → Pasir Gudang (heavy-lift vessel, ~30 days, weather window) and Pasir Gudang → Limbayong field (~10 days, wet tow, monsoon-sensitive).

Cost (AACE Class 5, -30 %/+50 %):

Line	Low (US*M*)|*High*(US*M*)	Class	Basis	
Hull acquisition (already contracted)	35.0	35.0	[M]	DNeX Bursa filing 20 Aug 2026
Hull tow Scotland → Pasir Gudang	8	15	[A]	Heavy-lift vessel benchmark; HLV rate ~US$200–400 k/day × 30–45 days; shared insurance
Class recert, hull blasting, coating	25	40	[A]	Benchmark US$5–10 k/dwt; 2007 hull, ~30 k dwt
Mooring system (replace 9 legs + anchors)	35	60	[A]	Per-leg cost US$3–5 M; chain + installation
New topsides (5,500–8,500 t @ US$15–20 k/t, integrated EPC)	80	170	[A]	FPSO Benchmarking 2000 cost/t; SBM/Keppler 2019; PaxOcean FPSO benchmark US$236 M for 30 k bopd, 5.5 kt (2024)
Gas train (slug-catcher, conditioning, metering)	40	80	[A]	150 mmscfd train benchmark
Power generation (gas turbines + WHR)	30	55	[A]	30–50 MW class
Water-injection package	25	45	[A]	75 kbwpd HP pumps + filters
Offloading system (SOFEC or shuttle)	25	50	[A]	New-build SOFEC CALM buoy or tandem off-take
Hull-side modifications, hook-up, commissioning	30	60	[A]	Yard labour + offshore hook-up
Project owner cost (EPCm, PMT, insurance, spares)	40	80	[A]	12–15 % of EPCI
Contingency (Class 5, ~30 % of base)	100	200	[A]	AACE 18R-97
Conversion total	**~475**	**~890**	[D]	Range
Recommended central estimate	**~600**		[A]	Mid-point of Class 5 range

> *Note: FACT_BASE §FACILITIES flagged a 2020 capex of US$600–800 M for "the FPSO" — this is consistent with the range above if the comparison was on a tanker-conversion benchmark; the Sevan-cylinder hull itself was US$35 M, so 2020 capex was the all-in processing/hull build, not the cost of the hull alone.*

Schedule (Class 5):

- Tow Scotland → Pasir Gudang: Q1 2027 (30–45 days, weather-window bound)
- Yard-cut to sail-away at MMHE: **18–24 months** (FPSO conversion benchmark, PaxOcean/MISC norms; Sevan cylinder deck is *easier* than a tanker conversion because there is no flare tower relocation, *harder* because the integrated deck and turret must be preserved)
- Sail-away to first oil: 3–4 months (tow + hook-up + commissioning)
- **Total: 24–30 months from FID**, putting first oil between **Q4 2028 and Q2 2029** if FID-day is 28 Aug 2026.

Brownfield integrity risk — the elephant in the room:

- The hull is **19 years old at sail-away** and **38 years old at end-of-field-life** if Limbayong produces 10 years. Class societies typically grant 5-year recertification cycles; 38 years is **beyond the design life of the original fatigue analysis** even though a Sevan cylinder sees lower cyclic loading than a monohull.
- The vessel was warm-stacked (Nigg, Scotland) — better than cold-stack (no anchoring cycles, no mooring fatigue) but **the topsides were never designed for tropical duty**, and the gas train is brand-new. The classic Sevan integrity watch-outs are: (i) **bilge-keel / column welds** in way of the cylindrical hull-to-deck joint — re-inspect 100 %; (ii) **turret bearing wear** after 4 years static; (iii) **riser balcony fatigue** in the new configuration.
- The cylindrical hull's *only* integrity advantage over a tanker-conversion is **no monolithic flare tower / no midship strength-rail modifications** — the Sevan design already integrates everything inside the 70 m disk, so there is no heavy structural cutting.

ALTERNATIVES EVALUATED

#	Option	Capex (US$M, Class 5)	Schedule	Technical risk	Verdict
A	**Newbuild cylindrical FPSO** (Sevan-type, 40 k bopd, 150 mmscfd, 600 k bbl, 75 k bwpd)	1,200 – 1,800	36 – 48 months from FID	Lowest integrity risk; design life 25 yr; long lead; only 1 qualified yard (Sevan Marine / Nanjing TPI)	**REJECT.** Capex is 2–3× conversion; schedule puts first oil at 2030+; the project is "top decile complexity" and "marginal" per SPE 210079; cannot bear newbuild premium.
B	**Converted tanker FPSO** (e.g., VLCC, single bottom, turret-moored)	700 – 1,000	24 – 30 months	Higher integrity risk on a 20-yr-old hull; topside integration on a 330 m × 58 m deck has long hook-up; mooring redesign for Sabah metocean; *no precedent* for 40 k bopd + 150 mmscfd + 75 k bwpd in Sabah	**REJECT.** MISC chose the cylinder deliberately — the cylindrical deck area (3,500 m² usable) is the *only* geometry that lets a 70 m-diameter hull host the 7–9 kt of integrated topsides Limbayong needs without major structural reinforcement. A tanker would force a smaller, split-module layout with longer offshore hook-up.
C	**Subsea tie-back to existing Sabah deepwater host**	400 – 700	30 – 42 months (SURF + 30–60 km flowline + riser balcony)	Kikeh is 35 km away, but **Kikeh is oil-only with no gas export** (no SSGP tie); 150 mmscfd Limbayong gas would have to be re-injected or stranded. Gumusut-Kakap (60 km) is a Semi-FPS, also oil-only and **already at capacity** post-PU-	**REJECT for primary development.** However, **KEEP as Phase 2** for the Bestari and L-B-P satellite prospects: tie-back to the new cylindrical FPSO is the right economics for the satellites, not the wrong way around.

#	Option	Capex (US$M, Class 5)	Schedule	Technical risk	Verdict
				A1 debottleneck. Malikai (30 km) is a TLP at 60 k bopd nameplate, with the wrong hull form for a tie-back manifold. All three would require **major topsides modifications at the host** to accept 40 k bopd + 150 mmscfd of new fluid.	
D	**Conversion of Excalibur at MMHE Pasir Gudang** (recommended)	475 – 890 (central 600)	24 – 30 months	Moderate: 19-yr hull, new gas + WI trains, storage short by 50 % (manageable with 6-day shuttle cycle), turret recert, mooring redesign.	**PROCEED** with a Step-2 rebid of topsides to close the 40 k bopd gap and a credible gas-export route to the SSGP.

Why the cylinder is the right call — the Sevan geometry concentrates topsides mass over the moonpool/turret, gives a low-motion hull (better uptime in Sabah's monsoon and typhoon swells), and is the only FPSO form MISC owns in this size class. The fact that the hull is **used** is not a defect — it is the *only* way a marginal field gets an FPSO at all. Industry norm: 40–50 % of new FPSO projects in SE Asia / NW Australia in 2024–2026 are used-hull conversions (PaxOcean, MMHE, Keppel order books).

Why storage is a non-fatal gap — Limbayong's 139 MMbbl oil at 40 k bopd is 9.5 years of plateau; a 6-day shuttle cycle = 110 shuttle calls over field life = ~US$1.1 B of shuttle-tanker capex+opex (at US$10 M/yr charter × 110/12 yr). A SOFEC buoy alongside would push storage to 600 k bbl but adds US$150–250 M and 8–12 months. **Recommend: shuttle-tanker offtake at 6-day cycle for first 5 years; review SOFEC at 50 % field-life if oil price supports it.**

GAS HANDLING & EVACUATION ROUTE

Spec: 150 mmscfd of non-associated gas at peak, ~54.7 bcf/yr at design (per FACT_BASE §VOLUMES). 790.9 bcf remaining gas reserves (FACT_BASE, GEM 2023) at 54.7 bcf/yr = **14.5 yr gas plateau**, longer than the oil plateau.

Route recommendation: dedicated subsea gas export pipeline from the FPSO to the Sabah–Sarawak Gas Pipeline (SSGP) at the Kimanis landfall, ~140–180 km of 16″–20″ pipeline, with a pipeline-end-manifold (PLEM) tie-in. The SSGP has been operating since 2014, is 500 km Kimanis → Bintulu (MLNG), 750 mmscfd class, and currently has spare capacity given the Kebabangan / Kamunsu East decline curves.

Why not (a) re-inject all gas: reservoir pressure support is required (the wells are deepwater toe-thrust, compartmentalised, water-injection-supported per FACT_BASE), but a 150 mmscfd gas-handling train is already spec'd. Flaring would breach PETRONAS zero-routine-flare; reinjecting 100 % requires extra compression and reduces sales-gas NPV.

Why not (b) send to Bintulu LNG via SSGP directly: 500 km transit with multiple offtake points; capacity is shared with Petronas upstream gas (Kebabangan, Bintulu Clastic, others). The Limbayong volume (54.7 bcf/yr) is meaningful — ~7 % of SSGP throughput — so a **dedicated tie-in + booking** would be needed and a pipeline capacity agreement with PETRONAS Energy Malaysia.

Why not (c) re-inject and re-inject, no export: leaks 790 bcf of revenue; gas-to-power for Sabah domestic has been the policy since 2014 (Borneo Post 2013).

Decision: Build the export pipeline. Indicative capex for a 16″–18″, 150 km subsea line is US$250–400 M (subsea 7″ equiv benchmark ~US$1.5–2.0 M/km × 1.5 for 16–18″ × 150 km + PLEM + SSGP tie-in skid). This is **not in the S4 conversion capex above**; it belongs in SURF / midstream. The FPSO topsides need only the slug catcher, gas conditioning, and metering skid (already in the US$40–80 M gas-train line).

Evacuation route (offshore personnel): Crew change by helicopter to a Sabah onshore base (Kota Kinabalu or Labuan, ~120 km), S-92 or H175 class on helideck sized to current CAP 437. Emergency evacuation: lifeboats to a stand-by vessel, then to shore. The cylindrical hull has **360° access around the deck** which materially improves evacuation geometry vs. a monohull FPSO with a bow turret.

KEY JUDGEMENTS

1. **The cylindrical form is correct; the size is short.** Sevan geometry is the only credible answer for a marginal Sabah deepwater field; the project lives or dies on a credible **Step-2 topsides rebid** that closes the 40 k bopd / 150 mmscfd / 75 k bwpd gap.
2. **Storage gap is solvable with shuttle-tanker cadence, not a SOFEC buoy.** 6-day cycle is operationally fine for 9.5 years; the SOFEC decision can be deferred to 50 % field-life.
3. **The hull age is the *manageable* risk, not the project-killer.** 19 years at sail-away is within the range of class society recertification practice; the *real* risk is mooring-system redesign for Sabah metocean, not the cylinder itself.
4. **Gas export to SSGP-Kimanis is the gas-NPV unlock.** Without the export pipeline, 790 bcf of reserves is stranded gas; with it, the gas NPV roughly equals the oil NPV at current Malaysian regulated prices.
5. **FID has been taken; the alternative analysis is for the *next* FDP, not this one.** All three rejects above are correctly archived for the Bestari and L-B-P Phase 2 development plan.

WHAT WOULD CHANGE MY MIND

- If a credible bidder offers a **newbuild 70 m Sevan-type cylinder at ≤ US$900 M all-in with ≤ 30 months to first oil**, switch from conversion to newbuild (storage, gas, WI all spec-fit, no integrity compromise).
- If SSGP capacity is *fully booked* when PETRONAS Energy Malaysia confirms (expected Q1 2027), the gas export NPV collapses and the project reverts to **gas-reinjection + reduced oil plateau**; the 40 k bopd design rate then requires a 150 mmscfd gas-injection compression train (capex +US$120–180 M) and the project slips below commercial threshold.
- If the DNeX-attributed **60 m / 270 k bbl / 25 k bopd** numbers turn out to be the *post-2022 Avalon-deck-removal* as-built state (rather than a trade-press error), the capacity gap widens to ~38 % on oil and the Step-2 rebid is no longer optional — it is the project.
- If the independent ABS class survey at MMHE finds **hull thickness below class-acceptable** at the column-to-deck weld (the classic Sevan fatigue hot-spot), the hull is scrap and MISC must source a substitute cylinder (effectively ending the project as currently scoped).
- If **Malikai TLP topsides are decommissioned before 2030** (Shell exit announcements in 2025–2026 are tracked), then a tie-back to Malikai becomes technically feasible for Limbayong *if* the TLP can be re-topsides for 40 k bopd + 150 mmscfd — at ~US$600–900 M all-in, this becomes the **lowest-cost alternative** to the conversion and the recommendation must flip.

CONFIDENCE

Claim	Confidence	Why
Excalibur is cylindrical, built 2007, 70 m × 70 m, 300 k bbl, Sevan type	**HIGH**	Primary source (Bursa filing) + corroborating Sevan press
30 k bopd / 20 k bwpd as-built nameplate	**HIGH**	Sevan press 2008, primary

Claim	Confidence	Why
Oil capacity short of 40 k bopd (Limbayong spec)	**HIGH**	27.5 k vs 40 k = 31 % gap, arithmetically certain
Storage short of 600 k bbl (Limbayong spec)	**HIGH**	Geometry: cylinder's central storage ring is fixed at build
Gas train is essentially absent as-built	**MEDIUM**	Chestnut was oil-only; no published evidence of gas train
Conversion capex US$475–890 M, central 600	**MEDIUM**	Class 5, AACE 18R-97, benchmark-based
Conversion schedule 24–30 months	**MEDIUM-HIGH**	MISC/MMHE internal benchmark; Sevan deck is *easier* than tanker
Tie-back to Kikeh / Gumusut / Malikai is rejected	**HIGH** for primary; **MEDIUM** if Malikai decomms	Capacity and topsides analysis is robust; the only path to a flip is a Malikai decomm
SSGP-Kimanis tie is the right gas route	**MEDIUM-HIGH**	SSGP is operating, has class capacity, the route is the only one that monetises 790 bcf
Hull integrity risk is manageable, not fatal	**MEDIUM**	19 years is within class practice; the unresolved item is the column-weld UT result at MMHE — *must be tested before FID-spend is committed*

Overall verdict confidence: MEDIUM-HIGH for the conversion concept, **MEDIUM** for the cost / schedule, **HIGH** for the alternatives rejection. The single largest unverified input is the hull's post-Nigg mooring and column-weld condition — recommend that **MISC publish a class-condition report before the project passes 30 % design gate**, and that the FDP include a **US$80–120 M contingency line** specifically for hull remediation.

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— End S4 —

S5 — SUBSEA, SURF & INSTALLATION

Limbayong Deepwater Development · PETRONAS Carigali (PCSB) 100% · Blocks G/J/R, offshore Sabah Screen date: 2026-10-19 · Basis: USD real 2026, AACE Class 4 (−30%/+50%) Prepared for: F13 SOVEREIGN, FDP screening · Evidence classes: [M] measured/sourced, [D] derived, [A] assumed benchmark, [U] unknown

HEADLINE

The full 10-well Limbayong SURF is a ~US$0.68 bn scope (Class 4: US$0.48–1.02 bn), not a US$75–250 M scope. The 2026 TechnipFMC "significant" band is a *booking classification*, not a cost estimate, and cannot describe the full scope: it is 2.7× below even the best-case build-up at the top of the band. The 2021 "US$250–500 M" band, escalated to 2026 at 4.5%/yr, lands at US$623 M — within 9% of my bottom-up number. That convergence is the strongest signal in this section: **the 2021 award was approximately correctly priced for the scope, and the 2026 award is either a partial call-off, a phased booking, or excludes trees.** Any FDP economics built on "SURF ≈ US$250 M" understates subsea capex by roughly US$430 M.

Second headline: **the architecture that fits this field is not the cheapest one.** Three drill centres at 6.5–18 km, 900–1,200 m, waxy Sabah crude, and an operator commitment to tie back prospects at 18–30 km all point to multiplexed electro-hydraulic control, insulated flexible flowlines, lazy-wave flexible risers, and a single dynamic umbilical to a subsea distribution unit with spare capacity. Direct hydraulic is technically excluded at these offsets, not merely uneconomic.

NUMBERS

Parameter	Value	Unit	Class	Basis
Wells (total)	10	wells	[M]	PETRONAS media release, 1 Jun 2021
Well split assumed (7 prod / 2 WI / 1 GI)	7 / 2 / 1	wells	[A]	Inferred from "oil and non-associated gas"; no public well count by type
Water depth range	900–1,200	m	[M]	PETRONAS, 1 Jun 2021
Distance to shore (Kimanis / Sabah coast)	~120	km	[M]	PETRONAS, 1 Jun 2021
Future tie-back radius (stated prospects)	18–30	km	[M]	PETRONAS, 1 Jun 2021
Drill centres / manifolds	3 production + 1 WI	count	[A]	10 wells in 3 clusters; 2 spare 4-slot positions
Tie-back distance, DC-1 (4 wells)	6.5	km	[A]	Field-layout assumption; no public field map
Tie-back distance, DC-2 (3 wells)	12.0	km	[A]	As above
Tie-back distance, DC-3 (3 wells)	18.0	km	[A]	As above

Parameter	Value	Unit	Class	Basis
Well-count-weighted mean tie-back	11.6	km	[D]	(4×6.5 + 3×12 + 3×18)/10
Maximum step-out	18.0	km	[A]	Set at the low end of the stated 18–30 km prospect radius
Production flowline, 8” ID insulated flexible	36.5	km	[D]	3 lines × (6.5 + 12 + 18)
Water-injection flowline, 6” flexible	18.5	km	[D]	2 lines × (6.5 + 12)
Gas-injection flowline, 6” 15 ksi flexible	6.5	km	[D]	1 line to DC-1
Total flowline	61.5	km	[D]	Sum of above
Umbilical (1 dynamic + 3 static, STU + FO)	24.9	km	[D]	1.4 dynamic + 6.5 trunk + 5.5 + 11.5 infield
Total pipe/umbilical to install	86.4	km	[D]	61.5 + 24.9
Flexible lazy-wave risers (3 × 8”, 3 × 6”)	6	risers	[A]	One per flowline; no spare riser in base case
Subsea trees, 10 ksi horizontal, Subsea 2.0 CTO	10	each	[M]/[A]	Count [M]; configuration [A] per 2026 iEPCI scope
Tree unit cost	10.5	US$M	[A]	Floor US$6.1 M/tree from Petrobras 82-tree, US500*M* + *tender*; *Asia*10 − *offCTOpremium*‖4 − *slotproductionmanifoldunitcost*|22.0|*US*M
2-slot WI manifold	8.0	US*M*|[*A*]|*Low − pressure*, *single − service*‖*Subseacontrolsystem*(*MUXElH*, *dualSEM*, *MCS*, *HPU*)|48.0|*US*M	[A]	10 wells / 3 manifolds; 7% of total
Jumpers & connectors (15 off)	1.4	US$M each \| [A] \| 10 tree + 3 manifold + 2 interconnect \| \| Production flowline unit rate, 8" insulated flex \| 2.40 \| US$M/km	[A]	Pipe + 5-layer insulation + lay; range 1.8–2.8
WI flowline unit rate, 6” flex	1.20	US*M*/*km*|[*A*]|*Uninsulated*, *lowerpressure*‖*Umbilicalunitrate*|1.30|*US*M/km	[A]	STU with 1” gas-lift tubes + FO; range 0.9–1.7
Flex-lay / construction vessel day rate	330,000	US$/day	[A]	High-spec 250 t AHC benchmark >US80*k*/*dayhull*; *fullSURFspread*3–4 × *hull*‖*Heavy − liftvesseldayrate*|420,000|*US*/day
Trenching spread	150,000	US/*day*|[*A*]‖*Post − layjettrenching*, *ROV*‖*Vesseldays—flex − laycampaign*|150|*days*|[*D*]|44*lay* + 8*umbilical* +	**[D]**	Hardware 459.6 + install 89.9 + eng/PM 8% + contingency 15%

Parameter	Value	Unit	Class	Basis
		36*risers* + 30*tie − ins* + 12*transit/survey* + 20*weather*||*Vesseldays—heavylift*|20|*days*|[*D*]|3*manifolds* × 5*d* + 5*dcontingency*||*Vesseldays—trenching*|41|*days*|[*D*]|61.5*km*@1.5*km/day*|| * *TotalSURF*(*basecase*, *exgasexporttoshore*) * *|* * 682 * *| * **US**M**		
Class 4 range	478–1,024	US*M*|[*D*]| − 30|*SURFperwell*|68.2|*US*M/well	[D]	682/10
SURF per km of pipe installed	7.9	US*M/km*|[*D*]|682/86.4||2021*bandescalatedto*2026(*top*)|623|*US*M	[D]	500 × 1.045[5]
2026 award band, top	250	US$M	[M]	TechnipFMC “significant” = US$75–250 M
Ratio: build-up ÷ 2026 band top	2.7	×	[D]	682/250

FIELD LAYOUT & ARCHITECTURE

Layout concept [A]

No public field map, well trajectory, or discovery timeline exists for Limbayong. The layout below is an explicit assumption set, built to be internally consistent and to stress the cost model at a realistic step-out. It is not a claim about the actual field.

- **FPSO** at field centroid [A]; spread-moored or turret configuration unresolved [U] — the FPSO is being sourced as the used cylindrical hull *Excalibur* (US$35 M, DNeX→MISC, Aug 2026), and riser interface design depends on whether that hull carries a disconnectable turret. This is the single largest unhedged interface risk in S5.
- **DC-1 (Limbayong Main):** 4 wells, 4-slot manifold, 6.5 km.
- **DC-2 (Central):** 3 wells, 4-slot manifold, 12.0 km.
- **DC-3 (North/Satellite):** 3 wells, 4-slot manifold, 18.0 km.
- One 2-slot WI manifold at DC-1 serving both injectors.
- 2 spare manifold slots retained deliberately: PETRONAS stated the project “will pave the way for the development of the surrounding prospects within 18 to 30 km.” Spare slots and umbilical spare tubes are strategic optionality for that stated follow-on, not gold-plating.

Controls: multiplexed electro-hydraulic, not direct hydraulic [A]

Direct hydraulic is excluded on three independent grounds: 1. **Offset.** Direct hydraulic is practically limited to roughly 5 km and ~20 functions before response time and hydraulic lag become unacceptable. At 18 km with 10 wells it fails outright. 2. **Function count.** Each tree needs SCSSV, production master, wing, choke, annulus vent, crossover, plus sand/pressure/temperature monitoring. Ten wells × ~8 functions plus manifold valves exceeds any direct-hydraulic hose bundle that could be installed economically. 3. **Diagnostics.** No feedback path, no condition monitoring, no remote choke adjustment — unacceptable for a field whose operator has publicly flagged “top decile reservoir complexity” and marginal economics. Flexibility to re-choke individual wells without a vessel is worth more than the control-system saving.

Base case: dual redundant subsea electronics modules per manifold, 2 × 50% hydraulic supply, fibre-optic in the umbilical for distributed temperature/acoustic sensing on the 18 km production line (leak and slug detection). **All-electric actuation is technically available on Subsea 2.0 and would cut umbilical diameter and cost by roughly a third, but has no PETRONAS track record in Malaysia; treat as a post-FID value-engineering option, not the base case.** [A]

Flexible vs rigid flowlines

Base case: flexible, insulated, reel-layable for all production, WI and GI lines. Reasons: - 900–1,200 m and 18 km sits inside the comfortable flexible envelope; 8” ID at 10 ksi is standard product, not a development item. - Compliant risers follow directly from flexible flowlines — no flex-joint, no riser base manifold, no SLWR buoyancy modules. - Reel-lay installation is roughly 2–3× faster than J-lay at these lengths, and the Sabah weather window is the binding constraint. - Insulation (5-layer PP, 50–70 mm) integrates into the pipe wall; pipe-in-pipe would be needed to match U-values with rigid steel.

Rigid steel would only win if (a) the gas export line to shore proceeds at 120 km — a 12–16” rigid SLWR plus pipeline is a different procurement and vessel class — or (b) reservoir pressure/temperature exceeds flexible limits (unknown: no public BHT/BHP for Limbayong). Flag as a falsifier, not a base case.

Manifolds and jumpers

Three 4-slot production manifolds, 10 ksi, 1,200 m rated, with pigging loop and integrated sand detection. Fifteen jumpers: 10 tree-to-manifold, 3 manifold-to-flowline, 2 interconnect. All ROV-installable — **no saturation diving anywhere in this development**, which removes an entire cost and safety category and is a genuine advantage of the 900–1,200 m range.

Risers: lazy-wave flexible, not free-standing

- 3 × 8” production, 3 × 6” WI/GI, all lazy-wave flexible, plus 1 dynamic umbilical.
- **Free-standing riser rejected.** A free-standing hybrid riser needs a buoyancy can, a foundation, and a flexible jumper to the FPSO; at 1,200 m it costs roughly US$80–150 M per tower. Justified only where riser count is high, pressure is extreme, or the host has severe motion. Here it would add US$100 M+ to solve a problem flexible lazy-wave already solves.
- **Steel lazy-wave riser (SLWR)** retained only for the deferred 120 km gas export route.

Umbilical

Single 1.4 km dynamic section from FPSO to a subsea distribution unit, then static trunk and infield legs — 24.9 km total. This is cheaper than three parallel dynamic umbilicals (one per drill centre) and concentrates the fatigue-critical section in one place. Spare electrical and hydraulic cores and two spare 1” tubes are included for the 18–30 km prospect tie-backs.

COST BUILD-UP

Line item	Qty	Unit	Rate (US*M*)|*US*M	Class	
Subsea trees, 10 ksi horizontal, Subsea 2.0 CTO	10	each	10.50	105.0	[A]
Tree installation / landing (rig-assisted)	10	well	2.00	20.0	[A]
Production manifolds, 4-slot	3	each	22.00	66.0	[A]
WI manifold, 2-slot	1	each	8.00	8.0	[A]
Subsea control system (MUX E/H, dual SEM, MCS, HPU)	1	lot	48.00	48.0	[A]
Jumpers & connectors	15	each	1.40	21.0	[A]
Production flowline 8” insulated flexible	36.5	km	2.40	87.6	[A]
Water-injection flowline 6” flexible	18.5	km	1.20	22.2	[A]

Line item	Qty	Unit	Rate (US*M*)|*US*M	Class	
Gas-injection flowline 6” 15 ksi flexible	6.5	km	1.60	10.4	[A]
Umbilical (STU + FO, static + dynamic)	24.9	km	1.30	32.4	[A]
Flexible lazy-wave risers, 8”	3	each	7.50	22.5	[A]
Flexible lazy-wave risers, 6”	3	each	5.50	16.5	[A]
Hardware subtotal				**459.6**	[D]
Flex-lay / construction vessel	150	days	0.330	49.5	[D]
Heavy-lift manifold installation	20	days	0.420	8.4	[D]
Trenching spread	1	lot	9.00	9.0	[A]
Pre-commissioning & commissioning	1	lot	14.00	14.0	[A]
Mobilisation / demobilisation	1	lot	9.00	9.0	[A]
Installation subtotal				**89.9**	[D]
Engineering, PM, installation engineering, logistics (8%)				44.0	[D]
Contingency (15%)				89.0	[D]
TOTAL SURF				**682.5**	[D]
Class 4 range (−30%/+50%)				**478–1,024**	[D]

Cost shape. Trees 15.4%, manifolds 10.8%, controls 7.0%, flowlines 17.6%, umbilical 4.7%, risers 5.7%, installation 13.2%, engineering+contingency 19.5%. This is a *hardware-heavy, install-light* profile — the opposite of a long single step-out, where Crondall Energy benchmarks put flowlines at 45%, umbilicals 30%, SPS 25% of SURF. The difference is structural: three clusters and ten trees load the SPS end. Anyone benchmarking Limbayong against a single 50 km tie-back will get the split wrong.

Scope boundaries (deliberately excluded). Drilling and completions (separate); FPSO hull, conversion, moorings and topsides (separate); 120 km gas export pipeline and shore terminal (deferred, Phase 2 — would add an estimated US$250–450 M); operating spares and 2-year IPS; flow-assurance chemicals; decommissioning. **If the 120 km export line is inside the sanctioned scope, S5 is understated by 35–65%.**

Sanity check against the two disclosed bands.

	2021 award	2026 iEPCI award
Reported band	US$250–500 M [M]	US$75–250 M [M]
TechnipFMC booking taxonomy	“substantial” = US$250–500 M [M]	“significant” = US$75–250 M [M]
Escalated to 2026 (top of band)	US$623 M [D]	—
My bottom-up	US$682 M [D]	US$682 M [D]

	2021 award	2026 iEPCI award
Verdict	**Consistent** (9% gap)	**Inconsistent** (2.7× gap at band top)

The 2021 number was reported as a *value band attached to a real scope*, and it survives escalation almost exactly. The 2026 number is a *booking class* — TechnipFMC's own defined tier — and a 10-well, 61.5 km, 900–1,200 m SURF cannot be delivered for US$250 M on any unit-rate set I can defend. Most probable readings, with my weights:

1. **Partial call-off / phased booking (p ≈ 0.50).** The iEPCI is structured as a frame with initial call-off for the first drill centre (4 wells, SPS + short flowlines ≈ US$180–240 M); later phases booked as they are released. Consistent with "fast-track" language in the award.
2. **Trees supplied outside this contract (p ≈ 0.20).** PETRONAS increasingly procures subsea trees under a separate long-term frame agreement. Removing trees and tree installation (−US$125 M) gives US$557 M — still above the band, so this only works combined with (1).
3. **The estimate is inflated (p ≈ 0.15).** Would require a single-cluster field with ≤8 km step-outs and ~US$25 M/well SURF. Possible only if the real field is far more compact than the 18–30 km prospect statement implies.
4. **Amendment / reservation fee, main value booked elsewhere (p ≈ 0.15).** The 2021 contract may never have been formally terminated; the 2026 award could be a re-scope delta.

The number to carry into the FDP is US$682 M, not US$250 M. Even the most favourable reading (2) only reaches US$557 M.

SCHEDULE & VESSEL RISK

Critical path [A, anchored on [M] dates]

Milestone	Window	Basis
FID	28 Aug 2026	[M] Upstream
iEPCI award	15 Sep 2026	[M] TechnipFMC
Detail design / Subsea 2.0 configuration freeze	Q4 2026 – Q3 2027	[A] 9–12 months for CTO platform
Tree manufacture & FAT (10 off)	Q1 2027 – Q2 2028	[A] 12–16 months from PO
Flexible pipe manufacture (61.5 km + risers)	Q4 2026 – Q4 2028	[A] 18–24 months; **capacity-constrained**
Manifold fabrication (4 off)	Q2 2027 – Q2 2028	[A] 14–18 months
Umbilical manufacture	Q3 2027 – Q3 2028	[A] 12–14 months
Offshore installation campaign	**Q1 2029 – Q3 2029**	[A] Single-season campaign, 211 vessel days
Tie-ins, commissioning, RFSU	Q3 – Q4 2029	[A]
First oil	**Q4 2029 – Q1 2030**	[A] Must align with Excalibur conversion
Deferred gas export (if sanctioned)	2030+	[A]

The schedule is driven by **flexible pipe lead time, not by installation**. Flexible pipe capacity is the tightest link in the global SURF chain entering 2026 — new capacity is not expected to relieve the market until 2029, and Petrobras alone has tendered roughly 500 km. Limbayong's 61.5 km of flowline plus 6 risers will be competing for the same extrusion slots as far larger Brazilian and Guyanese programmes. **Ordering flexible pipe more than 6 months after FID puts first oil into 2030.** [M for market condition, A for the schedule consequence]

Vessel market and availability risk

- **Day rates are elevated but softening at the low end.** 2025 fixtures: 150 t AHC crane vessels often >US$60,000/day; 250 t AHC >US$80,000/day — at or beyond the previous peak. Short-term and spot fixtures varied almost four-fold top-to-bottom in 2025, so a 2029 campaign cannot be priced off today's spot. [M]
- **Supply is arriving.** ~30 firm newbuild 150/250 t AHC vessels are on order, >20% of the active fleet — a ratio comparable to the 2008–2015 build wave, mostly Chinese-built, mostly speculative and uncommitted. This should loosen 2028–29 availability but widen the rate spread by vessel age and specification. [M]
- **Top-tier concentration.** The Subsea 7–Saipem merger (Saipem7), expected to close H2 2026, leaves three Tier-1 SURF contractors and a combined market leader. For a single 211-day campaign in a remote basin, that is a thin bench: two credible bidders, not four. [M for merger, A for consequence]
- **Malaysian OSV market** is seen as stable-to-positive over 2026–28, with a leaner offshore cycle per PETRONAS Activity Outlook 2026–2028 — supportive for support vessels, but the flex-lay/reel-lay class is not an OSV and is priced globally. [M/A]
- **Weather.** Sabah offshore is less monsoon-exposed than Sarawak, but 1,200 m operations are swell-sensitive. Assume ~200 operable days/year and a hard March–October campaign window; every 30 days of slippage costs roughly US$10 M in standby and pushes RFSU past the next window. [A]
- **The real vessel risk is not rate — it is sequence.** A single-season campaign installs 61.5 km of flowline, 24.9 km of umbilical, 6 risers, 4 manifolds and 15 jumpers with one flex-lay vessel. There is no second-vessel redundancy. A 20-day mechanical failure in June 2029 costs a full year to first oil.

KEY JUDGEMENTS

1. **SURF is US$682 M (Class 4: US$478 M–1,024 M), or ~US$68 M/well.** The disclosed 2026 band understates the scope by at least 2.7×. Any economic model using US$250 M is not conservative — it is wrong by roughly US$430 M, which on a 270 MMboe field is ~US$1.6/bbl of undeclared capex.
2. **The 2021 and 2026 awards are not two estimates of the same thing.** 2021 was a scope-valued award; 2026 is a booking class. Treating them as a declining cost trend — "TechnipFMC cut the price in half" — is a category error and the most likely analytical trap in this FDP.
3. **Multiplexed E/H control is mandatory, not optional.** Direct hydraulic cannot reach 18 km with 80+ functions. The cost of MUX (US$48 M, 7% of SURF) buys remote choke management on a field the operator itself calls top-decile complex — that is the highest-value 7% in the budget.
4. **Flexible lazy-wave everywhere; free-standing riser is not a candidate.** FSHR would add US$100 M+ to solve a motion problem that compliant flexible risers already solve at 1,200 m with six risers.
5. **Three drill centres with two spare slots is the right architecture, and it is also the tie-back option premium.** PETRONAS has publicly committed to the 18–30 km prospect radius. Sparing slots and umbilical cores now costs perhaps US$15–25 M; retrofitting a manifold position later costs a vessel campaign.
6. **The binding constraint is flexible pipe, not the vessel market.** 2029 vessel availability looks better than 2025; 2028–29 extrusion slots do not. This inverts the usual SURF risk ranking and should drive the procurement plan.
7. **No diving anywhere.** 900–1,200 m means all-ROV installation and intervention. This is a real cost and safety advantage over shallow-water Malaysian analogues and should be protected in the design (ROV-friendly torque tools, no diver-assisted connectors).
8. **The FPSO interface is the largest unhedged item in S5.** Whether *Excalibur* carries a disconnectable turret determines riser count, dynamic umbilical length, and whether the field can be storm-disconnected. Until that is settled, the riser and umbilical numbers carry an additional ±15% of their own value.

WHAT WOULD CHANGE MY MIND

Falsifier	Effect on this section
Actual field is one compact cluster, max step-out ≤8 km	Flowline/umbilical quantities fall ~55%; SURF drops toward US$420–480 M and the 2026 band becomes almost credible. This is the single most important unknown.

Falsifier	Effect on this section
The 120 km gas export line is inside sanctioned scope	Add US$250–450 M; S5 total goes to US$0.93–1.13 bn.
Subsea trees confirmed under a separate PETRONAS frame agreement	Remove US$125 M; total ~US$557 M, and reading (2) in the sanity check becomes primary.
Reservoir BHT >120 °C or BHP >10 ksi	Flexible pipe may be disqualified; rigid flowlines + SLWR add 20–35% to flowline and riser lines.
Excalibur confirmed as spread-moored with no turret	Riser design simplifies (no disconnectable turret), dynamic umbilical shortens; modest cost reduction but a hard constraint on riser count.
Operator confirms phased development (4 wells first)	Cost profile splits into two campaigns; total rises 10–15% from remobilisation, but year-1 capex falls to ~US$300 M.
A second Limbayong-area FID (e.g. Bestari) shares the FPSO	Spare slots convert to revenue; per-barrel SURF falls materially.
Vessel day rates fall below US$250 k/day for flex-lay	Installation line drops ~US$12 M — not material to the verdict.

CONFIDENCE

Element	Confidence	Why
Well count, water depth, distance, prospect radius	**High**	Primary PETRONAS release, corroborated by three trade press reports
Award bands and their meaning as booking classes	**High**	TechnipFMC's own published taxonomy; PETRONAS release confirms scope but not value
Total SURF within ±30%	**Medium**	Quantities are assumed, not measured; unit rates are benchmark-derived, not quoted
Per-well and per-km metrics	**Medium**	Arithmetic is sound; both inherit the layout assumption
Layout (3 drill centres, tie-back distances)	**Low–Medium**	Explicitly assumed. No public field map, trajectory or well count by type exists
Control philosophy (MUX E/H required)	**High**	Physics and function count, not a cost judgement
Riser selection (lazy-wave flex, reject FSHR)	**High**	Standard for 1,200 m, 6 risers, FPSO host
Schedule to first oil Q4 2029–Q1 2030	**Medium**	Anchored on [M] FID and award dates; flexible pipe lead time is the swing
Vessel day rates	**Medium**	2025 crane-vessel rates are [M]; flex-lay spread multipliers are [A]
The 2026 band is not the full scope	**High**	2.7× arithmetic gap cannot be closed by any defensible unit rate

Overall S5 confidence: Medium. The cost *level* is defensible to Class 4. The layout is an assumption and the single largest source of error — deliberately flagged rather than dressed up.

SOURCES

1. PETRONAS, "PETRONAS Awards Limbayong Deepwater EPCIC Contract to TechnipFMC," media release, 1 Jun 2021. https://www.petronas.com/media/media-releases/petronas-awards-limbayong-deepwater-epcic-contract-technipfmc — 10 wells, 900–1,200 m, 120 km offshore Sabah, SURF scope, 18–30 km prospect radius.

2. TechnipFMC, “TechnipFMC Awarded iEPCI Contract for PETRONAS’ Limbayong Project Offshore Malaysia,” 15 Sep 2026 (press release; site Cloudflare-blocked, text via Business Wire and trade press). Subsea 2.0 CTO platform; full subsea scope.
3. MaritimeMagz, “TechnipFMC Wins $75M-$250M iEPCI Contract for Malaysia’s Limbayong Project,” 16 Sep 2026. https://maritimemagz.com/technipfmc-wins-75m-250m-iepci-contract-for-malaysias-limbayong-project/ — value band; Subsea 2.0 CTO; fast-track framing.
4. Offshore Source, “TechnipFMC Lands Subsea 2.0 iEPCI Deal for PETRONAS Limbayong Development,” 15 Sep 2026.
5. JPT, “TechnipFMC Awarded Limbayong Subsea Bounty,” 23 Feb 2021 — reported US$250–500 M.
6. Offshore Energy, “TechnipFMC wins SURF EPCI contract with Petronas,” 1 Jun 2021 — iEPCI/FEED history; prospect radius.
7. TechnipFMC contract-tier definitions: “significant” = US$75–250 M; “substantial” = US$250–500 M — confirmed across multiple TechnipFMC releases (e.g. StockTitan, 2026; Offshore Network, West Africa award).
8. Upstream, “Exclusive: Petrobras fires up $500 million-plus tender for 82 subsea trees,” 2024 — implied floor of ~US$6.1 M/tree.
9. Crondall Energy, “Subsea Flowlines — Advanced Flow Management & Reuse” — SURF cost split: long step-out flowlines 45% / umbilicals 30% / SPS 25%; short step-out 30/20/50.
10. OEdigital / Fearnley Offshore Supply, “Subsea Vessel Market Outlook ‘Robust’ for 2026,” 30 Dec 2025. https://www.oedigital.com/news/533915-subsea-vessel-market-outlook-robust-for-2026 — 150 t AHC >US$60 k/day, 250 t AHC >US$80 k/day; ~30 newbuilds, >20% orderbook ratio; Subsea 7–Saipem merger H2 2026; four-fold spot spread.
11. Bravo Energy, “Deep Water Intelligence — Issue 01: The Flexible Pipe Squeeze,” Jun 2026 — flexible pipe capacity constrained, relief not before 2029; Petrobras ~500 km tender.
12. PETRONAS, Activity Outlook 2026–2028; Kenanga sector note, 29 Jan 2026 — leaner Malaysian offshore cycle; OSV market stable-to-positive 2026–28.
13. Bursa filing via The Star, 20 Aug 2026 — DNeX/Ping Petroleum sells FPSO *Excalibur* to MISC for US$35 M cash.
14. Shell, “First Oil from Phase 4 of the Gumusut-Kakap/Geronggong-Jagus East Development,” 2025; Offshore Magazine, Gumusut-Kakap — 19 subsea wells, ~1,200 m, 120 km offshore Sabah: closest architectural analogue.

Assumption register [A] (not sourced, stated for challenge): well split 7P/2WI/1GI; three drill centres at 6.5/12/18 km; 8” production / 6” WI-GI line sizes; 10 ksi rating; 4-slot manifolds with 2 spare; single dynamic umbilical to SDU; six flexible lazy-wave risers; all unit rates and vessel spreads; 200 operable days/year; 8% engineering and 15% contingency.

S6 — COST & SCHEDULE (AACE CLASS 4)

Project: Limbayong Deepwater Development, Blocks G/J/R, offshore Sabah (~120 km NW, 900–1,200 m WD) **Operator:** PETRONAS Carigali Sdn Bhd (100% WI). **FID:** 20 Aug 2026. **Basis:** USD, real 2026 terms. Escalation shown as a separate line so the estimate stays in constant money. **Estimate class:** AACE International 18R-97 **Class 4** — accuracy −15% to −30% low / +20% to +50% high. The band used here is **−30% / +50%** (the broad end of Class 4, justified by the absence of a public FDP, no public well count, no public FPSO capacity, and a used-hull conversion on an unverified vessel). Maturity level: 30–50% of definitive definition.

> ***CONTRADICTION WITH FACT BASE — FLAGGED.*** *FACT_BASE.md line 33 records 2023 "commercially unviable as originally proposed standalone FPSO" and line 42–44 record Excalibur acquisition + FID in Aug 2026. That is the correct sequence and my schedule is built on it. But the FID was taken **before** the subsea ceiling was halved (US$250–500M in 2021 → US$75–250M in 2026) on the **same** iEPCI/Subsea 2.0 model. A smaller subsea commitment at FID, on a project that already failed once on cost, is not a scope optimisation — it is a **staged commitment with the expensive half (floater + wells) still unhedged**. That single fact drives both the cost risk and the schedule risk below.*

HEADLINE

Forward capex, FID to first oil: US$4.05 billion (P50 point estimate), AACE Class 4, range US$2.83–6.07bn (−30%/+50%). Capex/boe: US$15.0/boe on 270 MMboe. Capex per flowing barrel: US$101,000/bopd (US$62,300/boepd). Realistic first oil: Q4 2030 (P50), not the 2029 date implied by a 36-month FID-to-first-oil plan. P90 = Q3 2031.

Excluded from the US$4.05bn: (a) ~US$570M of pre-sanction sunk cost 2002–2026 — real money, already spent, does not belong in a forward FID number but does belong in the full-cycle breakeven; (b) FPSO lease/opex; (c) abandonment.

CAPEX SUMMARY

Point estimate at deterministic line-item level (no contingency, no escalation in the line items). Quantities are [A] where no public figure exists; rates are [A] benchmark rates cross-checked against the analogues in the BENCHMARKING table.

WBS	Element	USD million	Class	Basis
1	**Wells — drilling & completion (10 wells)**	**950.0**		
1.1	Rig — 10 × 95 days × US$450k/day	427.5	A	6th-gen DP drillship, SE Asia 2026–28 term market US$400–480k/d; 95 d/well = 55 d drill + 40 d complete, toe-thrust, 900–1,200 m WD, ~3,500 m TD. [D from A rates]
1.2	Completion & well services (sand control, gravel pack, upper completion, stimulation)	280.0	A	US$28M/well; 0.65× rig cost, typical deepwater frac-pack ratio

WBS	Element	USD million	Class	Basis
1.3	Subsea trees, wellheads, controls (10)	180.0	A	US$18M/well installed; Subsea 2.0 configure-to-order trees US$12–20M
1.4	Drilling CAM, logistics, shorebase, materials	62.5	A	~6.5% of wells subtotal
2	**SURF / subsea production system**	**640.0**		
2.1	Manifolds, jumpers, SDU, flying leads (4 manifolds)	190.0	A	US$25M/manifold + US$90M jumpers/connectors
2.2	Flowlines, risers, umbilicals — 60 km @ US$4.2M/km	252.0	A	8–10” production + gas-lift + water-injection + 2 power/comms umbilicals; in-field tiebacks only (FPSO on field)
2.3	Installation — SURF vessels, ROV, survey, pre-commissioning	140.0	A	~310 vessel-days @ US$450k/d
2.4	SURF engineering, PM, qualification (HIPPS/erosion, waxy crude)	58.0	A	~9% of SURF
3	**FPSO — cylindrical, converted used hull**	**755.0**		
3.1	Hull acquisition (Excalibur, ex-Ping Petroleum UK)	35.0	**M**	Bursa filing via The Star, 20 Aug 2026 — US$35M cash
3.2	Reactivation, class renewal, conversion & new topsides (40,000 bopd, 150 mmscfd, WI, 100 POB)	550.0	A	Malaysian yard conversion; 45–60% of a US$1.2–1.8bn newbuild equivalent. Cylindrical (Sevan-type) geometry carries a **yard-experience premium** — MMHE has not delivered a cylindrical conversion
3.3	Mooring system, suction anchors, turret/swivel	80.0	A	12–16 leg spread mooring, 1,000 m WD
3.4	Sailaway, tow to Sabah, offshore hook-up	90.0	A	8–10 week ocean tow + 90-day hook-up
4	**Gas export pipeline to shore (SOGT/Kebabangan corridor)**	**300.0**	A	~120 km, 24–26” wet-gas line + shore approach + metering. **Highest-uncertainty single line** — if FID scope is oil-only with gas deferred/reinjected, delete this line and add

WBS	Element	USD million	Class	Basis
				~US$120M compression
5	**Offshore construction support, survey, commissioning spreads**	**120.0**	A	Non-SURF marine spread, pipeline pre-commissioning, nitrogen/well-test spreads
6	Project management, engineering, PMC, owner's costs	331.8	D	12.0% × (1..5) = 12% × 2,765.0
7	Operations readiness — crew, spares, first fills, contracts	85.0	A	~2% of capex
8	Escalation, 2026→2030 spend profile	250.0	D	~4%/yr on the ~55% of capex spent after 2027; kept separate so items 1–7 stay in real 2026 USD
	Base estimate (deterministic)	**3,431.8**		Sum 1–8
9	Project contingency	617.7	D	18.0% × 3,431.8. Covers known-unknowns *inside* the point estimate (rig-rate spike, yard overrun, well count +2, reservoir-driven scope)
	TOTAL, P50 POINT ESTIMATE	**4,049.5**		

Memo — pre-sanction sunk cost, 2002–2026 (NOT in the total above): exploration/appraisal wells ~7 × US$60M = 420; 3D/3D-4C seismic ~60; three FEED/tender cycles (2019, 2021, 2022–23) ~90. **≈US$570M [A]**, of which a material share is unrecoverable under the PSC cost-recovery ceiling. Full-cycle capex including sunk = **US$4.62bn → US$17.1/boe [D]**.

TOTAL WITH RANGE

Case	USD billion	US/*boe*(270*MMboe*)|*US*/bopd (40,000)	Trigger	
Low (−30%)	2.83	10.5	70,900	10 wells deliver, hull survey clean, gas line deferred, rig at US$380k/d, no yard overrun
P50	**4.05**	**15.0**	**101,200**	Base case above
High (+50%)	6.07	22.5	151,900	14 wells, hull requires major steel renewal, gas line plus compression, +12-month schedule, Malaysian yard productivity at the observed 0.6–0.7×

Case	USD billion	US/*boe*(270*MMboe*)|*US*/bopd (40,000)	Trigger
			NW-European benchmark

AACE Class 4, −30%/+50%, explicitly. The asymmetry is real and is not a hedge: on a used-hull cylindrical conversion with a top-decile-complexity reservoir, the distribution has a fat right tail and almost no left tail below −30% (you cannot build 10 deepwater wells for less than US$700M in 2026).

SCHEDULE

Built FID→first oil on a **critical-path basis**, not a wish-list. Two chains matter: FPSO conversion, and the drilling campaign (which only becomes critical if the rig slips).

Phase	Start	End	Duration	Basis
FID	20 Aug 2026	—	—	[M] Upstream, 28 Aug 2026
SURF iEPCI award	15 Sep 2026	—	—	[M] TechnipFMC PR, 15 Sep 2026
Detailed design / EPC engineering	Sep 2026	Sep 2027	13 mo	[A] Three prior FEEDs de-risk this; 13 mo not 18 mo
Long-lead procurement (trees, manifolds, gas compressors, mooring chain, line pipe)	Oct 2026	Mar 2028	18 mo	[A] Tree/manifold lead time 18–22 mo ex-works; compressors 22 mo
Hull delivery to Malaysian yard (tow from UK)	Nov 2026	Feb 2027	4 mo	[A] 8–10 wk tow + 1–2 mo yard slot wait. **Yard slot is the first real risk: MMHE's 2027 slots are contested**
FPSO reactivation, class renewal, conversion, topsides	Feb 2027	Aug 2029	**31 mo**	[A] 24–30 mo conversion + 3–6 mo for special survey/steel renewal on an unverified-age hull
Sailaway, tow, offshore hook-up	Sep 2029	Nov 2029	3 mo	[A]
Subsea installation campaigns	Nov 2028	Aug 2029	10 mo	[A] Two campaigns; must precede FPSO arrival for in-field flowlines, follow for riser tie-ins
Drilling campaign — 10 wells	Apr 2028	Feb 2031	**35 mo**	[D] 10 × 95 d × 1.15 NPT = 1,093 rig-days. Rig slot Apr 2028 assumes a 2026-let term contract; **SE Asia 6th-gen availability 2027–28 is the tightest since 2014**
— first 5 wells (sufficient for first oil)	Apr 2028	Oct 2029	18 mo	[D] 546 rig-days

Phase	Start	End	Duration	Basis
Commissioning & systems completion	Jun 2030	Oct 2030	5 mo	[A]
FIRST OIL — P50	**Nov 2030**		**51 mo from FID**	[D]
First oil — P10 (operator's implicit case)	Jul 2029		36 mo	[A] Requires conversion in 22 mo, no hull surprises, rig on time
First oil — P90	Sep 2031		62 mo	[A] One slip of either critical chain

Critical path: FPSO conversion (Feb 2027 → Aug 2029) → hook-up → commissioning → first oil Nov 2030. **Near-critical:** rig availability. If the drillship is not contracted by Q1 2027, first oil moves month-for-month into 2031 — the wells chain has only ~4 months of float against the floater.

Why 51 months and not 36. The three Malaysian deepwater benchmarks all beat this: Kikeh 43 months (FID 2004 → first oil Aug 2007), Gumusut-Kakap ~44 months (FID 2011 → Nov 2014), Malikai 46 months (FID Feb 2013 → Dec 2016). Every one of those was a **newbuild or purpose-built floater with a clean hull**. Limbayong is a **1990s-vintage used cylindrical hull**, on a reservoir the operator itself calls *top decile complexity*, in a yard with no cylindrical-conversion reference. Pricing this at 36 months would be assuming Limbayong executes better than three projects that all executed well and still took 43–46.

BENCHMARKING

Reserves and rates as cited; capex figures are the best public number and several are scope-ambiguous — treated as [A] where the scope behind the number is not public.

Project	Capex US$bn	Reserves MMboe	Capacity bopd	Capex/boe	Capex per daily bbl	Overrun / slip
Kikeh (2007, 1,330 m)	1.4 [M]	400–700 [M]	120,000 [M]	2.0–3.5	11,700	Facilities figure only; excludes some drilling. Schedule held, cost grew materially vs. 2004 sanction — **~+25–35% [A]**
Gumusut-Kakap (2014, 1,200 m)	~2.7 [A]	400 [M]	120,000 [A]	~6.8	~22,500	First oil slipped ~12 mo; **~+20% [A]**. SURF alone US$825M [M] — one contract, one third of total
Malikai (2016, 500 m)	~2.2 [A]	134 [M]	60,000 [M]	~16.4	~36,700	FID Feb 2013 → first oil Dec 2016 vs. 2015 target: **~+12**

Project	Capex US$bn	Reserves MMboe	Capacity bopd	Capex/boe	Capex per daily bbl	Overrun / slip
						mo, ~+20% cost [A]
Rotan (2014, tieback to Kikeh)	~0.9 [A]	~40 [A]	~30,000 [A]	~22	~30,000	Tieback model; low capex absolute, high per-barrel. **~+15% [A]**
Limbayong (this estimate)	**4.05**	**270**	**40,000**	**15.0**	**101,200**	**+35% vs. FID basis, +15 mo vs. 36-mo plan [D]**

What the table actually says. Limbayong sits *between* Malikai (US$16.4/boe) and Gumusut-Kakap (US$6.8/boe) on capex per barrel of resource — which looks acceptable. It is **2.7× Malikai and 4.5× Kikeh on capex per flowing barrel**, and that is the metric that governs payback. The reason is structural: 270 MMboe drained through a 40,000 bopd / 150 mmscfd facility means a short plateau and a long decline tail, so every dollar of facility capital is spread over fewer high-rate days. **This is what "marginal" means in the operator's own SPE 210079 title.** The project is not expensive per barrel in the ground. It is expensive per barrel per day.

Overrun statistics — deepwater. IPA's 2012 industry study (published via SPE 153695 / OTC-Brazil 29804) found **78% of upstream projects in 2011 experienced either a cost overrun or a schedule slip** [M]. EY's 2014 analysis put **64% of multibillion-dollar, technically challenging oil and gas projects into overrun or delay** [M, EY release], with the industry-wide exposure above US$500bn [M]. IPA's consistent finding across vintages: **deepwater and arctic megaprojects are the worst-performing category**, with median cost growth in the +20–40% band and schedule slip of 12–24 months [A]. Applied to a US$4.05bn P50, that median experience alone puts the expected outturn at **US$4.9–5.7bn**, which sits inside — but in the upper half of — my +50% ceiling. That is the honest read: **the +50% case is not a stress case, it is the median historical outcome for this asset class.**

KEY JUDGEMENTS

1. **US$4.05bn is the number, and the −30% side is barely reachable.** Ten deepwater wells at US$95M each and a US$755M converted cylindrical FPSO are floor-bound. The credible range is really US$3.6–6.1bn; the −30% case (US$2.83bn) requires deferring the gas line *and* a clean hull survey *and* a soft rig market simultaneously.

2. **The subsea contract halving is the tell, and it is not good news.** Same contractor, same Subsea 2.0 model, ceiling cut from US$250–500M to US$75–250M between 2021 and 2026. That is not efficiency — Subsea 2.0 pricing did not halve. It means **less scope is being bought at FID**. My SURF line (US$640M) is ~2.5–8× the announced contract band, and the gap has to be reconciled either as separate tree frame agreements, a separate installation EPCI, or scope that is not yet contracted. Any of those three answers raises, not lowers, the risk that SURF is under-budgeted at sanction.

3. **The FPSO is the cost and schedule problem, not the wells.** A US$35M used cylindrical hull bought from a UK North Sea operator is a hull with an unknown steel-renewal liability, and a cylindrical conversion is a specialist job with essentially no Malaysian reference. I have carried 31 months and US$550M for conversion. A clean special survey could save US$80M and 4 months; a bad one adds US$200M and 9 months. **This is a binary I cannot resolve from public data — mark it [U] and manage it.**

4. **First oil Q4 2030, and the operator will say 2029.** The 36-month case requires beating Kikeh, Gumusut-Kakap and Malikai — all clean-hull projects — by 7–10 months. It is not credible. Budget the cash flow on 2030 and treat 2029 as upside.

5. **The gas line is the swing scope item.** US$300M for a ~120 km wet-gas export line is a third of the FPSO cost and is only justified if the 790 bcf is monetised at a price that clears the tariff. If gas is deferred, capex drops to ~US$3.75bn and capex/boe on liquids alone (139 MMbbl) rises to US$27/boe — which is where the project starts to look uninvestable. **The gas is not a bonus; on this cost base it is what makes the barrel economics work.**

6. **Full-cycle breakeven is roughly US$48–56/bbl [D]** at US$4.05bn forward + US$570M sunk, 270 MMboe, 15-year life, 10% discount rate, US$4–6/mmbtu gas. That is a thin margin against a US$65–75 deck for a project with a +50% cost tail. **The correct verdict from S6 alone is conditional GO only if the hull survey and the rig contract are both closed before the first major FPSO payment.**

WHAT WOULD CHANGE MY MIND

Observation	Effect on this estimate
Excalibur's actual age, class status and steel-renewal survey report	±US$200M on 3.2, ±9 months. **The single highest-value unknown in the model.**
Award of a drillship contract with a named rig and a 2027–28 slot	Removes the near-critical path; first oil confidence moves from 40% to 70% at Nov 2030
Disclosure of the well count (my 10 is an assumption from "10 deepwater wells" in the fact base)	Each well ±US$95M. 14 wells = +US$380M before contingency
A second SURF/installation contract award above US$250M	Confirms SURF is a US$600M+ scope and validates my line; a *single* US$75–250M award covering everything would cut my SURF line by US$300M and I would be wrong on the high side
Gas sales agreement signed, or explicit statement that gas is deferred	Confirms or deletes the US$300M line, and settles whether the project is a 270 MMboe or a 139 MMbbl project
Any statement of a 2029 first-oil target from PETRONAS	I would hold 2030 and treat the difference as unpriced schedule risk, not as a better project
Yard slot confirmed at MMHE (or a Chinese/Singapore yard) for Q1 2027	±3 months on the critical path

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CONFIDENCE

Number	Value	Confidence	Why
Total capex, P50	US$4.05bn	**Medium (55%)**	Line items are benchmark-built and internally consistent; the two largest uncertainties (well count, hull condition) are structural, not arithmetic
Range	US$2.83–6.07bn	**High (80%)**	AACE Class 4 band is definitional; the historical deepwater overrun distribution independently supports the +50% side
Capex/boe	US$15.0	**Medium-high (65%)**	Numerator medium, denominator [M] from GEM. Robust to ±20% capex
Capex per daily bbl	US$101,200	**Low-medium (45%)**	Driven by the 40,000 bopd figure, which the fact base itself flags as suspicious (design-capacity-vs-daily-rate ambiguity)

Number	Value	Confidence	Why
First oil	Q4 2030 (P50)	**Medium (60%)**	Critical path is well-understood; the variance is concentrated in one binary (hull) and one market (rig)
Sunk cost	~US$570M	**Low (30%)**	Bottom-up from well count and FEED cycles; no public disclosure of Malaysian PSC sunk cost

Where I am least reliable: the well count and the FPSO capacity. Both are [U] in the public domain, both move the answer by more than US$300M, and neither can be resolved without the FDP or a Wood Mackenzie subscription. Everything downstream of those two inputs inherits that weakness.

SOURCES

#	Source	Used for	Class
1	Upstream Online, “Petronas finally takes FID on Limbayong deepwater development,” 28 Aug 2026 — https://www.upstreamonline.com/field-development/petronas-finally-takes-fid-on-limbayong-deepwater-development/2-1-2036155	FID date; cylindrical FPSO via MISC	M
2	TechnipFMC press release, “TechnipFMC Awarded iEPCI Contract for PETRONAS’ Limbayong Project,” 15 Sep 2026 — technipfmc.com (page Cloudflare-blocked at retrieval; headline/date confirmed via secondary)	SURF award date, iEPCI model	M
3	MaritimeMagz, “TechnipFMC Wins $75M–$250M iEPCI Contract for Malaysia’s Limbayong Project,” Sep 2026	2026 subsea contract ceiling	M
4	Corporate Formula, “Limbayong: PETRONAS Cuts TechnipFMC’s Contract Ceiling in Half,” 19 Sep 2026 — corporateformula.substack.com	2021 US$250–500M “substantial” vs 2026 US$75–250M “significant”; same Subsea 2.0 scope model	M (analyst)
5	The Star / Bursa filing, 20 Aug 2026 (DNeX–MISC FPSO Excalibur, US$35M)	Hull acquisition cost	M
6	Offshore Technology, “Kikeh FPSO Development Project Profile” — offshore-technology.com	Kikeh capex US$1.4bn, 1,330 m WD, 120,000 bopd, 400–700 MMbbl, first oil Aug 2007	M
7	Offshore Technology, “Malikai Deepwater Oilfield Project” — offshore-technology.com	Malikai FID Feb 2013, first oil Dec 2016, 500 m WD, Block G, 60,000 bopd	M
8	Global Energy Monitor, “Gumusut-Kakap Oil Field (Malaysia)” — gem.wiki	GK FID, first oil 2014, 400 MMbbl recoverable, Block J, 5.80 N 114.41 E	M
9	Rigzone / Acergy, “SapuraAcergy JV Snags $825MM Gumusut-Kakap Devt Contract”	GK SURF contract value US$825M as a scope-share anchor	M

#	Source	Used for	Class
10	Oil & Gas Advancement, “Malikai Deepwater Oilfield Project”	26,000 t TLP, 17 wells, FEED 2009, FID Feb 2013, KBB tie-back	M
11	AACE International Recommended Practice 18R-97, *Cost Estimate Classification System*	Class 4 accuracy −15/−30% to +20/+50%; maturity 30–50% of definition	M
12	SPE 153695-PA / JPT, Merrow, “Oil and Gas Industry Megaprojects: Our Recent Track Record”	Upstream megaproject cost/schedule failure rates	M
13	OTC-Brasil 29804-MS, “Starting with the End in Mind: More Certain Field Development” (2019), citing IPA 2012 study	78% of upstream projects in 2011 faced cost overrun or schedule slip	M
14	EY via PR Newswire, “Oil and gas megaproject overruns to cost industry more than US$500b,” 14 Aug 2014	64% of multibillion-dollar technically challenging projects overran; >US$500bn exposure	M
15	PETRONAS Carigali, SPE 210079-MS (2022), “Limbayong: Decoding Industry Top Decile Reservoir Complexity for Marginal Deepwater Development”	“top decile complexity” and “marginal” as the operator’s own characterisation — the basis for the 18% contingency and the +50% tail	M
16	Global Energy Monitor, Global Oil & Gas Extraction Tracker (data year 2023) — via FACT_BASE.md	139 MMbbl liquids, 790.9 bcf gas, 14.53 MMbbl/y design capacity	M

Not obtainable within budget / behind paywall: Wood Mackenzie Limbayong asset report (US$3,720) — would resolve well count, plateau, FPSO capacity and unit costs; Upstream full FID article; FPSO Excalibur class records and age. These three would move my confidence on the total from 55% to ~75%. Marked [U] rather than estimated.

S7 — ECONOMICS & FISCAL

Scope: Contractor-side (PCSB 100% WI) pre- and post-tax cash flow for Limbayong, real 2026 USD, discounted 10% real. Model code and outputs: `evidence/s7_model.py`, `evidence/s7_metrics.py`. Every figure below is reproducible from those two files.

Red flag on the shared fact base (stated loudly, per rule 6): the TechnipFMC iEPCI award of **US$75–250M** for "full subsea scope" across **10 deepwater wells in 900–1,200 m** is 2–4× below any credible SURF cost for that scope. Either (a) the award covers a narrow, partial scope and the balance sits in a separate contract not in the public record, or (b) it is a headline number with options/EPCIC exclusions. I have modelled SURF at **$600M**, not $250M. If $250M were truly all-in, NPV10 rises by **$107M to $395M** and IRR to 13.4% — material, but not enough to change the verdict. This is still the largest single unverified value lever in the public record.

HEADLINE

Limbayong clears a 10% real hurdle and fails a 15% hurdle. It is a gas project wearing an oil project's clothes.

#	Metric	Value	Class
1	**NPV10** (real, post-tax, contractor)	**+US$288M** on US$2,900M capex — PV/capex = 0.10	[D]
2	**IRR** (post-tax, real)	**12.3%**	[D]
3	**UTC** (capex + opex, undiscounted, incl. abandonment)	**US$21.06/boe**	[D]
4	**Breakeven Brent** (NPV10 = 0, flat-equivalent)	**US$60/bbl**	[D]
5	**Brent required for 15% IRR**	**US$96/bbl** — above the top of my high deck	[D]
6	Reserves required for NPV10 = 0	233 MMboe = **86% of the 270.8 MMboe booked**	[D]
7	Reserves required for 15% IRR	325 MMboe = 120% of booked	[D]
8	Gas contribution to NPV	Oil-only counterfactual (no gas, no pipeline, capex $2,250M) is **NPV10 −$29M / IRR 9.7%**; gas is worth **+$317M**, i.e. **110% of total NPV**	[D]
9	Peak funding (max cumulative negative)	US$2,784M, reached 2029	[D]
10	Government + PETRONAS host take	45.6% of gross revenue	[D]

Verdict: at a 10% real hurdle — the right hurdle for a 100%-state-owned operator with a Sabah gas mandate — this is a **GO by US$288M**, a 10% margin on capital employed that a single 30% reserve shortfall erases. At a 15% hurdle it is a **NO-GO at every Brent price in the 2027–2035 forward range**. FID was taken because PCSB is not optimising for IRR; it is buying 790 bcf of Sabah gas and 139 MMbbl of liquids with a US$35M second-hand hull and a tax incentive. Judge it on that basis, not on a corporate hurdle rate it cannot meet.

ASSUMPTIONS

All money is **real 2026 USD, unescalated**; discount rate **10% real**. A nominal model at 2% inflation and 12.2% nominal WACC reproduces these results to within ~2%. Costs are **AACE Class 4 (−30%/+50%)**; no estimate here is better than Class 4.

Parameter	Value	Unit	Class	Basis
Reserves — liquids	139.0	MMbbl	[M]	GEM Global Oil & Gas Extraction Tracker (data year 2023), crude + condensate remaining recoverable
Reserves — gas	790.9	bcf	[M]	Same source, 22,398.75 MMm^3
Reserves — total	270.8	MMboe	[D]	139.0 + 790.9 × 0.16667 (6 Mcf/bbl)
Liquids plateau	40,000	bopd	[M]→[D]	GEM design capacity 14.53 MMbbl/y ÷ 365 = 39,808 bopd; rounded to 40,000. Cross-checks against a 139 MMbbl / 15-yr field life
Gas plateau	150	mmscfd	[M]	GEM design capacity 1,550.4 MMm^3/y = 54.7 bcf/y
First oil	2030	year	[A]	FID 28 Aug 2026 + 41 months. FPSO conversion of an existing hull (24–30 mo) runs in parallel with a 10-well, 2-rig drilling campaign (30 mo). Range 2029–2031; 2029 is modelled as the acceleration case
Field life	2030–2045	years	[A]	16 producing years; PSC production period 25 yr, so no contractual truncation
Ramp	45% / 85% / 100%	of plateau	[A]	Two-year ramp, standard for a 10-well subsea tie-back with phased well hook-up
Decline — liquids	17	%/yr	[A]	Exponential, after 4-yr plateau. Toe-thrust turbidites with solution-gas drive and water injection typically 15–25%
Decline — gas	13	%/yr	[A]	After 5-yr plateau; gas plateau set by 150 mmscfd plant limit, not by deliverability

Parameter	Value	Unit	Class	Basis
Capex — drilling & completion	850	$M	[A]	10 wells × $85M all-in (dry-hole risk carried pre-FID; range $60–120M/well for 900–1,200 m, 4–6 km tie-back). Benchmark: Sabah deepwater development wells $60–110M
Capex — subsea production system + iEPCI	600	$M	[A] (band [M])	TechnipFMC iEPCI $75–250M [M] **plus** trees, manifolds, umbilicals, flowlines and installation campaign excluded from that band = $350M. See red flag above
Capex — FPSO	485	$M	[D]/[A]	Excalibur purchase US$35M [M: DNeX Bursa filing 20 Aug 2026] + $450M conversion/upgrade/life-extension for 40 kbopd + 150 mmscfd [A]. Used cylindrical hull, 1990s vintage — condition is [U]
Capex — gas export pipeline	650	$M	[A]	120 km, 26″ insulated wet-gas line + beach/tie-in at ~$5.4M/km. Without this line the 790 bcf cannot be sold — it is not optional
Capex — owners, PMT, pre-FID, contingency	315	$M	[A]	~12% of the $2,585M technical scope
Capex — total	2,900	$M	[D]	850 + 600 + 485 + 650 + 315. **$10,710/boe**
Capex phasing	5 / 25 / 40 / 26 / 4	% in 2026/27/28/29/30	[A]	S-curve, FID Aug-2026, first oil 2030
Abandonment	200	$M	[A]	FPSO demob, well P&A contribution, subsea recovery; spent 2045
Opex — fixed	95	$M/yr	[A]	FPSO operations, marine spread, subsea inspection, insurance, onshore support. Benchmark $80–120M/yr for a 40 kbopd + 150 mmscfd deepwater hub
Opex — variable	4.00	$/boe	[A]	Chemicals, well intervention, consumables, tariff-linked items
Opex — implied unit	7.6 / 9.6 / 20.8	$/boe	[D]	$7.6/boe at plateau (201 ÷ 26.56 MMboe in 2032); $9.6/boe life average; $20.8/boe in 2045 as

Parameter	Value	Unit	Class	Basis
				fixed cost spreads over 5.7 MMboe
Liquids price differential	−1.00	$/bbl vs Brent	[A]	Sabah light sweet (Kimanis-type) normally trades at or slightly above Brent; −$1.00 charged for CO_2/quality and netback to a Malaysian export point
Gas price	6.50	$/MMBtu	[A]	Wellhead netback below the Malaysia Reference Price (MRP Q4-2025 = RM36.64/MMBtu ≈ $8.7 [M: Gas Malaysia notice]) and below the RM30/MMBtu (~$7.1) domestic power-sector cap [M: The Edge, 2023]. Range $3.9–9.1
Royalty	10.0	% of gross	[M]	Malaysia DW PSC terms table: royalty 10%
Research cess	0.5	% of gross	[M]	Same table; **not** cost-recoverable
Export duty — crude/condensate	10.0	% of crude revenue	[M]	Federal export duty 10% on petroleum crude exported; Sabah crude is an export stream
Sabah State Sales Tax — gas	5.0	% of gas revenue	[M]	Sabah State Sales Tax Enactment 1998; natural gas 5% effective 1 Apr 2020 [M: MPM “Other Governing Laws”]
Cost recovery ceiling — oil	70	% of post-royalty oil revenue	[M]	Malaysia DW PSC: oil 70%, gas 80% (UDW oil 75%)
Cost recovery ceiling — gas	80	% of post-royalty gas revenue	[M]	As above
Profit split — oil	50% to 100 MMbbl cum., 60% thereafter	% of profit oil to contractor	[M]	Malaysia DW PSC volume-based split: ≤100 MMbbl 50%, 100–200 MMbbl 60%, >200 MMbbl 70%
Profit split — gas	50	% of profit gas to contractor	[M]	≤2 tscf 50%, 2–4 tscf 60%; Limbayong is 0.79 tscf, so 50% throughout
Petroleum income tax	38	% of chargeable profit	[M]	Petroleum (Income Tax) Act 1967 (Act 543)
Capital allowance	100% of QCE over 10 yr, straight line	—	[M]/[A]	PITA Second Schedule; 10-yr write-off for petroleum QCE. No initial allowance assumed — mildly conservative

Parameter	Value	Unit	Class	Basis
Investment allowance	60% of QCE, capped at 70% of statutory income/yr, carry-forward	—	[M]	PITA (Investment Allowances) Regulations 2013, reg. 12 — **Deepwater Project (>200 m water depth)** is a named qualifying project. 900–1,200 m qualifies. Approval by MOF on MPM verification is assumed granted [A]
PSC vintage assumed	Malaysia **Deepwater PSC (1993 terms, as published in the 2013 MPM/PMU fiscal tables)**	—	[A]/[U]	Limbayong's PSC text is not public. The block was awarded in the deepwater rounds (G/J/R area, Sabah), so DW terms apply. **2018 DW enhancements** (Upstream, 27 Nov 2018) are *not* quantified — they would move terms in the contractor's favour, so my NPV is, if anything, conservative on this axis
PCSB working interest	100	%	[M]	Welligence; 100% WI means no partner dilution and no carried-interest relief
Discount rate	10	% real	[A]	Standard real hurdle for NOC upstream screening; sensitivity to 15% given explicitly

Price deck — Brent, real 2026 USD

Year	Low	Base	High	Basis
2027	70	**84**	95	Base 2027 = EIA STEO, released 6 Oct 2026: Brent $84/b in 2027 [M]
2028	62	**80**	95	Base: post-conflict normalisation; STEO shows 2026 at $96 on a Middle East war premium [M] which is *not* extrapolated
2029	58	**77**	92	Base: gradual convergence to long-run marginal cost
2030	55	**76**	90	Base: plateau onset; long-run anchor
2031	55	**75**	90	Long-run $75 real = midpoint of EIA AEO reference ($70–80/b real) and IEA WEO STEPS; also sits just above US shale full-cycle breakeven ($60–70)

Year	Low	Base	High	Basis
				which is the structural floor
2032	55	**75**	90	
2033	55	**75**	90	
2034	55	**75**	90	
2035	55	**75**	90	
2036–2045	55	**75**	90	Flat real

Low deck = demand-peak/OPEC-share-war scenario: 2025 realised Brent was $69 [M: STEO], so $55 is a further 20% real decline sustained for two decades. High deck = sustained OPEC+ discipline plus recurring supply disruption. **The base deck is not a forecast; it is the number the project has to survive.**

PRODUCTION PROFILE USED

Built independently of every other section, from the two design capacities and two reserve numbers. Shape: 2-year ramp (45%/85%), 4-year liquids plateau at 40 kbopd, 5-year gas plateau at 150 mmscfd, then exponential decline (17%/yr liquids, 13%/yr gas). The profile is then **scaled so cumulative volumes land exactly on 139.0 MMbbl and 790.9 bcf** — no rounding drift into the reserves.

Stream	Plateau	Plateau years	Cumulative	Recovery check
Liquids	40,000 bopd (14.60 MMbbl/yr)	2032–2035	139.0 MMbbl	Matches [M] exactly by construction
Gas	150 mmscfd (54.7 bcf/yr)	2032–2036	790.9 bcf	Matches [M] exactly by construction
Total	**65,000 boe/d (23.7 MMboe/yr)**	2032–2035	**270.8 MMboe**	

Peak year 2032–2035 at 26.56 MMboe. Production ends 2045 at 5.7 MMboe/yr (tails out, not truncated). Note the asymmetry that drives the whole valuation: **liquids plateau lasts 4 years, gas plateau lasts 5 and the gas tail is long** — gas is 48% of boe but 34% of gross revenue and the majority of late-life cash flow.

CASH FLOW SUMMARY

US$M, real 2026. Revenue is gross field revenue. Pre-tax CF = contractor entitlement (cost oil + profit share, net of cess) − opex − capex; royalty, export duty and SST are already removed inside entitlement.

Year	Prod (MMboe)	Revenue	Opex	Capex	Pre-tax CF	Tax	Post-tax CF	Cum.
2026	0.0	0	0	145	-145	0	-145	-145
2027	0.0	0	0	725	-725	0	-725	-870
2028	0.0	0	0	1,160	-1,160	0	-1,160	-2,030
2029	0.0	0	0	754	-754	0	-754	-2,784
2030	12.5	737	145	116	256	9	247	-2,537
2031	22.9	1,348	187	0	758	53	705	-1,833
2032	26.6	1,569	201	0	898	69	828	-1,004
2033	26.6	1,569	201	0	898	69	828	-176

Year	Prod (MMboe)	Revenue	Opex	Capex	Pre-tax CF	Tax	Post-tax CF	Cum.
2034	26.6	1,569	201	0	792	57	735	559
2035	26.6	1,569	201	0	531	34	498	1,056
2036	24.1	1,384	191	0	463	66	397	1,454
2037	20.4	1,167	177	0	433	54	379	1,832
2038	17.4	986	164	0	357	25	331	2,164
2039	14.8	833	154	0	292	1	292	2,455
2040	12.6	704	145	0	238	91	148	2,603
2041	10.7	595	138	0	193	73	120	2,723
2042	9.1	503	131	0	155	59	96	2,819
2043	7.8	426	126	0	123	47	76	2,895
2044	6.6	361	122	0	96	36	59	2,954
2045	5.7	306	118	200	-127	28	-155	2,799
Total	**270.8**	**15,624**	**2,603**	**3,100**	**3,571**	**772**	**2,799**	

Where the gross revenue went (US$M, % of gross): Royalty 1,562 (10.0%) · Export duty 1,029 (6.6%) · Sabah SST 267 (1.7%) · Research cess 78 (0.5%) · PIT 38% 772 (4.9%) · PETRONAS host profit share 3,414 (21.8%) · **Contractor entitlement 9,274 (59.4%)** — of which cost recovery 5,503 + profit share 3,849 − research cess 78. **Total government + PETRONAS take = 7,122 = 45.6% of gross.** Contractor keeps 54.4% of gross but spends 36.5% of gross on capex + opex, which is why the NPV margin is thin.

Fiscal mechanics that visibly bind: - The **70% cost recovery ceiling binds in 2030–2033**. Unrecovered cost carries forward; full payout of the 2026–2030 spend takes until 2033. This is why the pre-tax CF steps down in 2035 once the pool empties and the contractor is on pure profit share. - The **oil profit split steps 50% → 60%** when cumulative liquids pass 100 MMbbl (2037). Small, real, and it is the only progressive element that works in the contractor's favour here. - **Tax is back-end loaded, not avoided:** 2030–2039 pays almost nothing because of the investment allowance; 2040–2045 pays at close to the full 38% marginal rate. Total PIT is $772M with the incentive versus $1,433M without it — the investment allowance abates **$661M** of tax and defers the rest.

METRICS

Metric	Value	Arithmetic
NPV10	**+US$288M**	Σ post-tax CF / 1.10^(t−2026). Cross-check by hand: PV of outflows 2,330; PV of inflows 2,624; net 294 (±6 from rounding)
IRR	**12.3%**	Bisection on the 20-year post-tax stream: FID −145/−725/−1,160/−754 then +247… −155
Payback (undiscounted)	**2034**	Cumulative turns positive during 2034: −176 at end-2033, +559 at end-2034. **8 years after FID, in the 4th full producing year**
UTC	**US$21.06/boe**	(3,100 capex incl. 200 abandonment + 2,603 opex) ÷ 270.8 MMboe = 21.06
Capex-only unit	US$11.45/boe	3,100 ÷ 270.8
Opex-only unit	US$9.61/boe	2,603 ÷ 270.8

Metric	Value	Arithmetic
Breakeven Brent (NPV10 = 0)	**US$60/bbl flat-equivalent**	Deck multiplier 0.796 → 2030–35 reference price $75 × 0.796 = $59.7
Brent for 15% IRR	**US$96/bbl**	Multiplier 1.284 → $75 × 1.284 = $96.3
Brent for 10% IRR	**US$60/bbl** (identical to the NPV10 = 0 price — that is the definition of IRR)	Deck multiplier 0.796
Reserves breakeven (NPV10 = 0)	**233.2 MMboe** (119.7 MMbbl + 681 bcf)	Scale factor 0.861
Reserves for 15% IRR	**325 MMboe**	Scale factor 1.200
Capex for 15% IRR	US$2,338M (US$8,634/boe)	Needs a 19% underspend against a Class 4 estimate — i.e. the −30% end of the range must be realised in full, and then some
Gas price for NPV10 = 0	US$3.86/MMBtu	Gas floor; below the domestic cap by $3.2
Opex for NPV10 = 0	+70% (fixed $161M/yr)	Late-life opex is the second-weakest link after reserves
PV/capex	0.10	288 ÷ 2,900
Investment-allowance value	**+US$353M NPV, +2.8 pts IRR**	Without the 60% IA, NPV10 = **−$65M** and IRR = 9.5%. **The project is uninvestable on a 10% hurdle without this incentive**

Basis bridge — why the contractor view and the consolidated view differ

A sibling section in this pack (`S7X-economics-calibrated.md`) quoted **NPV10 ≈ US$1.1bn at 17.3% IRR** and called that the "PETRONAS-consolidated basis". That is not a contradiction of the US$288M above; it is a different boundary, and the difference is the whole story of Malaysian PSC economics:

Boundary	NPV10	IRR	What it counts
Contractor (PCSB as PSC contractor) — the number in this section	**+$288M**	**12.3%**	Cost recovery + contractor profit share, less all capex, opex and 38% PIT
PETRONAS as host (profit oil/gas share)	+$1,304M	—	PETRONAS' 50%/40% of profit after cost recovery; no capital at risk
PETRONAS consolidated (host + contractor)	**+$1,592M**	**19.8%**	Σ undiscounted $6,213M
Government (royalty + duty + SST + cess + PIT)	+$3,708M undiscounted	—	Accrues to the Federal and Sabah governments, not to PETRONAS

Read this table carefully, because it is the political economy of the field in four lines: **the first dollar of consolidated value is the host's, not the investor's.** Limbayong works at 19.8% consolidated and 12.3% contractor-side. A foreign partner — Shell, who discovered it in 2002 — would earn the contractor line and would be right to walk. PETRONAS can proceed precisely because it collects both lines. That, not the oil price, is why this project waited for PCSB to take 100% and for a US$35M hull to appear.

SENSITIVITY

Each row re-runs the full fiscal stack. ΔNPV is against the US$288M base.

Rank	Case	NPV10 (*M*)|*ΔNPV*(M)	IRR	
1	**Reserves −30%**	**−384**	**−672**	6.8%
2	Reserves +30%	852	+564	16.3%
3	**Capex +30%**	**−135**	**−423**	9.2%
4	Capex −30%	644	+356	16.9%
5	**Oil price −25% ($56)**	**−70**	**−358**	9.4%
6	Oil price +25% ($94)	612	+324	14.7%
7	**Gas price −40% ($3.90)**	**+5**	**−283**	10.0%
8	Gas price +40% ($9.10)	548	+260	14.2%
9	**Opex +30%**	170	−118	11.4%
10	Opex −30%	403	+115	13.1%
11	**Schedule +18 mo, +8% capex**	197	−91	11.8%
12	Schedule −12 mo (first oil 2029)	382	+94	13.0%

Tornado ordering, largest swing first: reserves (−672) > capex (−423) > oil price (−358) > gas price (−283) > opex (−118) > schedule (−91).

This ordering is the finding, not the table. **The two largest value drivers are both cost-and-volume, not price.** Reserves and capex together swing $1.1B of NPV — nearly four times the base NPV. A price-only sensitivity would flatter this project; the operator's own SPE 210079-MS calls the reservoir "top decile complexity" and the development "marginal", and the model agrees with the operator, not with the FID.

Note on the schedule number: a pure time-shift is almost NPV-neutral (deferred capex offsets deferred revenue at 10%), which is why +18 months on its own costs only $8M. The **real** schedule risk is the cost escalation and FPSO day-rate exposure that a slip drags with it — modelled here as an 8% capex adder, giving −$91M. Read row 11 as the lower bound of schedule damage, not the estimate.

Deck outcomes: Low ($55 flat) → NPV10 **−$96M**, IRR 9.2%. Base ($75) → +$288M, 12.3%. High ($90 flat) → +$547M, **14.2%** — still short of 15%.

The three questions asked directly

What oil price makes this a 15% IRR? US$96/bbl Brent, flat, real, for the whole 2027–2045 path ($75 × 1.284). That is 28% above base and 7% above my high deck. In the plausible 2027–2035 range, **Limbayong does not earn 15% at any price**.

What reserves number makes it break even? 233 MMboe (119.7 MMbbl + 681 bcf) — **86% of the booked 270.8 MMboe**. You can lose 14% of the reserves and still return 10%. For 15% IRR you need **325 MMboe, 120% of booked**. The band between "still stands" (86%) and "works properly" (120%) is narrow, and the booked number is a 2023 regulatory estimate for a field with no public well data.

Is the gas monetisation material or cosmetic? Material and decisive — it is the project. Removing gas (and the $650M pipeline) leaves NPV10 at **−$29M and IRR at 9.7%**: oil alone does not clear a 10% hurdle. Gas adds **$317M of NPV, 110% of the total**. Every one of the following is true simultaneously: the FPSO is justified by liquids throughput, the *value* is justified by gas, and the gas only works because it can be sold into a Sabah market at $6.50/MMBtu rather than at an LNG netback struck after liquefaction, shipping and the 5% state sales tax. The gas price floor is **$3.86/MMBtu** — below that the project is NPV-negative. At $10.22/MMBtu it reaches a 15% IRR, but that is above the RM30/MMBtu (~$7.1) domestic cap and is therefore not achievable into the Malaysian market.

KEY JUDGEMENTS

1. **This is a 10%-hurdle project priced by a company that does not need 15%.** NPV10 of $288M on $2.9B of capital is a 1.10× PV/capex ratio. For a listed IOC that is a destroyer of value against a 15% cost of capital. For PCSB — 100% WI, sovereign balance sheet, explicit Sabah gas mandate — 12.3% real clears. FID is therefore rational *as policy*, not as portfolio economics. Say that plainly to the CEO: this is a strategic gas supply decision with an oil co-product, not an oil development with a gas by-product.

2. **The project's investability is manufactured by a tax incentive.** Remove the 60% investment allowance under the PITA (Investment Allowances) Regulations 2013 and NPV10 falls to **−$65M**. $353M of the $288M NPV — more than all of it — is the fiscal incentive. That is a defensible policy choice (deepwater, >200 m, marginal field, national gas supply) but the FDP should not be presented as commercially robust without naming this. If MOF declines or trims the IA, the project is dead on a 10% hurdle.

3. **Reserve risk dominates everything, and the public data cannot bound it.** −30% reserves → NPV −$384M. The field has a 2002 discovery, a 2014 oil confirmation, no public well data, no public seismic, and an operator paper whose title contains "Top Decile Reservoir Complexity" and "Marginal". A 14% shortfall breaks even. For a toe-thrust, faulted, compartmentalised turbidite with a 10-well development plan, a 14% shortfall is not a tail risk — it is close to the base case for this play type. **The single most valuable thing PCSB can do pre-first-oil is appraise, not accelerate.**

4. **Capex intensity of $11.45/boe is the number that will be quoted against this project.** That is roughly 1.7× Gumusut-Kakap (~$6,900/boe) and closing on Malikai (~$14,000/boe) — the two Sabah deepwater benchmarks that bracket "acceptable" and "cautionary tale". UTC of $21.06/boe follows from it. The driver is only 270.8 MMboe of reserves against $2.9B of capital; a 30% reserve upgrade alone would cut capex intensity to $8.24/boe and lift IRR to 16.3%. **UTC is the cleanest single metric for communicating why this field waited 24 years from discovery to FID** — and why the 2021–2023 standalone newbuild concept was cancelled.

5. **The $35M FPSO is the best capital decision in the project and the least analysed.** Buying a used cylindrical hull for 1.2% of total capex fundamentally changes the economics versus the 2021–2023 standalone newbuild concept that was cancelled as "commercially unviable". But a used 1990s hull carries condition, life-extension and uptime risk that a Class 4 estimate cannot see. The $200M abandonment and $450M conversion are judgements, not quotes. If the hull needs $650M rather than $450M, that is roughly **−$85M of NPV** — 30% of the project's margin.

6. **45.6% total take is light for Malaysia and I do not fully believe it.** Published Malaysia PSC splits typically show 60–70% government take. My 45.6% reflects DW terms (70%/80% cost recovery, 50% profit split, 10% royalty, no supplementary payment modelled) plus a 60% IA. If the actual Limbayong PSC carries a supplementary payment, a higher profit share to PETRONAS, or no investment allowance, contractor NPV falls by a further $200–350M. **This is the single largest fiscal uncertainty and it is unresolvable from public sources.**

7. **Breakeven at $60 is a 20% cushion, not a comfortable one.** With Class 4 capex (−30%/+50%) and reserves that could plausibly be −30%, the joint probability of falling below breakeven is not small. A portfolio manager should treat $60 as the *central* breakeven and $75–80 as the *robust* one — the price at which the project survives a simultaneous 20% reserve shortfall and 20% capex overrun.

WHAT WOULD CHANGE MY MIND

Would make me more positive: - **SURF scope confirmed at $250M all-in.** Worth **+$107M of NPV** (→ $395M) and IRR to 13.4%. Note the structural point: **each $1 of capex only costs the contractor $0.43 of NPV**, because cost recovery, the investment allowance and the capital allowance return more than half of it. That is also why the capex tornado is smaller than the reserves tornado — shortfalls are punished far harder than overruns. - **A reserve audit or appraisal well result above 325 MMboe.** That is the 15%-IRR threshold. One good step-out well at the L-B-P trend could do it. - **Confirmation that Limbayong's PSC carries the 2018 deepwater enhancements.** Those were explicitly designed to improve DW returns; if they include a higher cost recovery ceiling or better profit split than the 1993 table, my NPV is understated by an unknown but material amount. - **A reduced or nil royalty in the actual PSC.** Some Malaysian deepwater vintages departed from the standard 10%. Removing 5 percentage points of royalty is worth **+$120M of NPV** (verified by re-run: 5% royalty → NPV $408M); nil royalty is worth +$239M. I model the published 10% because that is what the MPM table says; the PSC text itself is [U]. - **Gas**

price above $8/MMBtu contracted with a Sabah offtaker. Each $1/MMBtu on gas is roughly $100M of NPV at these volumes. $10.22/MMBtu delivers 15% IRR. - **Capex below $2,340M** (a 19% underspend). That requires roughly $560M of savings — conversion at $350M, drilling at $70M/well, pipeline at $500M and SURF at $500M, all at once. It is the full −30% Class 4 outcome, not one or two good contracts. Possible; not plannable.

Would make me more negative: - **Any reserve downgrade below 233 MMboe.** The project is then NPV-negative at base deck. - **Loss or reduction of the 60% investment allowance.** NPV goes to −$65M immediately. This is the highest-severity single-point fiscal risk in the model. - **A supplementary payment or a profit-split step-down appearing in the actual PSC text.** - **FPSO conversion above $550M, or first oil slipping beyond 2031 with cost escalation** (not mere deferral — deferral alone is nearly free). - **A gas offtake priced at an LNG netback (~$4.50/MMBtu) rather than domestic Malaysian pricing.** That takes NPV to **+$74M at 10.6%** — the project becomes an oil-equivalent development that barely clears 10%, and most of the 790 bcf is stranded value.

CONFIDENCE

Element	Confidence	Why
Fiscal mechanics (royalty 10%, cess 0.5%, export duty 10%, PIT 38%, 70%/80% ceilings, volume-based profit split, 60% IA)	**High**	Each is anchored to a published Malaysian instrument or MPM fiscal table. The *applicability* of the 1993 DW table to this specific PSC is the weak link, not the rates
Production profile shape	**Medium**	Plateau is anchored to published design capacity (40 kbopd, 150 mmscfd); ramp, decline rates and 2030 first oil are judgement. Profile shape changes NPV by ±15% at constant reserves
Reserves	**Medium-low**	139 MMbbl / 790.9 bcf are a third-party tracker's figure for a field with no public well data. The 86% breakeven threshold means this is the binding uncertainty
Capex $2,900M	**Low-medium**	AACE Class 4 by construction. Every line except the $35M FPSO and the $75–250M iEPCI band is a benchmark, not a quote. ±30/50% is honest; ±50% is arguable
Opex $95M/yr + $4/boe	**Low-medium**	Benchmark for a 40 kbopd/150 mmscfd deepwater hub. Late-life opex on a used hull is genuinely unknown
Gas price $6.50/MMBtu	**Medium**	Sits between the domestic power cap ($7.1) and a netback after processing. But no GSA is public and CO_2 content could force a discount I cannot quantify
NPV10 = +$288M	**Low-medium**	Correct arithmetic, Class 4 inputs, and it sits only 10% above zero. The honest range is **−$400M to +$850M**, i.e. the sign is not robust
IRR = 12.3%	**Low-medium**	±3.5 points across the reasonable input range. "Not quite 15%" is the robust conclusion; "12.3%" is not
Breakeven $60/bbl	**Medium**	More robust than NPV because it is a ratio — but it inherits the same reserve and capex uncertainty. True range $50–$72

Element	Confidence	Why
Gas is material, not cosmetic	**High**	The oil-only counterfactual (−$29M NPV, 9.7% IRR) is a structural result, not a marginal one. It survives wide changes in gas price and pipeline cost

Single sentence of confidence: I am confident in the *shape* of the answer — a marginal, gas-led, incentive-dependent deepwater development that clears 10% and fails 15% — and not confident in the *precision* of any one number in it.

SOURCES

1. **Petroleum (Income Tax) Act 1967 (Act 543)** — 38% petroleum income tax; Second Schedule capital allowances; consolidated text as at 1 Jan 2026. hasil.gov.my.
2. **Petroleum (Income Tax) (Investment Allowances) Regulations 2013, P.U.(A) 120** — 60% investment allowance on QCE for qualifying projects including **Deepwater Project (>200 m)**; exemption capped at 70% of statutory income. Text via Moore Stephens tax flash; guidelines via MOF "Guidelines on Application of Tax Incentives for Upstream Petroleum Industry".
3. **Malaysia Petroleum Management (PETRONAS), Fiscal Terms** — PSC taxonomy, DW/UDW terms by water depth, R/C structure. petronas.com/mpm.
4. **Malaysia PMU / APEX 2013 booklet, "Malaysia — Petroleum & Investment"** — the DW PSC terms table used here: royalty 10%, export duty 10% (exported oil only), PIT 38%, research cess 0.5%, cost ceiling DW oil 70% / gas 80%, UDW oil 75%, volume-based profit split (oil 50/60/70% at 100/200 MMbbl; gas 50/60/70% at 2/4 tscf), PCSB participation min 15%, production period 25 yr.
5. **Oil & Gas Journal, 22 Nov 1993** — SK-A/SK-B deepwater licence terms, corroborating the 70–75% cost recovery ceilings and volume-based profit splits for Malaysian DW acreage.
6. **MPM, "Other Governing Laws"** — Sabah State Sales Tax Enactment 1998: crude, natural gas, LNG, condensates all at **5%**, effective 1 Apr 2020.
7. **IMF, "Natural Resource Taxation in the Asia-Pacific Region", 2015** — federal **export duty 10%** on petroleum crude exported from Malaysia.
8. **EIA Short-Term Energy Outlook, released 6 Oct 2026** — Brent $96/b 2026, **$84/b 2027**; 2025 actual $69/b. eia.gov/outlooks/steo.
9. **PETRONAS Energy & Gas Trading, "Gas Pricing & Tariff"** — Malaysia Reference Price (MRP) mechanism (LNG FOB weighted average, 6–7 month lag to Brent).
10. **Gas Malaysia Berhad notice, Q4 2025** — MRP = **RM36.64/MMBtu** (≈US$8.7 at 4.2 RM/USD).
11. **The Edge Malaysia, 28 Aug 2023** — domestic piped gas to the power sector capped at **RM30/MMBtu** for the first 1,000 mmscfd.
12. **Upstream, 27 Nov 2018** — PETRONAS introduced enhancements to deepwater PSC terms (content paywalled; existence and date confirmed). Not quantified in this model — upside.
13. **The Star / Bursa Malaysia filing, 20 Aug 2026** — DNeX agrees to sell FPSO Excalibur to MISC for **US$35M** cash.
14. **TechnipFMC press release / MaritimeMagz, 15 Sep 2026** — iEPCI award, full subsea scope, **US$75–250M**.
15. **Global Energy Monitor, Global Oil & Gas Extraction Tracker (data year 2023)** — 139.0 MMbbl liquids, 790.9 bcf gas, design capacity 14.53 MMbbl/y and 1,550.4 MMm³/y.
16. **SPE 210079-MS (2022), "Limbayong: Decoding Industry Top Decile Reservoir Complexity for Marginal Deepwater Development"** — PETRONAS Carigali authors; the operator's own characterisation of reservoir complexity and project marginality.
17. **Welligence** — PCSB 100% working interest, operatorship.

Model: `evidence/s7_model.py` *(engine),* `evidence/s7_metrics.py` *(metrics, breakevens, tornado). Reproducible with* `python3 s7_metrics.py` *from that directory. Pure stdlib — no external dependencies.*

S7X — ECONOMICS, RECALIBRATED AGAINST EXXONMOBIL'S MODELING TOOLKIT

HEADLINE

The ExxonMobil Modeling Toolkit is not a field-development economics model — it is an equity-analyst toolkit for modelling XOM's corporate earnings. But it publishes two numbers that let us calibrate ours, and applying them **overturns the verdict in my first memo.** My earlier conclusion (NPV10 +US$61M, 10.5% IRR, "not a value project") used the **wrong fiscal basis**: I applied a foreign contractor's terms to a field PETRONAS owns 100%. Corrected, Limbayong returns **NPV10 +US$1.11bn and 17.3% IRR at US$70 Brent on a PETRONAS-consolidated basis** — and it does not work for any foreign partner at any realistic oil price. That is the real strategic fact.

WHAT THE TOOLKIT ACTUALLY IS

Document	What it contains	What it is not
Modeling Basics (Jul 2026, 16pp)	How to model XOM's segmented earnings and cash flow	Not a project DCF
Sensitivities & Key Assumptions (2Q26, 12pp)	Annual earnings sensitivities, reference price deck, unit earnings	Not a fiscal model
Interactive Analyst Center	5yr historical + guidance, downloadable	Corporate only

Method published on p5: **Volume × Unit earnings ± (price difference × sensitivity) = earnings**, then **+ DD&A − cash capex = cash flow**. That is a corporate trickle-down, not a field development cash-flow model. There is no IRR, no NPV, no PSC, no production decline curve, no project WBS. [M]

THE TWO NUMBERS THAT MATTER

1. Brent earnings sensitivity. XOM discloses: *increase of $1/bbl Brent = +$700M annual Upstream earnings*, based on ~4.3 Moebd 2024 production. [M: Sensitivities p3; Modeling Basics p8]

```
$700M / (4.3 Mboe/d × 365) = $0.446 per boe, per $1/bbl Brent
```

44.6 cents of every Brent dollar reaches XOM's net income. 55.4% is taken by governments, hosts, and price-linked entitlement effects. That is the single most useful benchmark in the document — it is a real, disclosed, audited effective-take number from the largest integrated in the world, across a portfolio of PSCs, concessions, and OPEC terms.

2. Reference price deck. XOM's constant-price basis [M: Sensitivities p4]: 2024 = **$65/bbl Brent flat real**, $3.00/mmbtu Henry Hub, $6.50/mmbtu TTF 2030 = **$75.40/bbl Brent**, $3.50 HH, $7.50 TTF (2.5% inflation, 2024 reference year)

This is not a forecast. XOM says so explicitly. But it is the deck the largest listed integrated uses to compare projects internally — which makes it a far more defensible screening deck than today's US$105 spot.

3. Unit earnings at $65 Brent: ~$10/oeb [M: Modeling Basics p8]. XOM makes about US$10 per barrel of oil equivalent at US$65 flat. That is an integrated, tax-paid, portfolio-wide number.

THE CALIBRATION TEST

My model's implied government take was **42.1%**. XOM's disclosed effective take is **55.4%**. A **13.3-point gap** — my model was flattering the project.

	Effective take	NPV10 @ $70	IRR
My original model	42.1%	+$61M	10.5%
Calibrated to XOM's 55.4%	54.6%	**−$649M**	**3.7%**

Read that again. **Calibrating to XOM's disclosed sensitivity turns a marginal project into a dead one.** That is what a real benchmark does — it does not confirm you, it corrects you.

THE BASIS ERROR — AND WHY IT FLIPS THE ANSWER

Limbayong is **PETRONAS Carigali, 100% working interest, operator** [M: Welligence]. There is no foreign partner. So "government take" is PETRONAS paying PETRONAS. Applying contractor fiscal terms to it is a category error — and I made it.

Basis	Who is the investor	NPV10 @ $70	IRR @ $70	Brent needed for 10% IRR
Contractor (foreign PSC holder)	An external company	**−$531M**	5.0%	**$96/bbl**
PETRONAS consolidated	PETRONAS itself	**+$1,113M**	17.3%	**$45/bbl**

> *Limbayong is not an oil project that happens to be owned by PETRONAS. It is a PETRONAS balance-sheet project that only exists because PETRONAS can be its own host.* ***No foreign partner would fund this field at any oil price below US$96.*** *That single fact explains the twenty-four years of delay better than any geological story.*

PETRONAS-BASIS ECONOMICS (the decision basis)

Assumptions: 5% Sabah state sales tax on crude (real external leak), PITA 38% on taxable income, 12-year straight-line DD&A, opex $12/boe, gas $6/mmbtu, 66,000 boe/d plateau for 6 years then 14%/yr decline, capex $3.17bn spread 2027–2030, first oil 2029.

Brent	NPV10 (capex $3.17bn)	IRR	NPV10 (capex $5.0bn)	IRR
$50	+$227M	11.6%	−$910M	5.7%
$60	+$672M	14.5%	−$444M	8.0%
$65	+$893M	15.9%	−$215M	9.0%
$70	**+$1,113M**	**17.3%**	**+$11M**	**10.1%**
$80	+$1,551M	19.9%	+$461M	12.0%
$90	+$1,987M	22.4%	+$907M	13.9%
$105	+$2,640M	26.1%	+$1,569M	16.5%

Breakevens on PETRONAS basis (capex $3.17bn):

Hurdle	Brent required	Max capex at $70 Brent
10% IRR	**$45/bbl**	$5,018M ($18.2/boe)
15% IRR	**$62/bbl**	$3,605M ($13.1/boe)
20% IRR	**$80/bbl**	$2,746M ($10.0/boe)

This is the crux. The project clears 15% at $62 Brent *if* capex holds at $3.17bn. It fails at $70 Brent if capex reaches $5.0bn. So the entire investment case rests on one number nobody in public has: **the real capex.**

CAPEX IS THE ONLY VARIABLE THAT MATTERS

Capex	Capex/boe	NPV10 @ $70	IRR
$2.5bn	$9.1	+$1,505M	22.0%
$3.17bn	$11.5	+$1,113M	17.3%
$4.0bn	$14.5	+$618M	13.3%
$5.0bn	$18.2	+$11M	10.1%
$6.0bn	$21.8	−$605M	7.7%

Global deepwater benchmark is typically **US$15–25/boe**. My $11.5/boe central estimate sits *below* the industry range — which is suspicious, and is the single largest uncertainty in this document. If Limbayong lands at industry-typical $18/boe, capex is $5.0bn and the project returns **10.1% — exactly the cost of capital.**

SENSITIVITY — WHAT MOVES THE ANSWER

Variable	NPV10 swing	Note
Capex $3.17bn → $5.0bn	**−$1,102M**	The dominant risk
Brent $70 → $55	−$650M	
Plateau 66k → 45k boe/d	−$683M	
Delay 2 years	−$879M	
Gas price $6 → $3/mmbtu	−$373M	
Opex $12 → $18/boe	−$335M	

Note the ordering has changed from my first memo. **Capex is now first, not third** — because once the fiscal basis is corrected, the project has enough margin that price matters less, and the un-known capex matters more.

WHAT THE TOOLKIT GIVES US THAT WE DID NOT HAVE

1. **A disclosed effective-take benchmark (55.4%).** Every future PETRONAS project screen should be run through it before anything else. It is the cheapest sanity check available.
2. **A defensible reference deck ($65 flat real / $75.40 by 2030).** Screen at $65, not at spot. If it does not work at $65, it is a price bet.
3. **The Volume × Unit earnings decomposition.** Cleaner than my cash-flow stack: it separates *performance* (volume, unit cost) from *price* (sensitivity). Two different management problems, two different owners.
4. **XOM's breakeven definition**: "the oil price required for our cash flow to cover dividends and cash capex." That is a **cash-coverage** definition, not an NPV definition. It is a better question than IRR for a sovereign-owned operator, because PETRONAS does not have shareholders to satisfy — it has a national budget to fund.

WHAT OUR MODEL DOES BETTER THAN THE TOOLKIT

Capability	XOM Toolkit	This model
Project-level DCF / IRR / NPV	✗	✓
PSC cost recovery and profit split	✗ (consolidated only)	✓
Production profile and decline	Guidance only	✓ explicit
Capex WBS by facility	✗	✓
Fiscal regime modelling	✗	✓
Disclosed, audited effective take	✓	calibrated to it

Capability	XOM Toolkit	This model
Reference price deck with inflation basis	✓	adopted

The toolkit is a **corporate earnings bridge**, not a **project gate**. Using it as a gate would be as wrong as using a project DCF to model XOM's share price. But using its sensitivity to calibrate a project model is exactly right, and that is what this section did.

KEY JUDGEMENTS

1. **Limbayong is value-accretive to PETRONAS and unavailable to anyone else.** At $70 Brent and $3.17bn capex: 17.3% IRR consolidated; 5.0% for a foreign partner who would need $96 to reach 10%. Both numbers are true at once. [D]
2. **My first verdict was wrong on basis, and I am retracting it.** The corrected number is +$1.11bn, not +$61M. The error was applying contractor terms to a 100%-owned field. [D]
3. **Capex is the binding unknown.** $3.17bn clears 15%; $5.0bn clears 10%. Public data cannot resolve which. [D]/[U]
4. **At XOM's own reference deck of $65 flat real, the project returns 15.9%.** That is the honest screening result — above hurdle, not spectacular. [D]
5. **Gas is not cosmetic.** It is 39% of boe and worth ±$373M of NPV on a $3 price swing. An unsigned gas sales agreement is a US$373M hole in the sanction. [D]

WHAT WOULD CHANGE MY MIND

- Real capex above $5.0bn (project becomes an IRR-equals-WACC asset)
- Plateau below 55,000 boe/d sustained (NPV falls below $300M)
- Gas price below $5/mmbtu contractually locked
- A farm-out proposal: this model says no external partner can be found at these terms, so any farm-out is a value transfer, not a risk transfer

CONFIDENCE

Economics method: **HIGH**. Fiscal basis correction: **HIGH** — the 100% WI fact is unambiguous. Capex central estimate: **LOW** — it sits below global deepwater benchmarks and is the dominant uncertainty. Reservoir deliverability: **LOW** — no public well data.

SOURCES

1. ExxonMobil Modeling Toolkit — Overview of Modeling Toolkit, 7pp, accessed 19 Oct 2026 [M]
2. ExxonMobil Modeling Basics, July 2026, 16pp — p5 method, p6 Upstream, p8 unit earnings [M]
3. ExxonMobil 2Q 2026 Sensitivities and Key Assumptions, 31 Jul 2026, 12pp — p3 Brent sensitivity +$700M/$1, p4 constant price basis and marker table [M]
4. Global Energy Monitor, Limbayong Gas Field, data year 2023 — 139 MMbbl, 790.9 bcf [M]
5. PETRONAS media release, 1 Jun 2021 — 10 wells, 900–1,200 m, 120 km offshore Sabah [M]
6. Welligence Energy Analytics, 31 Aug 2026 — PCSB 100% WI, discovery 2002 [M]
7. Crude Truth, "Petronas and What a Well costs in Malaysia", 28 Jul 2026 — PSC fiscal architecture, EPT 70% cost recovery, legacy 40% [M]
8. Sabah state sales tax on petroleum, 5% — applied as external leak [A]

S8 — RISK, DECISION & WHAT IT TAKES TO MAKE IT GO

ERRATUM — issued by HERMES, 19 Oct 2026, after independent verification. The claim "*MMHE RM952M MISC contract, 19 Oct 2026*" (appears in R-17, C-1, the Confidence table and the Sources list) is **WITHDRAWN**. The Edge Malaysia article carrying that figure is dated **2011**, not 2026 — "19 Oct 2026" is the page's render date. Its two contracts are *MT Onozo → FPSO Cendor* (Terengganu, Petrofac) and *SS Tenaga Satu → FSU Lekas* (Melaka). Neither concerns Excalibur and neither concerns Limbayong. Verified against the article body, which cites "*financial year ended March 31, 2011*". **Consequence:** the MMHE conversion cost is therefore *entirely unknown* from public data, which makes risk R-17 and condition C-1 *weaker*, not stronger, than stated below. Nothing else in this section is affected.

> ***Issued:** 2026-10-19 · **Series:** Limbayong FDP screen · **AACE Class:** 4/5 (−30%/+50%) on CAPEX. All USD real 2026 unless stated. Base price deck: Brent **US$75/bbl**, gas RM30/MMBtu ≈ **US$7.32/MMBtu**, MRP-floored. Discount rate **10% real** unless stated. 38% PITA, 10% cash payment. First oil 2029 (base) / 2030 (slip). A **25,000-bpd, 19-yr-old cylindrical hull** is being asked to do a **40,000-bpd/150-MMs-cfd** job, and 36% of forecast gross revenue is gas whose export route is **not publicly defined**. That is the single fact every other risk pivots on. Risks are listed with [M]easured, [D]erived, [A]ssumed, [U]nknown tags.*

HEADLINE

- **NPV@10% base case: ~US$0.3 B (positive but thin).** P50 combined point case (~118 MMbbl, capex 2.9 B, Brent 72, gas RM30, +12 mo) is **−US$0.75 B**. P90 ~ **−US$2.0 B**. Monte-Carlo P(NPV<0) ≈ **80%** at AACE Class 4 inputs.
- **Project IRR ≈ 12.1%** at base; **5.6%** at P50; **17.0%** at upside. **It does not clear a 15% real hurdle** under any central case. To clear it you need either **CAPEX ≤ US$1.8 B** (NPV@15% = +$215 M, IRR 17.4%) or **≥180 MMbbl recoverable** (NPV@15% = +$33 M, IRR >15%). Both thresholds sit **outside** the published design basis (CAPEX is already ~US$2.45 B) and the public volume range (139 MMbbl base, 97 P90). At 160 MMbbl the project still fails the 15% test (NPV@15% = −$144 M, IRR 13.8%).
- **Breakeven Brent @10% ≈ US$65/bbl.** At the screen base of US$75 the project is **viable but barely**; the EMV-weighted register pulls it to negative expected NPV.
- **The two cycle-killers are (1) the Excalibur/hull capability gap and (2) gas export.** Both are resolvable, but neither has been resolved at FID.
- **First oil 2029 is implausible.** 2027 sanctions implies <24 months to first oil; industry norm for a 10-well + 1,000-m DW + converted-FPSO development is **36–48 months** after FID. A 12-month slip to 2030 cuts NPV ~$200 M; a 24-month slip flips the base case negative.
- **Recommendation: HOLD** the sanction of irrevocable capex (drill rig, long-lead SURF orders, MMHE conversion kick-off) until the **FIVE CONDITIONS** below are resolved. FID on 28 Aug 2026 was a *political* decision; the *physical* decision is still in front of PETRONAS, MMHE, MISC and TechnipFMC. Implement a structured 6-month Assurance Programme; re-judge at Go/No-Go gate 1 in April 2027.
- **Confidence: 60%** in the FIVE-CONDITIONS framing and the HOLD recommendation; **35%** in any single cost or volume number (the public record is thin and the hull identity is inferential, not contractually public). See CONFIDENCE table.

RISK REGISTER

Risks are ranked by absolute EMV contribution. *Impact* = the marginal change in base-case NPV@10% (US$ M) and the schedule slip (months) if the risk materialises, holding everything else at base. *Owner* is the named party most able to act.

Mitigations are specific, not platitudes.

Risk	Category	P	Impact USD	Impact months	Mitigation	Owner
R-01 Cylindrical Excalibur cannot meet 40 kbopd/150 MMscfd design basis — published 25 kbopd/0.5 Mt oil processing [M], 19 years old, fitted with water injection that may not run at 75 kbwpd	Facilities / Process	0.45	−US$550 M	+18	Engineering re-base: 2-phase topsides upgrade or alternative hull. Pre-sanction “Process Sizing & Equipment List” review. Owner do not commence MMHE Module E (topside fabrication) before ToPS verification.	MISC (hull) / PCSB (overall)
R-02 Gas export route undefined (no pipeline, no FLNG decision public) — gas = 36% of forecast gross at US$75; if stranded → 5.37/MMBtu or zero, −$200–500 M NPV	Facilities / Commercial	0.65	−US$350 M	+12	Mandate FEED for 2 options: (a) 120-km pipeline to Kebabangan/Kimanis cluster, (b) FLNG. Award by Q1 2027. Tie-in capex excluded from FID until route approved.	PCSB Upstream / Gas Malaysia
R-03 Recoverable oil < P50 (118 MMbbl) — public range 139 (base)–97 (P90); -30% volumes flips base negative	Subsurface	0.40	−US$535 M	0	Independent Reserves Audit (RPS/Ryder Scott). 4D repeat seismic over 2027–28. Pilot well with 2-year production logging before 6× template expansion.	PCSB Subsurface
R-04 Reservoir compartmentalisation in toe-thrust sands (SPE 210079 “top-decile complexity”)	Subsurface	0.55	−US$267 M	+6	Pre-emptive FDP 4D-monitoring and step-out appraisal well #11 (~$90 M). Modular “tie-back ready” SURF kit.	PCSB Subsurface
R-05 Drilling capex overrun +25% (10-well deepwater + 1,000 m)	Drilling	0.60	−US$212 M	+5	Two-rig tender for cost competition; benchmark against Gumusut-Kakap unit well cost; staged contract with go-fail gates.	PCSB Drilling
R-06 Wellbore instability / MPD underperformance (sand, clay, abnormal pressure)	Drilling	0.45	−US$140 M	+4	Dedicated geomechanics cell per Limbayong-2; real-time WOB/ROP limits; managed-pressure drilling package optionality.	Drilling sub-committee
R-07 DW rig availability & day-rate spike 2027–28 (GoM 2026: $479–650 k/d; SE Asia lower)	Drilling / Market	0.40	−US$90 M	+8	Lock 2-rig letter-of-intent 2026 Q4; SE Asia long-term rig (Sapura, Valor, DD) preferred.	PCSB Drilling

Risk	Category	P	Impact USD	Impact months	Mitigation	Owner
R-08 Excalibur hull / class / RBWI fails survey; unplanned dry-dock	Facilities	0.50	−US$120 M	+7	Independent class + mooring integrity survey 2026 Q4 before MMHE conversion kick-off. Pre-buy long-lead spare turret/mooring chains.	MISC / DNeX
R-09 Sour/dirty gas (CO2/H2S) → treating + metallurgy	Subsurface / Process	0.30	−US$180 M	+5	PVT lab re-analysis on existing Limbayong-2 samples; specify SS-clad trains from day 1.	PCSB Process
R-10 PCSB first-time deepwater operator capability gap (vs Shell, Murphy)	Execution	0.45	−US$220 M	+9	Secondment of 8–12 Shell/Equinor deepwater discipline leads under 3-yr contract; embed MISC operations team in PCSB.	PCSB HOS / PCSB Upstream
R-11 27-month FID-to-first-oil unachievable (industry norm 36–48 mo)	Schedule	0.85	−US$206 M	+12	Revise first-oil target to 2030; restate investor messaging; staged AFE for sub-packages.	PCSB PMO
R-12 Brent < US$60 sustained (>12 months)	Market	0.35	−US$465 M	0	Hedge 60% of oil for 2 yrs from first oil via costless collar; PSC “shut-in” trigger in PITA.	PCSB Commercial / Treasury
R-13 Gas price capped at MRP RM31.92/MMBtu, no indexation, PETRONAS nominator constraint	Market / Commercial	0.45	−US$225 M	0	Long-term GSPA (10 yr) at MRP × 1.15 indexation; nominate minimum 100 MMscfd.	PCSB Commercial / Gas Malaysia
R-14 PETROS / Sabah state ownership / participation claim (after 2024 Sarawak case)	Fiscal / Political	0.30	−US$300 M	+6	Pre-emptive MoU with Sabah CM, PETROS and PETRONAS; legal opinion on PSC perimeter before committal capex.	PCSB Legal / MPM
R-15 PITA / cost-recovery ceiling renegotiated downward	Fiscal	0.20	−US$180 M	0	2026 PSC law review watching brief; commercial sensitisation runs at 35% PITA.	PCSB Commercial
R-16 Carbon / flaring penalty / CCS obligation post-2027 Carbon Levy (MY Carbon Levy)	Fiscal / ESG	0.30	−US$120 M	+4	Zero-routine flaring design; LPG recovery in 2030 capex add-on; CCS-ready.	PCSB ESG / Facilities
R-17 MMHE Pasir Gudang yard slot /	Facilities / Schedule	0.50	−US$95 M	+8	Dual-yard option (MMHE +	MISC Project

Risk	Category	P	Impact USD	Impact months	Mitigation	Owner
conversion capacity conflict (MMHE 2026 already has ~RM2 B backlog incl. RM952M MISC contract 19 Oct 2026)					Boustead/heavy industries); 24% schedule buffer.	
R-18 No farm-out partner; 100% PCSB carry in capex	Partner	0.50	−US$150 M	+6	Active 2027 Q1 farm-out data-room (target: Shell, Murphy, ENI SEARAH). Modest probability of a Tier-1 at the right number.	PCSB Commercial
R-19 EIA / DOE / marine biodiversity licence delay (coral, marine mammal)	Regulatory	0.25	−US$60 M	+6	EIA submitted within 60 days of FID; community Sabah engagement programme.	PCSB HSE / MPM
R-20 SURF / Subsea 2.0 delivery & installation congestion (TechnipFMC also has Petronas 2026 backlog)	Facilities	0.35	−US$80 M	+5	Subsea supply-warranty bonds; dual source trees/manifolds.	PCSB SURF
R-21 Sand control / completion productivity shortfall	Subsurface	0.35	−US$150 M	+4	Frac-pack vs. open-hole gravel-pack design trial on first 2 producers; real-time downhole gauge.	PCSB Completions

EMV ADJUSTMENT

Unweighted sum of (P × Impact): −US$2,094 M across the 21 risks. Adding this to the base-case NPV of **+US$313 M** gives an **expected (risk-adjusted) NPV of ≈ −US$1.8 B** at the AACE Class 4 risk surface. This is *more* negative than the Monte-Carlo mean of −$727 M (below) because the register stacks every risk at full severity simultaneously, whereas the Monte-Carlo samples them at their stated probabilities; the register number is therefore a *stress-case* upper bound on downside, and the MC mean is the better central estimate.

Category roll-up (Σ P×impact):

Category	EMV $M	% of total
Facilities (R-01 hull, R-02 gas, R-08 hull-class, R-11 schedule, R-17 yard, R-20 SURF)	−786	38%
Subsurface (R-03 volume, R-04 compartment, R-09 sour gas, R-21 sand)	−467	22%
Market (R-12 Brent, R-13 gas price)	−264	13%
Drilling (R-05, R-06, R-07)	−226	11%
Fiscal / Political (R-14 PETROS, R-15 PITA, R-16 carbon)	−162	8%

Category	EMV $M	% of total
Execution (R-10 PCSB capability)	−99	5%
Partner / Regulatory (R-18 farm-out, R-19 EIA)	−90	4%
Total	**−2,094**	**100%**

Facilities is 38% of the register and is entirely self-inflicted — it is the cost of buying a 19-year-old hull and hoping the topsides can be made to work. That is a choice, not geology.

Largest EMV contributors: **R-01** hull/process gap (−$248 M), **R-02** gas export (−$228 M), **R-03** volume (−$214 M), **R-11** schedule (−$175 M), **R-12** Brent floor (−$163 M), **R-05** drilling overrun (−$127 M), **R-13** gas price (−$101 M), **R-14** PETROS (−$90 M). The top five alone sum to −$1,028 M EMV, i.e. ~49% of the total register exposure.

Two caveats the reader must carry. (1) The impacts are *marginal* (each measured from base holding the others at base); because risks are correlated — a gas-route failure (R-02) and a hull/process failure (R-01) are the same physical problem seen from two angles — the register over-counts the downside. (2) **R-18** (no farm-out) has no direct project-NPV impact; it is a *balance-sheet* risk to PCSB (US$2.4 B of 100% carry) and shows up in the register as strategic, not financial. Both caveats push in the same direction: **the register is a stress bound; the Monte-Carlo mean is the central expectation.** Neither is a positive number.

Monte-Carlo (20,000 paths, decorrelated drivers, log-normal CAPEX):

Statistic	NPV@10% (US$ M)
P10	−1,839
P50	−736
P90	+394
Mean	−727
P(NPV<0)	**80.5%**

Ratchet sensitivity (base = $313 M, NPV@10%):

Lever	Δ NPV $M	Abs NPV $M
+12 months schedule	−206	+107
+24 months schedule	−393	**−80** (flips negative)
−30% recoverable oil (97 MMbbl)	−535	−222
+25% CAPEX (US$3.06 B)	−541	−229
Gas stranded at RM22/MMBtu	−332	−19
Brent floor US$55 sustained	−619	−306
Gas route secured @1.15 × MRP	**+218**	+531
CAPEX −US$500 M via farm-out carry	**+442**	+755
Opex US$16 → US$22/boe	−396	−83

Read the asymmetry: the two levers that *create* value are **commercial** (gas pricing, partner carry) and are worth +$218 M and +$442 M respectively. Every **technical** lever is negative. That is the signature of a project whose value sits in its contracts, not its rocks.

Independent conclusion: at the **public** central case, the project is **marginally viable**. The **risk-adjusted** central case is **uneconomic**. The fact that PETRONAS sanctioned it anyway is consistent with a **non-cash NPV** that internalises strategic value (Malaysian contractor, capability-building for PCSB, Gentari-pivot optionality) — but a private contractor at the same numbers would not have sanctioned.

FIVE CONDITIONS

The five conditions that *must* be true for Limbayong to clear a disciplined hurdle (**NPV@15% real > 0, IRR > 15%, and P(NPV<0) < 35%** at the AACE Class 3 ±20% risk surface). None of the five is publicly established today.

#	Condition	Status	Evidence	To Resolve	Cost	Duration	Owner
C-1	The chosen FPSO can deliver the 40-kbopd/150-MMscfd design basis (i.e. Excalibur 25-kbopd hull is upgraded, replaced, or the design is re-based to match)	**UNKNOWN**	MISC bought Excalibur 25-kbopd/300-kbl Sevan SSP300, 2007-built [M, iMarine 20 Aug 2026]. Offshore-Technology "60,000 bpd" 2021 design basis; current BOD says 40 kbopd/150 MMscfd. MMHE RM952 M MISC contract 19 Oct 2026 [M, Edge] covers *some* conversion but topsides upgrade scope is undisclosed.	Independent Process Sizing Review + FEED-level Equipment List by Class Society (DNV/ABS). Re-confirm or re-base. If capex delta > $200 M, abort FID.	US$8–12 M	4–5 months (Complete by Apr 2027)	MISC / PCSB Facilities
C-2	A firm gas export route (pipeline to Kebabangan/Kimanis *or* FLNG) is selected, FEED complete, with a signed HoA with Gas Malaysia / PETRONAS Gas *or* a sanctioned FLNG order — with 1.15 × MRP price indexation and ≥100 MMscfd nomination	**UNKNOWN**	No public statement of gas route. Upstream 28 Aug 2026 "cylindrical FPSO supplied by MISC" silent on export. SPE 210079 abstract does not state route.	Two-parallel-route FEED (1× pipeline, 1× FLNG) + commercial term-sheet. Issue an ITT for engineering, EPC and shipowner.	US$15–20 M (FEED) + US$25–40 M (FEED for FLNG)	6 months (by May 2027)	PCSB Upstream / PCSB Commercial
C-3	An independent Reserves Audit re-states recoverable oil P50 ≥ 135 MMbbl *and* P90 ≥ 100 MMbbl, with compartmentalisation risk explicitly modelled	**UNKNOWN**	SPE 210079 (2022) calls reservoir "top-decile complexity, marginal development" [M, title]. Public base 139 MMbbl, P90 97 MMbbl [D	Independent Reserves Audit (RPS Energy, Ryder Scott, or Netherland Sewell); 4D seismic feasibility; pilot well 12-month test.	US$3–5 M (audit) + US$90–120 M (appraisal well)	9–14 months (audit Q1 2027; appraisal Q3 2027)	PCSB Subsurface

#	Condition	Status	Evidence	To Resolve	Cost	Duration	Owner
			from public range].				
C-4	A farm-out / strategic partner is signed at ≥30% WI with carry ≥$700 M (or equivalent), OR PCSB issues internal commitment at the lower IRR (≤12%) and accepts the strategic-value thesis	**FALSE**	100% PCSB WI [M, Welligence]. No public farm-out campaign.	2027 Q1 data-room; bilateral with Shell/Murphy/ENI. If no Tier-1, accept PCSB internal committee (MSS-rationale; partial state-carrier view).	US$4–6 M (process)	6–9 months (by Q3 2027)	PCSB Commercial / MPM
C-5	A credible 36-month FID-to-FO schedule (first oil **Q4 2030 or Q1 2031**) is baselined, MMHE conversion + TechnipFMC iEPCI + drilling 10 DW wells all on a single integrated schedule with shared milestones, and the project is publicly re-stated at that schedule	**FALSE**	27-month 2027-12 → 2029 FO implied; Upstream 28 Aug 2026 silent on FO date. Wood Mackenzie cited 2028 (per Fact Base).	Integrated Master Schedule with critical-path identification; P50 and P90 disclosed; quarterly steering-committee until first oil.	US$2–3 M (planning)	3 months (by Feb 2027)	PCSB PMO + MISC + TechnipFMC

Reading: three of five are UNKNOWN (C-1 hull, C-2 gas, C-3 reserves) and **two are FALSE on the public record** (C-4 partner, C-5 schedule). Zero of five are TRUE. The FID on 28 Aug 2026 was therefore a *commitment to find out*, not a *commitment that the answers are known*. That is a defensible strategy **only if** the next 6 months deliver a Gate-1 Go/No-Go that can still abort irrevocable capex — and, on the public evidence, there is no sign such a gate has been defined.

PRE-FID ASSURANCE PROGRAMME

The FDP-quality study the public record *cannot* supply — required to bring C-1 to C-4 from UNKNOWN/FALSE to TRUE — and what it costs vs. the institutional norm.

Workstream	Scope	Cost (US$ M)	Duration	Owner
Independent Process Sizing Review + Class Soc survey on Excalibur	Topside redesign feasibility, mooring/turret integrity, equipment list, re-cap	8–12	4–5 mo	MISC / DNV
Gas export FEED x2 (pipeline + FLNG) + term-sheet	120-km pipe route, riser, onshore tie-in; FLNG deck + mooring	30–50	6 mo	PCSB Upstream / PCSB Commercial
Independent Reserves Audit + PVT re-analysis	Recompute 2P/3P; compartmentalisation model; gas composition	4–6	3–4 mo	PCSB Subsurface
Appraisal well #11 (Bestari step-out)	Drilling, logging, MDT, 60-day test, PVT	90–120	9–12 mo	PCSB Drilling
4D baseline seismic	Acquire 600 sq km 4D over 2027 (after first	25–40	6–8 mo	PCSB G&G

Workstream	Scope	Cost (US$ M)	Duration	Owner
	production)			
Integrated Master Schedule, risk-loaded	Critical path, Monte-Carlo FO date	2–3	3 mo	PCSB PMO + MISC + TechnipFMC
Farm-out data-room + commercial process	Bilateral Tier-1 partners	4–6	6–9 mo	PCSB Commercial
Independent third-party review (Operator A)	12-week FDP-grade audit, gate review	3–5	3 mo	PCSB + external
TOTAL		**~US$170–240 M**	**~14–18 mo critical path**	

Institutional FDP norm for a 270-MMboe, 1,000-m DW, 10-well project with a converted FPSO and floating gas export: - 12–18 months integrated FDP study (FEED + GeoSciences + Process + Commercial + Project Services) - US$80–150 M direct + US$200–400 M of internal staff cost - Three formal gate reviews (GOR, GOC, GOL) - 1,500–2,500 engineering drawings - An independent operator peer review

Comparison. What PETRONAS appears to have done in practice (the 28 Aug 2026 FID with the 27-month FO): an aggressive parallel-tracking of FEED, hull acquisition (MISC Excalibur), SURF award (TechnipFMC) and class-society engagement, with no public reserves audit, no public gas route, and no public schedule rebaseline. The cost *saved* is ~US$100–200 M of front-end study money, but the *risk* is to put ~US$1.5–2.5 B of irrecoverable capex on the back of an internal-only FDP.

Verdict on Assurance Programme: the *missing* US$170–240 M of independent FDP work is a **premium for confidence**, not a *necessity* if PETRONAS is willing to accept the negative expected NPV in exchange for strategic optionality. **The honest answer is that FID was not, technically, complete**; what was sanctioned was a *structured commitment to complete it* with a built-in 6-month window in which to abort. The Assurance Programme above is the work needed to **close that window** in a way a sovereign board can defend to public scrutiny.

RECOMMENDATION

HOLD — formalise the **6-month Assurance Window** between FID and Go/No-Go Gate 1 (target April 2027). In that window:

1. **Stop MMHE Module E** (topside fabrication) until **C-1** is closed. (Estimated opportunity cost: US$30–50 M of re-work if FEED is wrong; loss of 6 months schedule.)
2. **Re-baseline first oil to Q4 2030** as the *contracted* plan. Public messaging: “FID 2026; first oil 2030; this is a 4-year development, not a 27-month one.”
3. **Launch the gas-export ITT** in October 2026; award FEED December 2026.
4. **Issue a competitive farm-out data-room** by January 2027 to Shell, Murphy, ENI.
5. **Authorise the Reserves Audit** October 2026; completion Q1 2027.
6. **Establish a sovereign Board-level Risk Committee** with monthly progress reports on C-1 to C-5.

Go/No-Go Gate 1 — April 2027: - **GO** if C-1, C-2, C-3 are TRUE at low-cost and C-4 is in advanced negotiation. - **NO-GO** (i.e. cancel MMHE conversion, return TechnipFMC iEPCI) if **any** of C-1, C-2, C-3 returns FALSE. - **REVIEW** if C-4 is FALSE only (project proceeds, but with a written acknowledgement of NPV erosion).

This is not a “do not sanction” recommendation. It is a *sanctioned* recommendation that the public case for the *commitment to spend ~$2.4 B* depends on a small set of well-defined facts that have not yet been established. If they are established as TRUE by April 2027, the project clears the hurdle. If they are not, the right call is a structured unwind that protects the Malaysian fabrication chain (MMHE, MISC, TechnipFMC) by routing the assets to other PETRONAS projects (Gumusut-Kakap phase-2, Kelidang, Rotan and similar satellite tie-backs) — not a forced no-go.

CONFIDENCE

Claim	Confidence	Why
Five conditions are *the right* five	55%	A different framing (e.g. carbon price, partner economics) could rank them differently.
Base NPV ≈ +US$300 M, IRR ≈ 12%	50%	AACE Class 4 inputs are wide; opex and CAPEX are judgement, not source. Monte-Carlo mean is −US$727 M.
P50 NPV negative	65%	Volume cap and CAPEX inflation in the public record both skew negative.
First oil 2029 implausible	75%	Industry norm vs published schedule gap is material.
FID was *politically* correct	70%	24-year discovery, MISC/TechnipFMC/MMHE chain, PCSB first DW leadership — all push to a decision.
The hull Excalibur is wrong for the design	55%	Stated 25 kbopd vs 40 kbopd; not yet contradicted by MISC.
Gas route is the single biggest commercial risk	70%	36% of gross revenue; no public route; pipeline 120 km is a real number.

Overall confidence in the FIVE-CONDITIONS framing and the HOLD recommendation: 60%. The framing is a *defensible operator*'s view, not a *right* view. A second independent reviewer at AACE Class 3 could legitimately conclude the project clears at base if (a) the hull upgrade is in fact feasible at <$200 M, (b) gas pricing is indeed >1.0 × MRP, and (c) the schedule slip is <12 months.

SOURCES

Public sources (cited).

- [M] iMarine, "DNeX to Sell Idle FPSO Excalibur to MISC for $35 Million", 20 Aug 2026 — hull spec (300,000 bbl storage, 25,000 bpd, 19-yr old, built 2007).
- [M] Teekay, "Hummingbird Spirit FPSO" — original 270 k bbl / 25 kbopd / 47 POB.
- [M] The Edge Malaysia, "MMHE lands jobs worth RM952m from MISC", 19 Oct 2026 — conversion contract value.
- [M] Offshore-Technology, "Limbayong Deep-water Development Project, Offshore Sabah, Malaysia", 15 Jun 2021 — 40,000 bopd design basis, $600–800 M FPSO capex, 10 wells.
- [M] The Star, "DNeX proposes US$35mil sale of FPSO Excalibur to MISC", 20–21 Aug 2026.
- [M] Upstream, "Petronas finally takes FID on Limbayong deepwater development", 28 Aug 2026.
- [M] SinorigOffshore, 26 Aug 2026 — MISC to bring Excalibur back to Malaysia for conversion; Limbayong link.
- [M] Upstream, "Limbayong highlights continuing post-covid toll on project economics", 24 Mar 2023 — peak 40,000 bopd, viable at $50/bbl.
- [M] TechnipFMC PR / MaritimeMagz 15 Sep 2026 — iEPCI US$75–250 M award.
- [M] Welligence — operator 100% PCSB.
- [M] PETRONAS Production Sharing Contract reference terms (MPM) — 70% cost-recovery ceiling, 38% PITA, 10% cash payment, profit-sliding scale.
- [M] Malaysia Reference Price (MRP) — RM31.92/MMBtu (Mar 2026), RM33.05/MMBtu (Q2 2026).
- [M] Global Energy Monitor (2023) — 139 MMbbl + 22,398.75 MMm³.
- [M] ResearchGate / SPE 210079 (2022) — "Limbayong: Decoding Industry Top Decile Reservoir Complexity for Marginal Deepwater Development", PCSB authors.
- [A] Industry benchmark — 10-well DW drilling ~US$1.0–1.4 B; SURF ~US$150–300 M; topsides new-build 60 kbopd US$600–800 M; converted Sevan up-cycle 150–350 M.

Internal models.

- S8 cashflow (this document) — base NPV@10% +US$313 M; IRR 12.1%; P(NPV<0) 80.5%.
- Monte-Carlo 20,000 paths; log-normal CAPEX (σ 26%); volume {97, 118, 139, 160} sampled by 40/35/18/7; Brent log-normal around US$72 (σ 30%); gas {RM22, 30, 38} by 45/35/20.

Dead paths / things deliberately not assumed.

- No public PSC perimeter, so PETROS risk (R-14) is *political* probability not a *legal* claim.
- No public new seismic 2024–2026; any “4D” assumption is forward-looking.
- No public contract for the hull upgrade scope; assumed subset of MMHE RM952 M.
- No public schedule with milestones; “first oil 2029” treated as marketing, not engineering.